

Overall Report

Total Average

Looking across all **9 dataset families × 3 noise levels = 27 benchmark cases**, the result is more nuanced than STC is worse than the existing algorithms.

Method	Mean ARI over 27 cases
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STC / SHATTER	0.6650
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HDBSCAN	0.6215
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GMM (BIC)	0.6201
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DBSCAN	0.6046
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UPGMA	0.6002
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Ward	0.5931
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k-means	0.5918
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OPTICS	0.5894
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So, by **raw mean ARI**, STC comes out highest.

Family Summary

The average hides a very strong dependence on cluster geometry. The family summaries show that clearly.

Family	STC mean ARI	Best primary method	Interpretation
Density bridge	0.7862	k-means 0.9404	STC competitive, but several baselines better
Different densities	0.8641	k-means 0.9860	STC excellent until high noise

Family	STC mean ARI	Best primary method	Interpretation
Elliptical Gaussian	0.6396	UPGMA 0.9851	Clear STC weakness
Moons	0.6251	DBSCAN 0.7268	STC strong for non-convex geometry, density baselines stronger
Nested structures	0.8768	STC 0.8768	Clear STC strength
Noise / outliers	0.8763	GMM 0.8951	STC highly competitive
Rings	0.6869	STC 0.6869	Clear STC strength
Spherical blobs	0.5956	k-means 0.9420	Clear STC weakness
Spirals	0.0340	HDBSCAN 0.0407	All methods essentially fail

Noise Level Difference

Noise level STC mean ARI

Low **0.8461**

Intermediate **0.6834**

High **0.4654**

So STC is very strong overall at low noise, remains competitive at intermediate noise, but deteriorates substantially under high noise.

For comparison, at high noise GMM averages about **0.602**, k-means 0.548, Ward 0.536, and UPGMA 0.530. Thus, conventional methods become stronger overall under the strongest perturbation.

The outlier experiment is important because STC performs surprisingly well there. Its mean ARI is **0.8763**, second only slightly to GMM at 0.8951. Even at high contamination STC retains ARI 0.7319.

So, the weakness is not simply “STC cannot handle noisy data.”

STC appears relatively robust to isolated outliers, but less robust when noise directly perturbs the continuous local structure of the clusters.

Some Findings

Geometry Dependence

STC is particularly good when the cluster definition is based on a complicated shape rather than compactness around a centre. The best examples are **rings** and **nested structures**. For rings, STC reaches mean ARI 0.6869 while HDBSCAN, the next strongest baseline, reaches only 0.4256; most classical methods are essentially unable to recover the concentric structures.

Nested structures are even more convincing: STC reaches **0.8768**, while the next-best primary baseline, HDBSCAN, reaches 0.6093. STC remains strongest even at intermediate and high noise.

Moons support the same idea, although less strongly. At low noise STC obtains ARI **0.9804**, and at intermediate noise it still substantially outperforms k-means, GMM, UPGMA, and Ward. DBSCAN/HDBSCAN/OPTICS are more robust there.

So, there is real evidence supporting the **shape-truthful** motivation.

Where STC is Bad

On conventional compact structures, especially **spherical blobs and elliptical Gaussians**, STC performs considerably worse.

For spherical blobs, STC's mean ARI is 0.5956 compared with around **0.94** for k-means and GMM.

For elliptical Gaussian clusters, STC reaches only **0.6396**, while k-means, GMM, UPGMA, and Ward are around **0.98**.

That is not especially surprising scientifically: those datasets satisfy the assumptions of the conventional methods quite well. STC's additional geometric flexibility does not provide an advantage there.

Spirals Datasets

The spiral experiment does not support STC, but it also does not favour another method. We need to update spiral datasets.

Q is not sufficient

Several experiments show that STC's internal quality measure can disagree with the known truth.

The clearest example is low-noise spirals:

- ARI = **0.0727**
- Q = **0.8469**
- coverage = **0.9947**

So, an internally coherent density partition can still be completely different from the intended clusters. This strongly supports using ARI/NMI alongside the internal STC score during evaluation.

Density-bridge family

1. Low noise

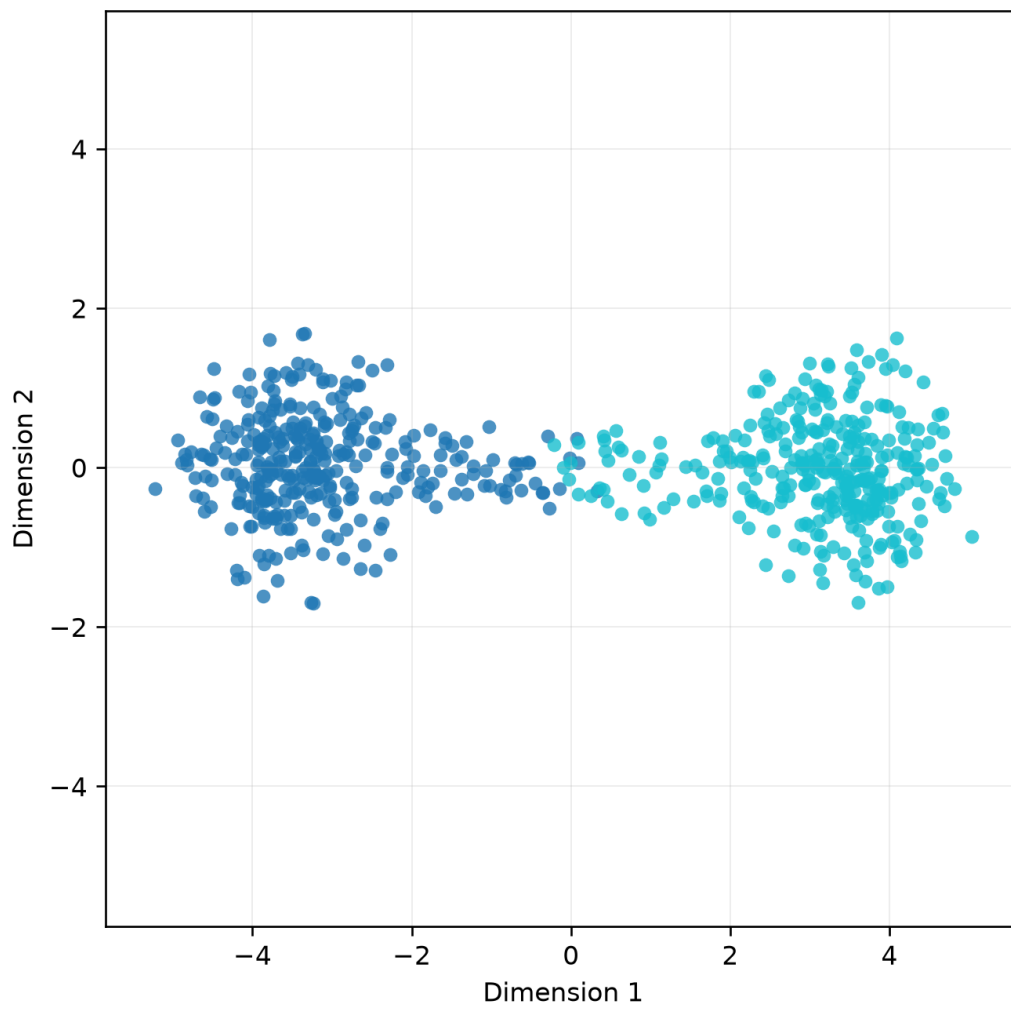
Here STC performs reasonably well:

True K=2

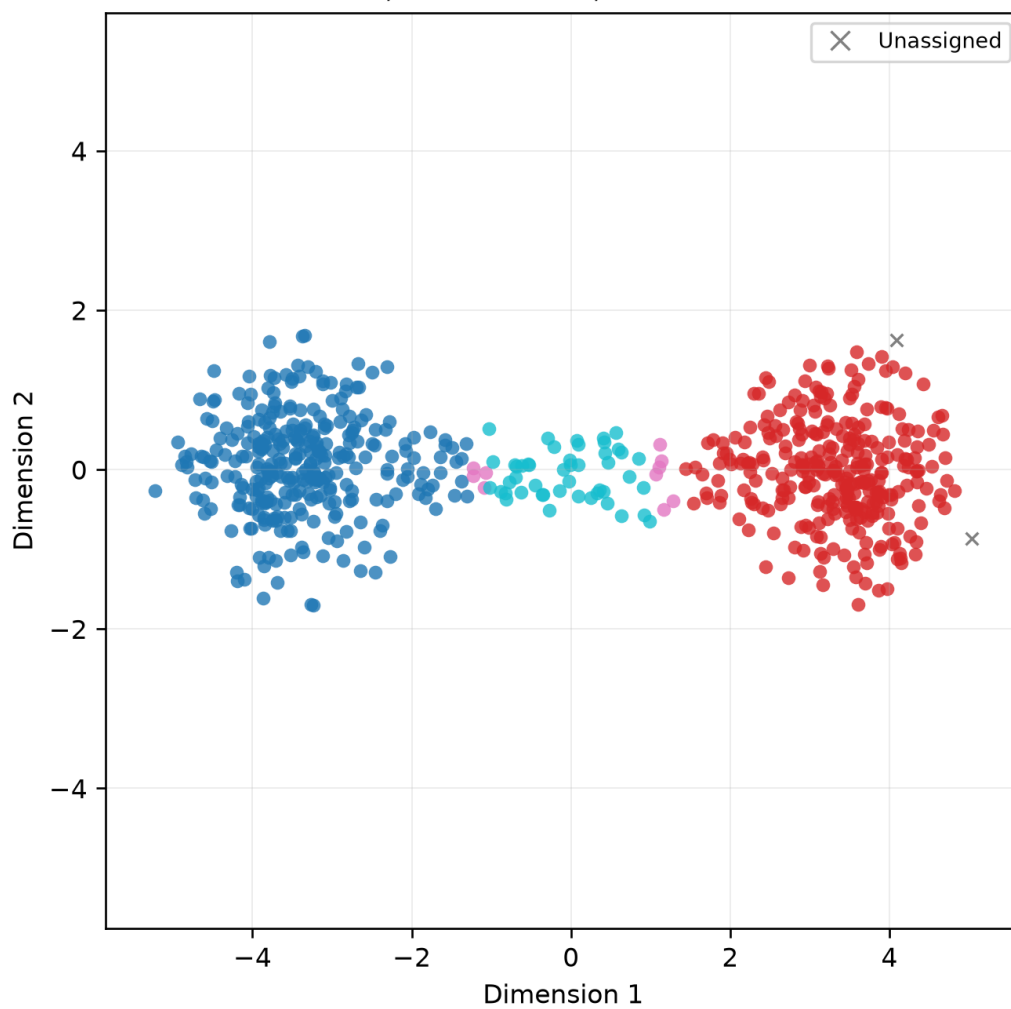
- **STC: ARI 0.8291, NMI 0.7548, K=4**
- DBSCAN: ARI 0.8076, NMI 0.7270, K=2
- HDBSCAN: ARI 0.7650, NMI 0.6817, K=3
- OPTICS: ARI 0.6955, NMI 0.6829, K=2
- **k-means: ARI 0.9639, NMI 0.9260, K=2**
- GMM-BIC: ARI 0.7261, NMI 0.6959, K=3
- UPGMA: ARI 0.8317, NMI 0.7818, K=2
- Ward: ARI 0.8317, NMI 0.7818, K=2
-

K-means performs best. STC has reasonably strong ARI/NMI but overclusters with **K=4 instead of K=2**.

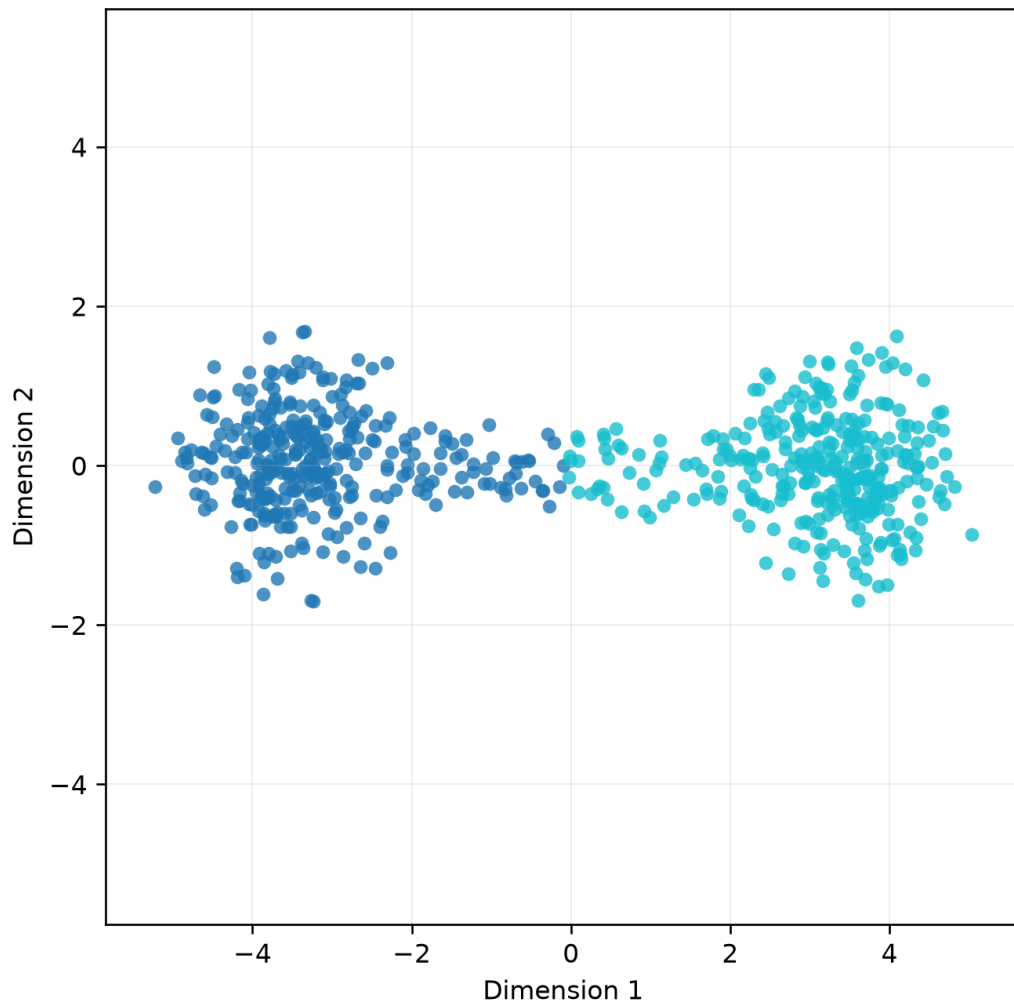
Ground truth
K=2



SHATTER partition
K=4, ARI=0.8291, NMI=0.7548



Best competitor: k-means (elbow K)
K=2, ARI=0.9639, NMI=0.9260



2. Intermediate noise

This is where STC starts falling behind:

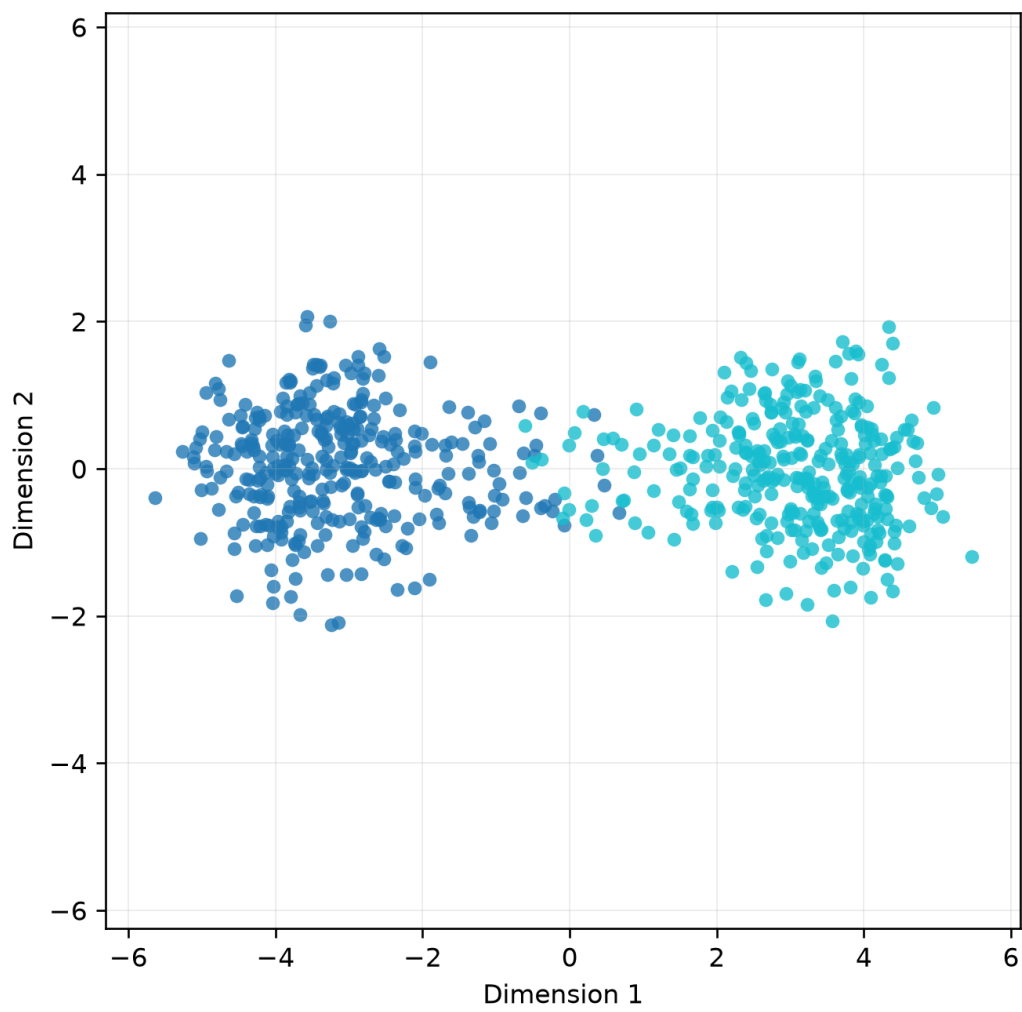
True K=2

- **STC: ARI 0.7630, NMI 0.6720, K=8**
- DBSCAN: ARI 0.8209, NMI 0.7373, K=2
- HDBSCAN: ARI 0.6569, NMI 0.5974, K=7
- OPTICS: ARI 0.6578, NMI 0.6477, K=2
- **k-means: ARI 0.9461, NMI 0.8961, K=2**
- GMM-BIC: ARI 0.7866, NMI 0.7430, K=3
- UPGMA: ARI 0.9285, NMI 0.8723, K=2
- Ward: ARI 0.9168, NMI 0.8726, K=2

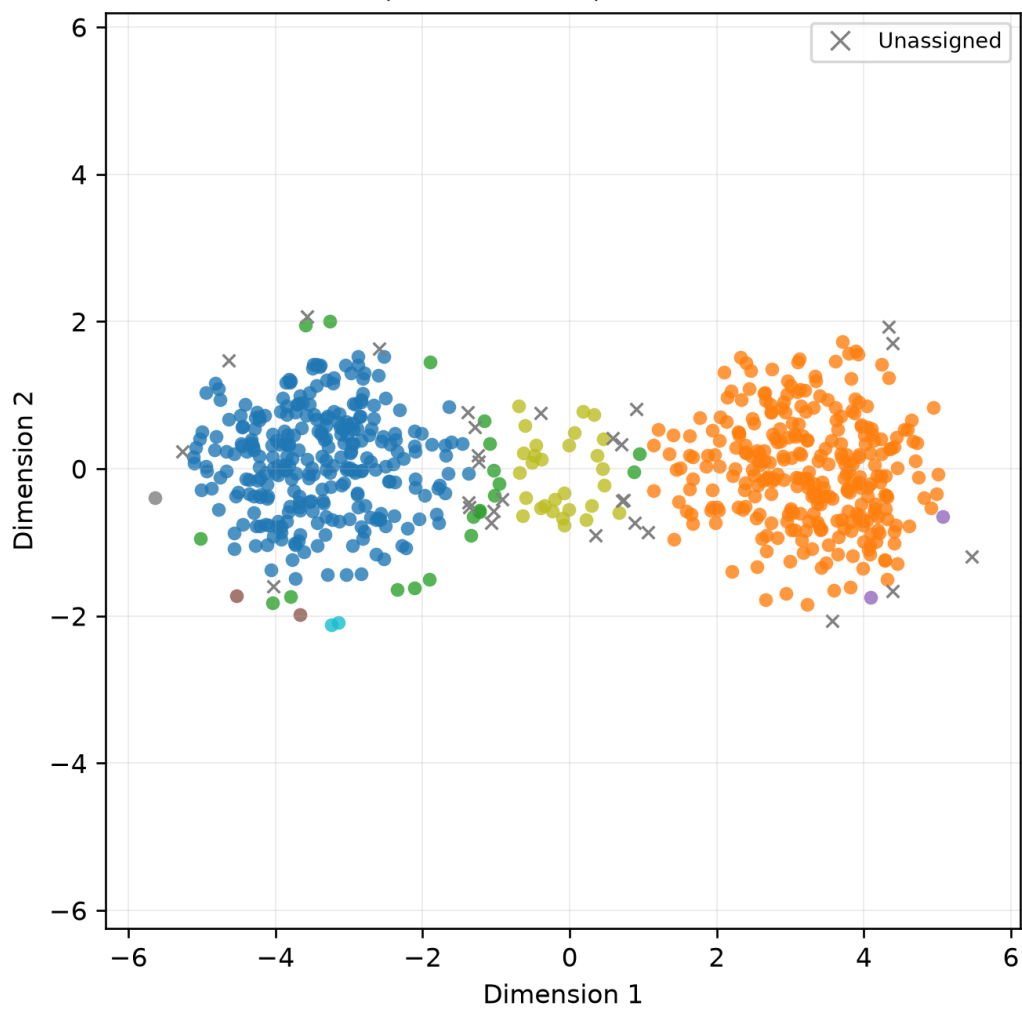
K-means performs best. STC still retains moderate agreement with the ground truth, but overclusters strongly with **K=8 instead of K=2**.

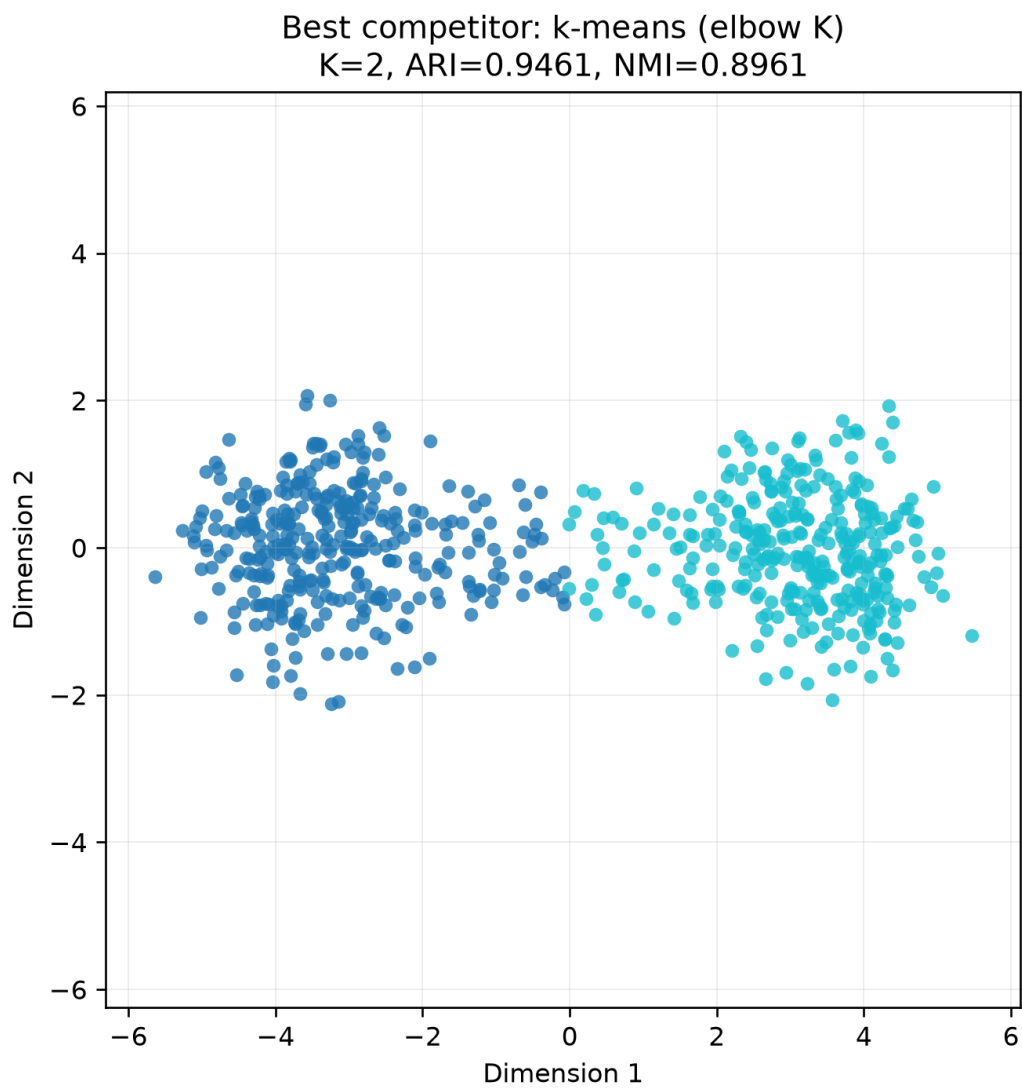
So, five of the seven primary baselines beat STC here; only HDBSCAN and OPTICS are worse

Ground truth
K=2



SHATTER partition
K=8, ARI=0.7630, NMI=0.6720





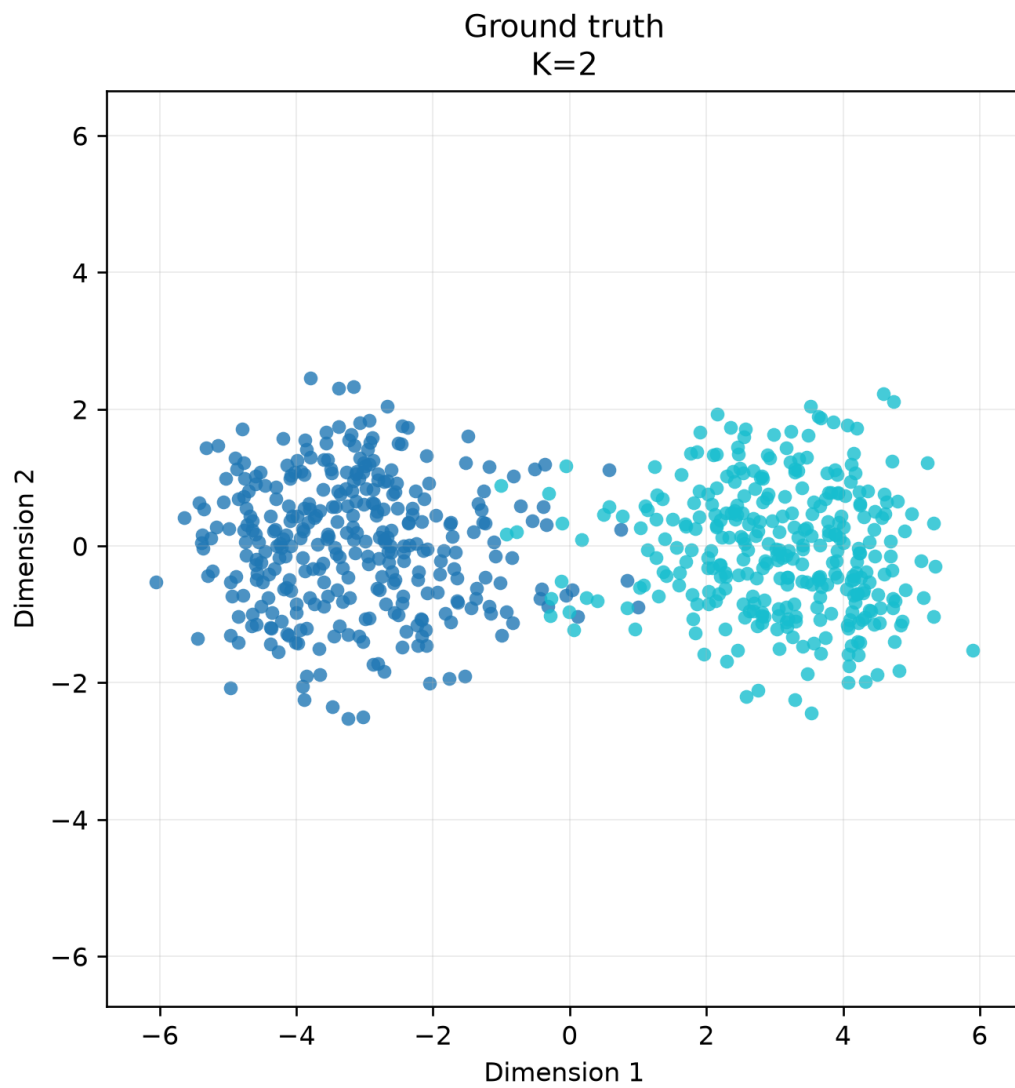
3. High noise

The overclustering problem becomes much stronger:

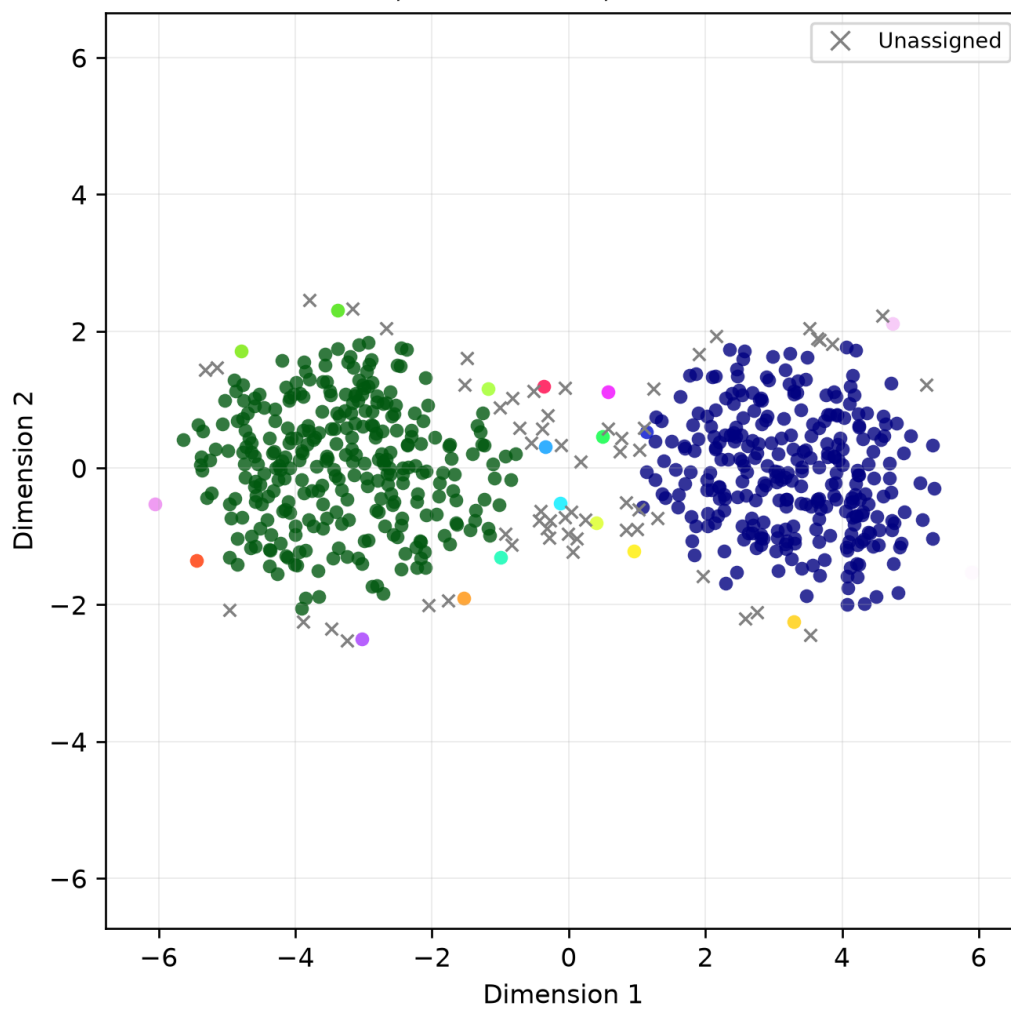
True K=2

- **STC: ARI 0.7666, NMI 0.6744, K=21**
- DBSCAN: ARI 0.8400, NMI 0.7350, K=2
- HDBSCAN: ARI 0.7919, NMI 0.6864, K=3

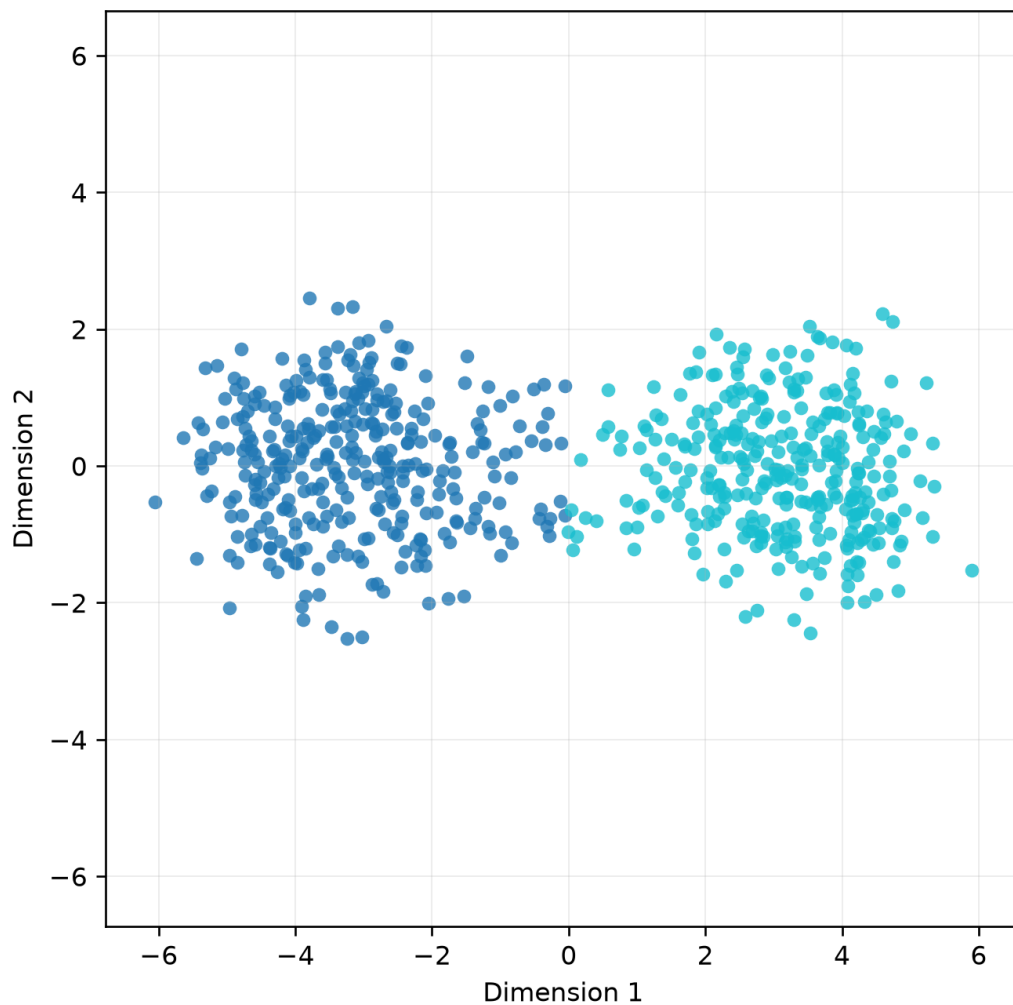
- OPTICS: ARI 0.8624, NMI 0.7753, K=2
- **k-means: ARI 0.9110, NMI 0.8441, K=2**
- GMM-BIC: ARI 0.8995, NMI 0.8326, K=2
- UPGMA: ARI 0.8709, NMI 0.8072, K=2
- Ward: ARI 0.8995, NMI 0.8279, K=2



SHATTER partition
K=21, ARI=0.7666, NMI=0.6744



Best competitor: k-means (elbow K)
K=2, ARI=0.9110, NMI=0.8441



Report

K-means performs best. STC still gives moderate ARI/NMI, but severely overclusters with **K=21 instead of K=2**.

A more important limitation is STC's increasing over-fragmentation. Although the ground truth contains two clusters, STC predicts 4 clusters at low noise, 8 at intermediate

noise, and 21 at high noise. Its ARI remains relatively stable despite this fragmentation—0.8291, 0.7630, and 0.7666—but the growing discrepancy in cluster count indicates that STC increasingly subdivides the underlying structures as the data become noisier.

Different Density

1. Low noise

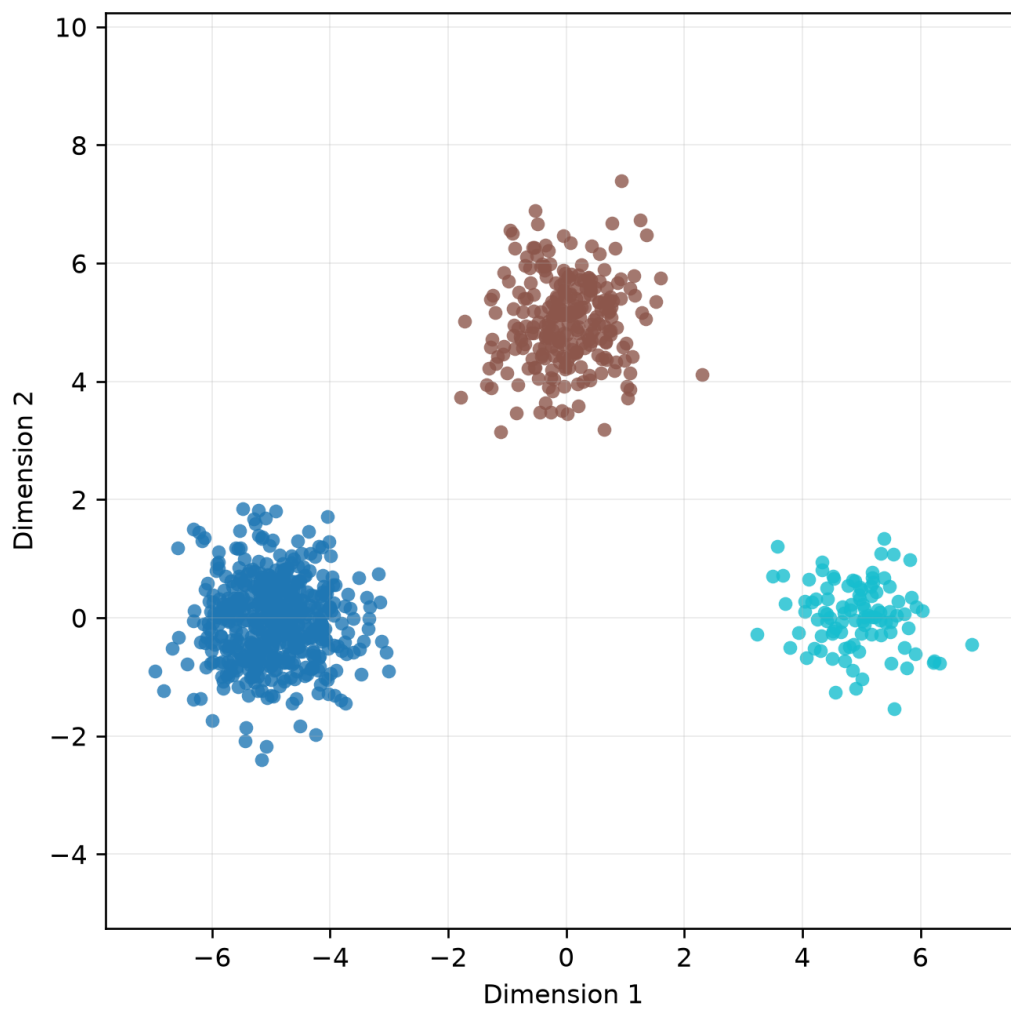
Here STC performs almost perfectly:

True K=3

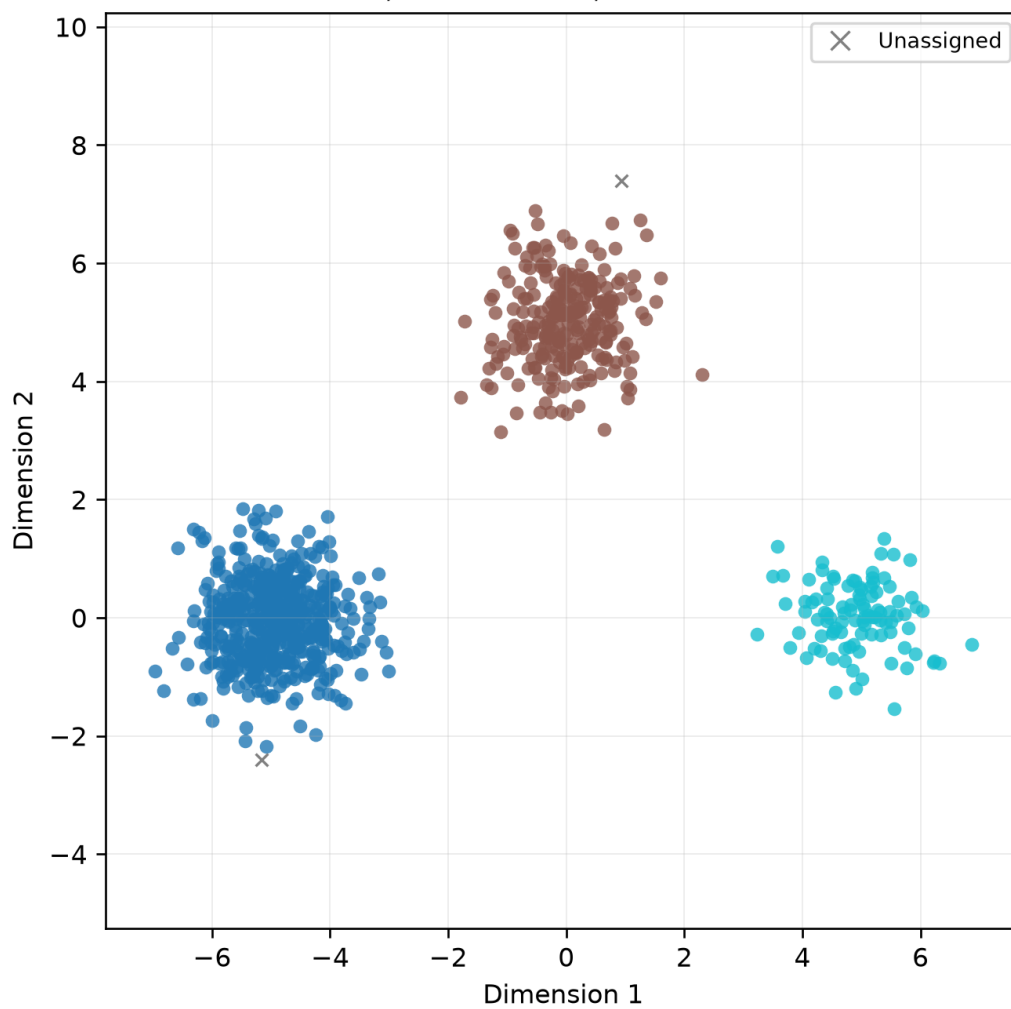
- **STC: ARI 0.9958, NMI 0.9904, K=3**
- DBSCAN: ARI 0.9950, NMI 0.9798, K=3
- HDBSCAN: ARI 1.0000, NMI 1.0000, K=3
- OPTICS: ARI 1.0000, NMI 1.0000, K=3
- **k-means: ARI 1.0000, NMI 1.0000, K=3**
- GMM-BIC: ARI 1.0000, NMI 1.0000, K=3
- UPGMA: ARI 1.0000, NMI 1.0000, K=3
- Ward: ARI 1.0000, NMI 1.0000, K=3

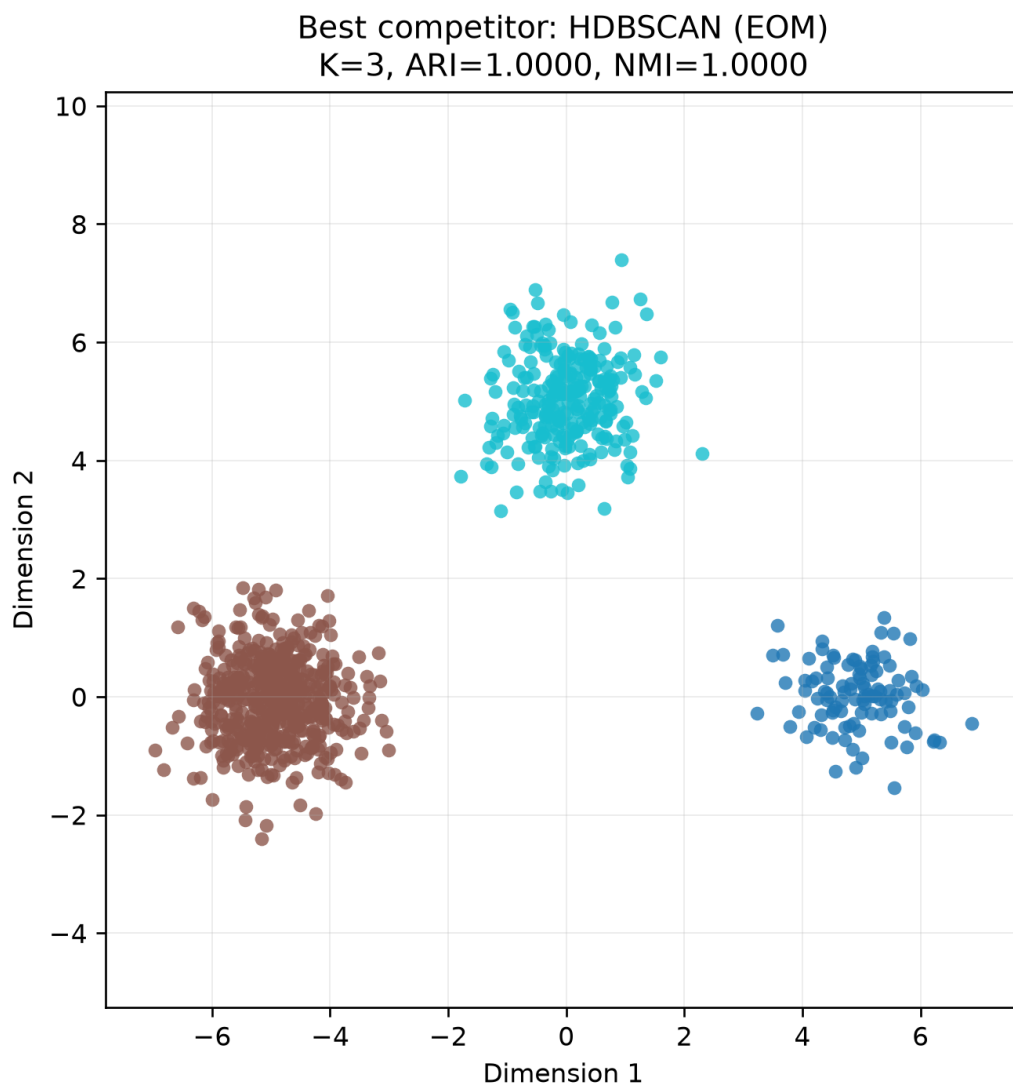
Most methods recover the structure perfectly. STC also performs extremely well and correctly identifies **K=3**.

Ground truth
K=3



SHATTER partition
K=3, ARI=0.9958, NMI=0.9904





2. Intermediate noise

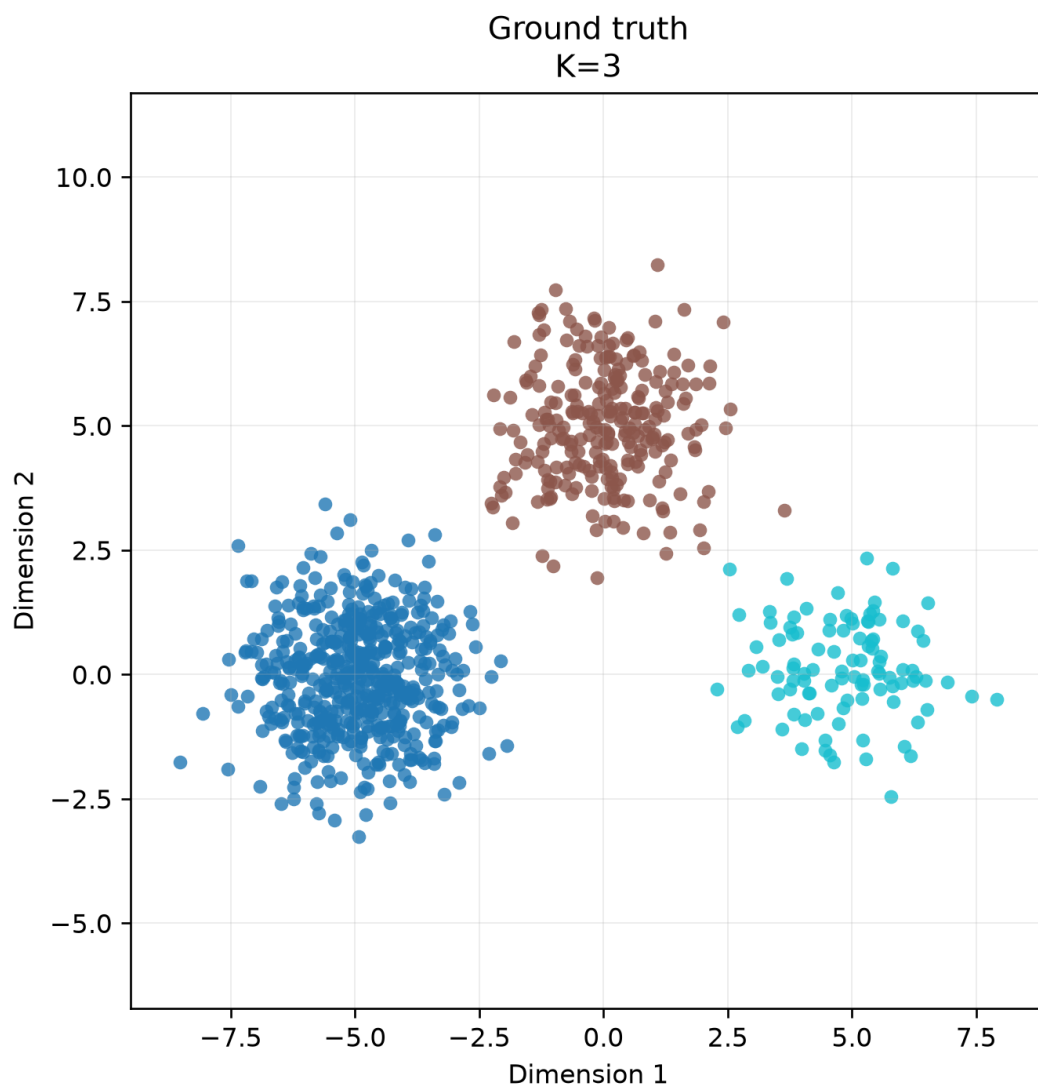
STC remains very strong:

True K=3

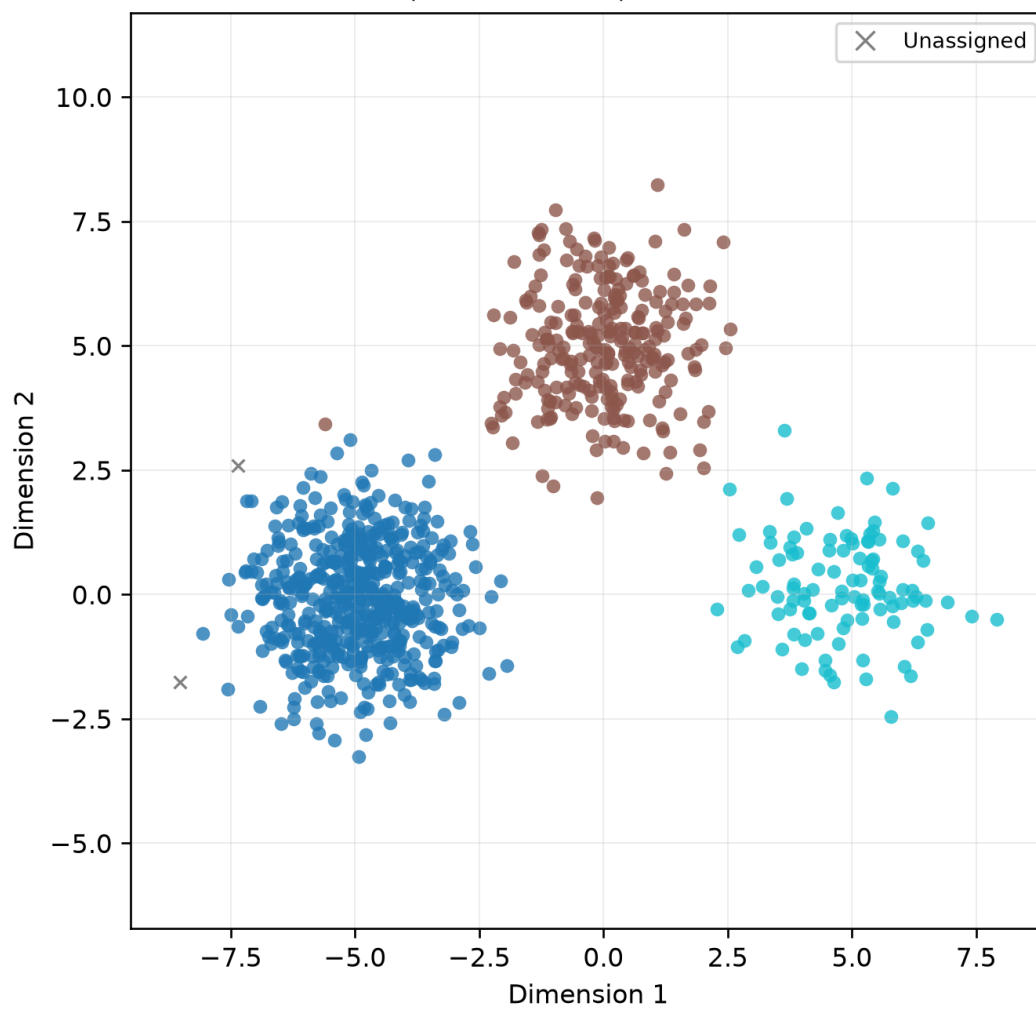
- **STC: ARI 0.9883, NMI 0.9755, K=3**
- DBSCAN: ARI 0.9911, NMI 0.9762, K=3

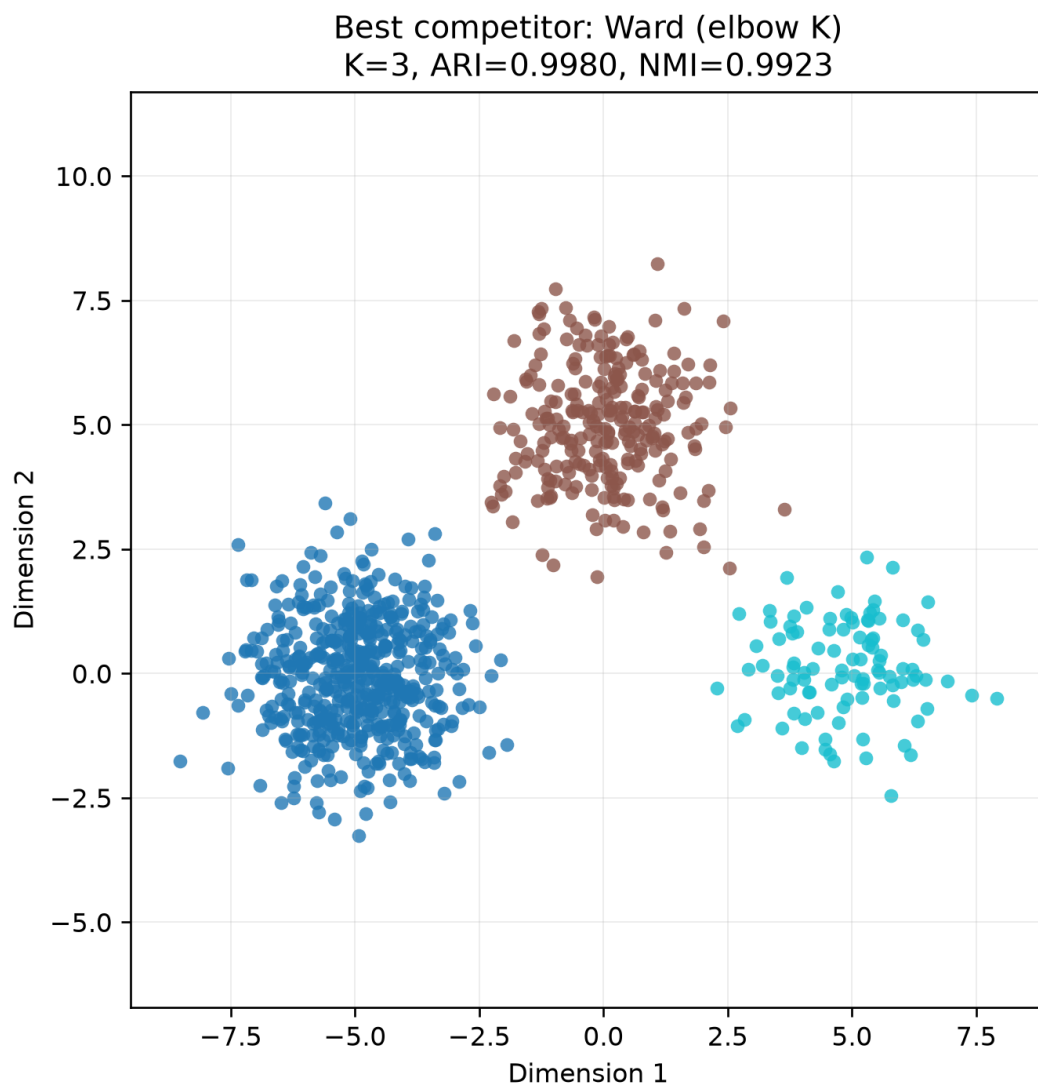
- HDBSCAN: ARI 0.9886, NMI 0.9648, K=3
- OPTICS: ARI 0.9980, NMI 0.9923, K=3
- **k-means: ARI 0.9980, NMI 0.9923, K=3**
- GMM-BIC: ARI 0.9980, NMI 0.9923, K=3
- UPGMA: ARI 0.9980, NMI 0.9923, K=3
- Ward: ARI 0.9980, NMI 0.9923, K=3

All methods remain very accurate. STC correctly identifies **K=3**, with only a small ARI/NMI difference from the strongest baselines.



SHATTER partition
K=3, ARI=0.9883, NMI=0.9755





3. High noise

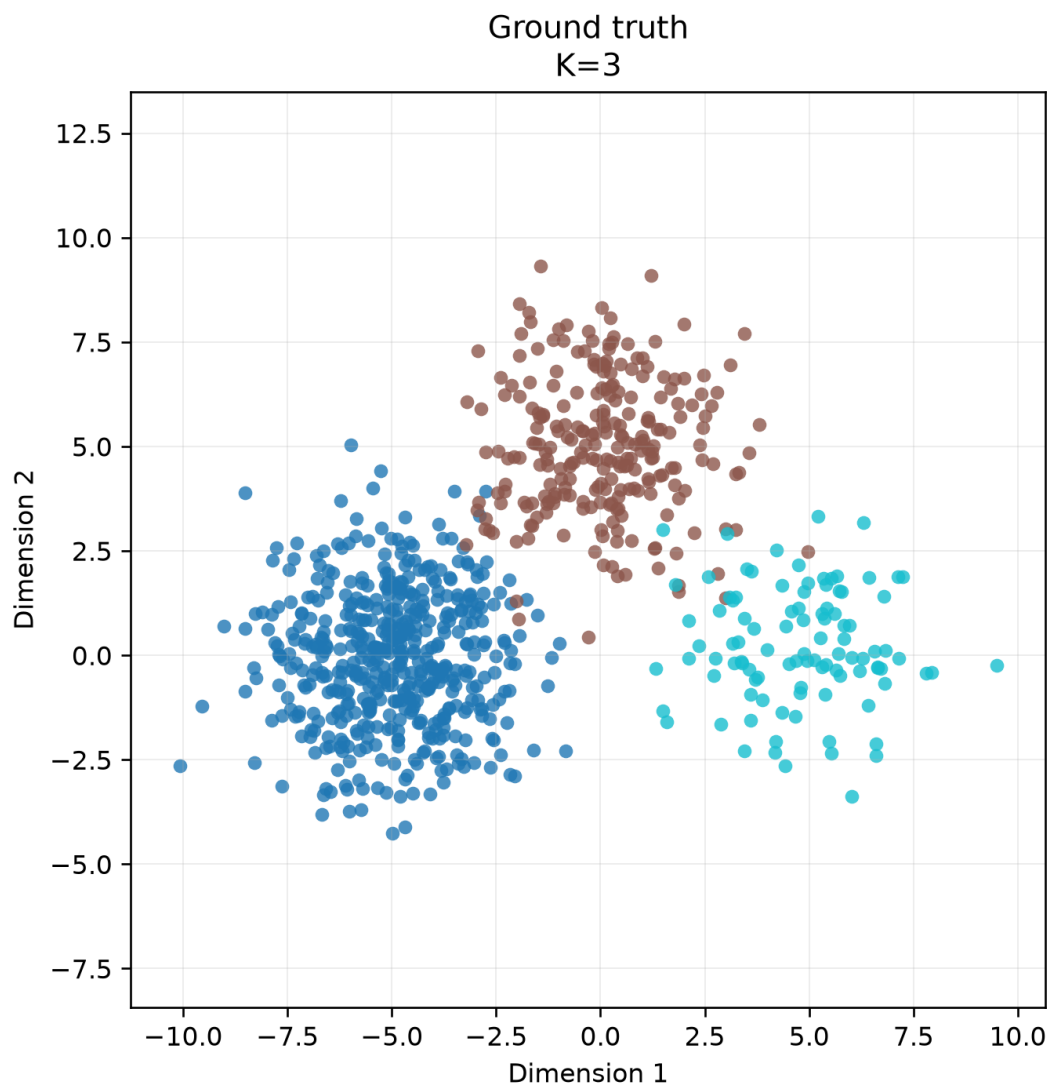
Here STC performance drops substantially and strong overclustering appears:

True K=3

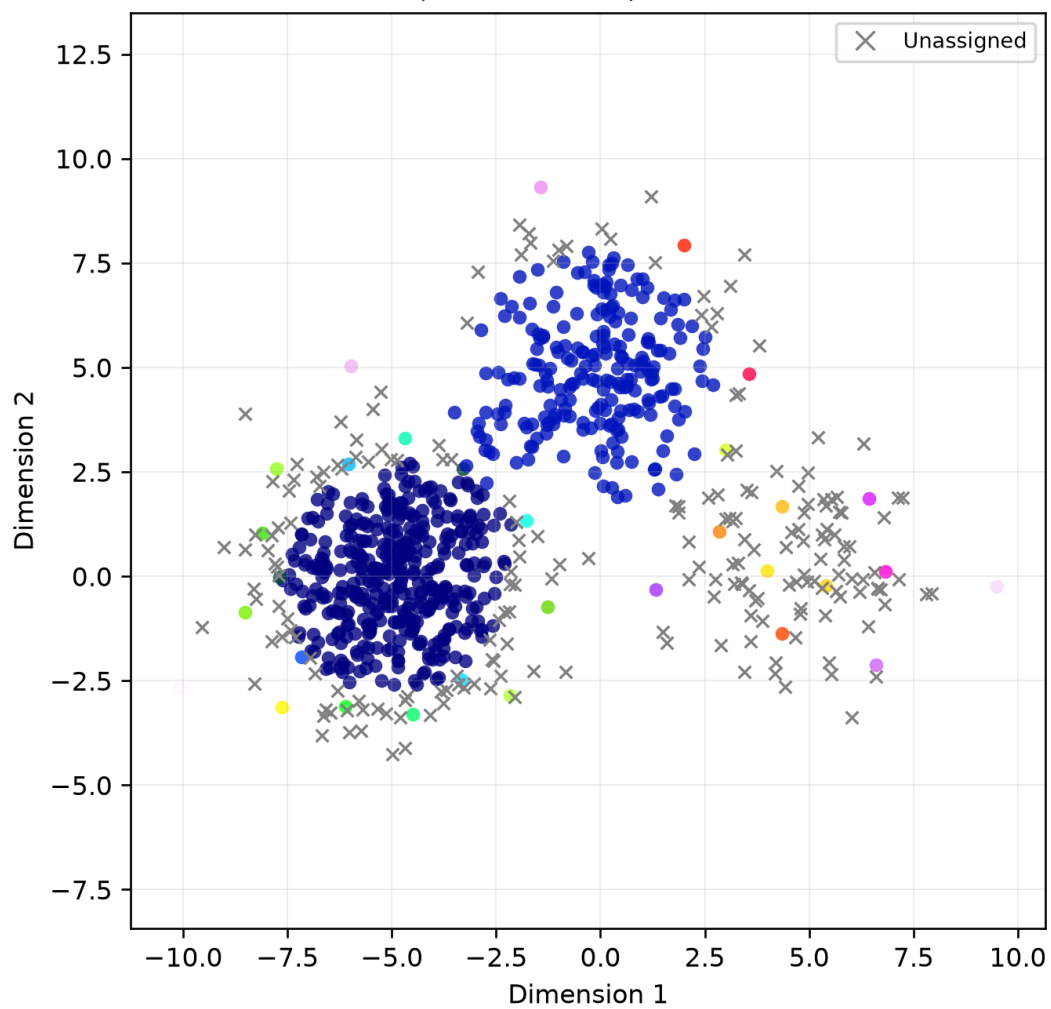
- **STC: ARI 0.6082, NMI 0.5800, K=33**
- DBSCAN: ARI 0.7780, NMI 0.6745, K=5
- HDBSCAN: ARI 0.1671, NMI 0.3585, K=27
- OPTICS: ARI 0.3309, NMI 0.4069, K=2

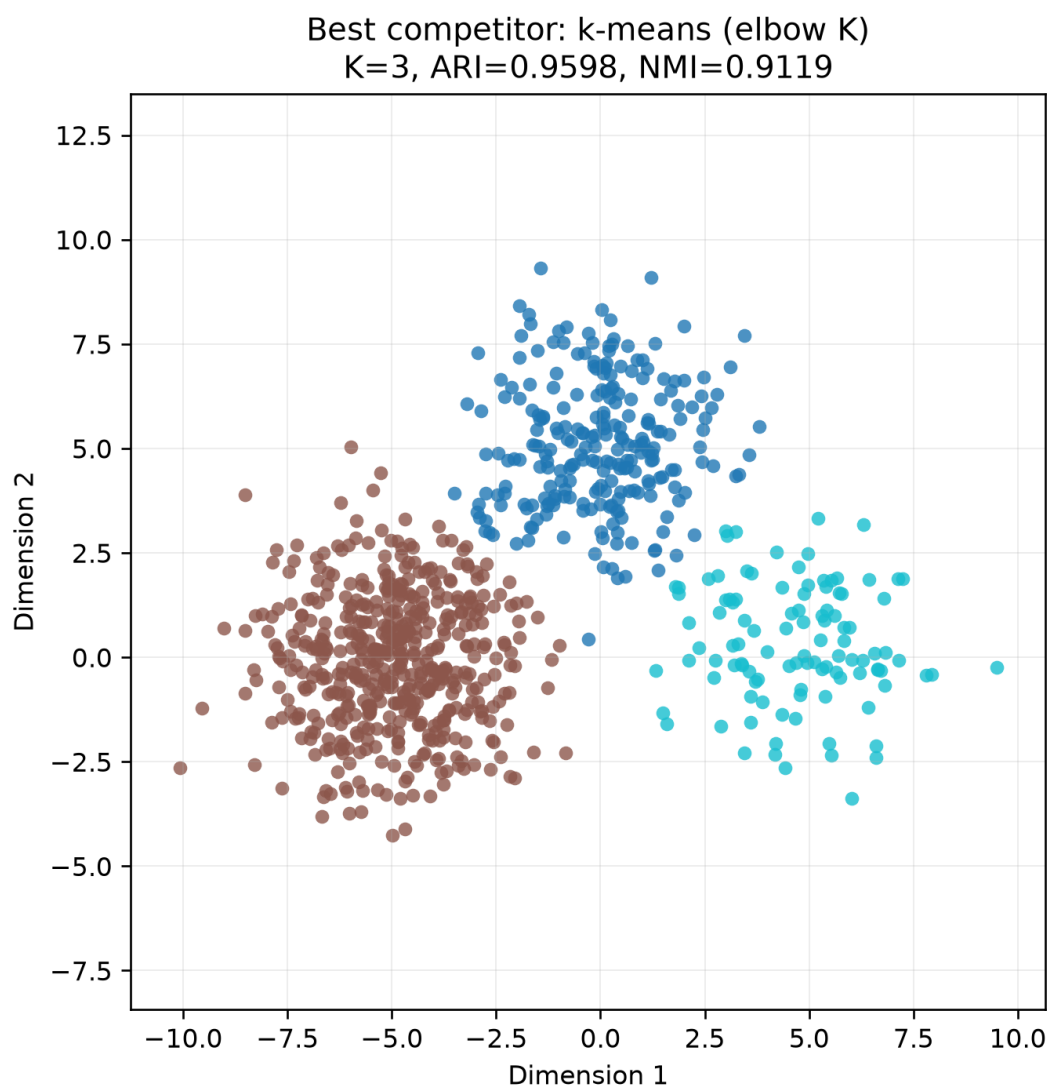
- **k-means: ARI 0.9598, NMI 0.9119, K=3**
- GMM-BIC: ARI 0.9436, NMI 0.8935, K=3
- UPGMA: ARI 0.9467, NMI 0.8928, K=3
- Ward: ARI 0.9188, NMI 0.8670, K=3

K-means performs best. STC deteriorates strongly under high noise and severely overclusters with **K=33 instead of K=3**.



SHATTER partition
K=33, ARI=0.6082, NMI=0.5800





Report

STC performs extremely well on clusters with heterogeneous densities under low and intermediate noise, achieving ARI values of 0.9958 and 0.9883 while correctly recovering all three ground-truth clusters. This indicates that variation in cluster density alone does not substantially impair STC. Under high noise, however, performance deteriorates sharply: ARI falls to 0.6082 and the predicted cluster count rises from the ground-truth $K = 3$ to $K = 33$. Most conventional baselines remain substantially more stable under this condition. The experiment therefore identifies strong noise, rather than density heterogeneity itself, as the principal limitation for STC. The high-noise result

again demonstrates STC's tendency toward excessive fragmentation when local density structure becomes strongly perturbed.

Elliptical Gaussian

1. Low noise

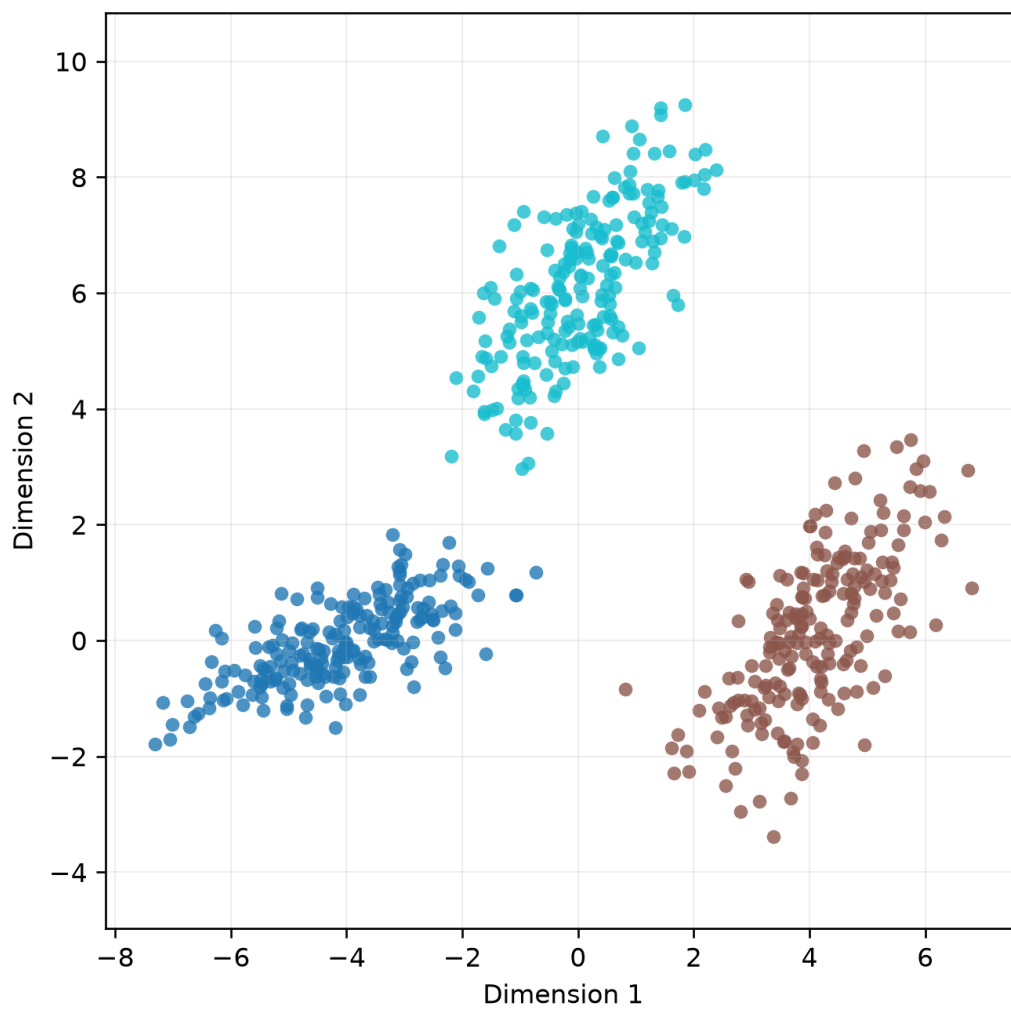
STC performs reasonably well but already shows some overclustering:

True K=3

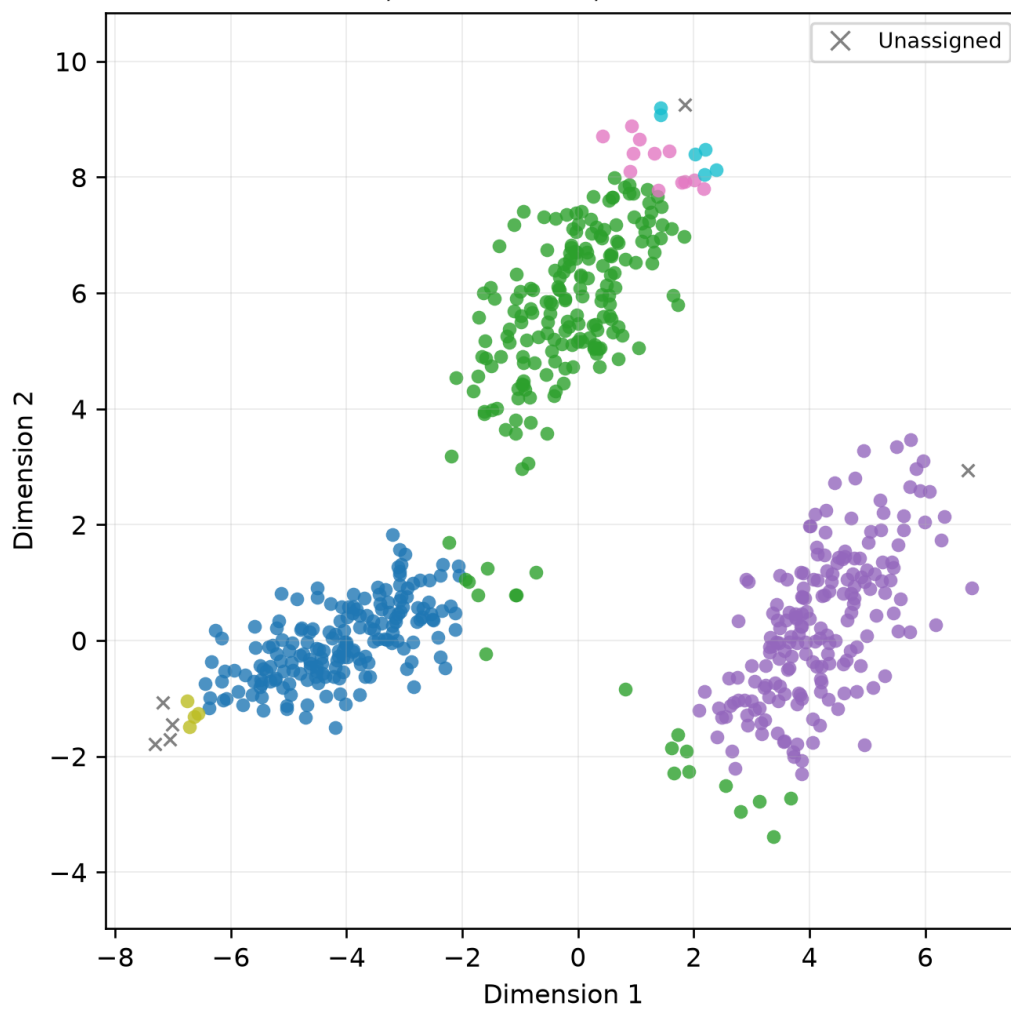
- **STC: ARI 0.8368, NMI 0.8005, K=6**
- DBSCAN: ARI 0.9975, NMI 0.9952, K=3
- HDBSCAN: ARI 1.0000, NMI 1.0000, K=3
- OPTICS: ARI 1.0000, NMI 1.0000, K=3
- **k-means: ARI 1.0000, NMI 1.0000, K=3**
- GMM-BIC: ARI 1.0000, NMI 1.0000, K=3
- UPGMA: ARI 1.0000, NMI 1.0000, K=3
- Ward: ARI 1.0000, NMI 1.0000, K=3

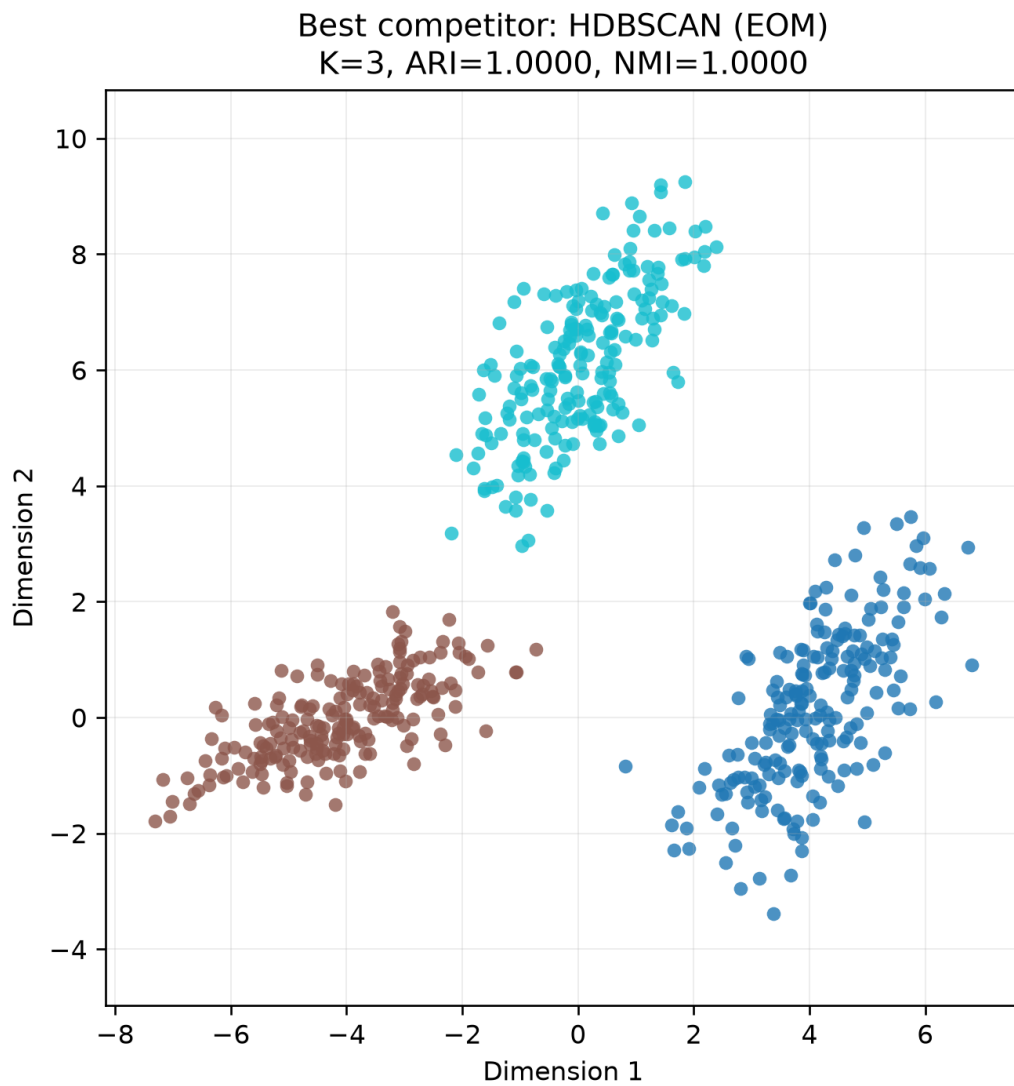
Most baselines recover the structure perfectly. STC still achieves reasonably strong ARI/NMI but overclusters with **K=6 instead of K=3**.

Ground truth
K=3



SHATTER partition
K=6, ARI=0.8368, NMI=0.8005





2. Intermediate noise

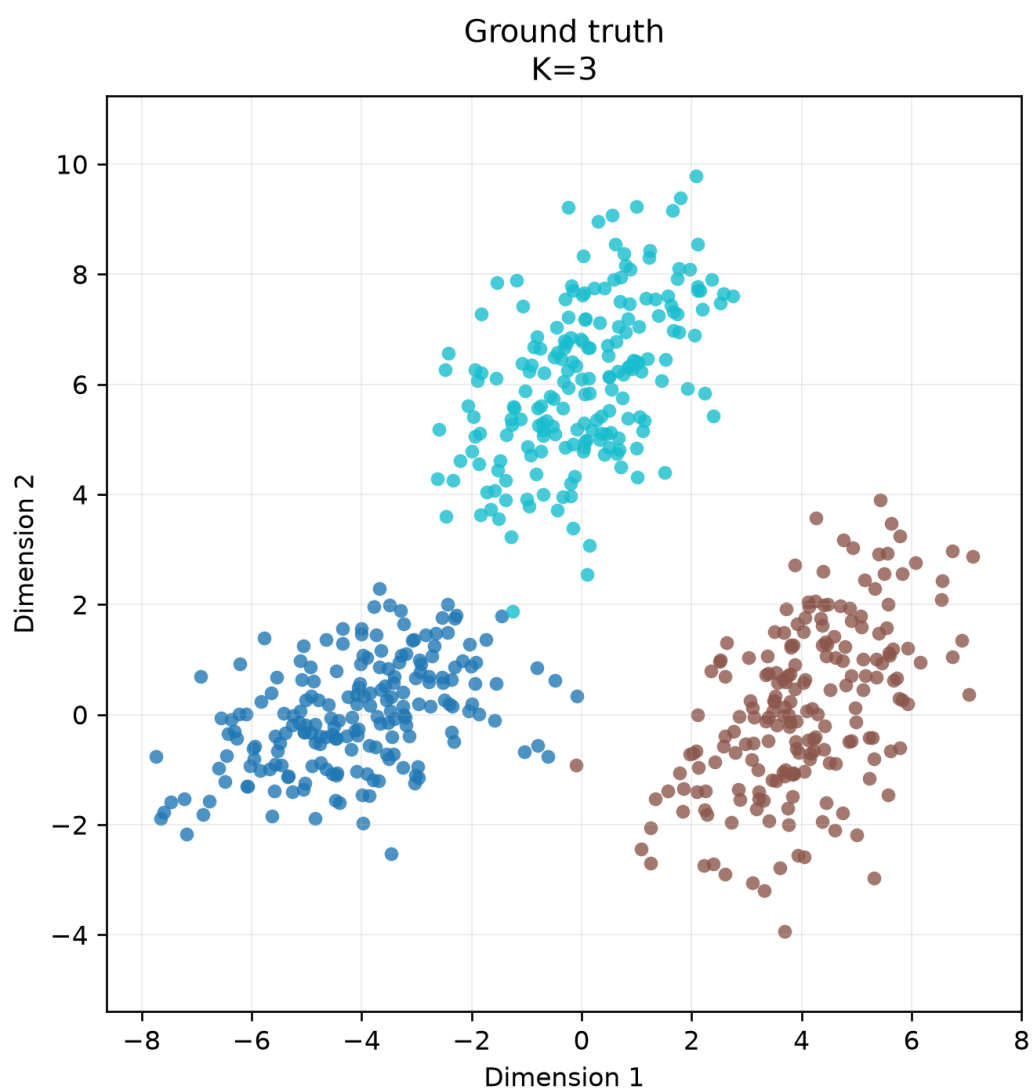
STC performance drops considerably and overclustering becomes much stronger:

True K=3

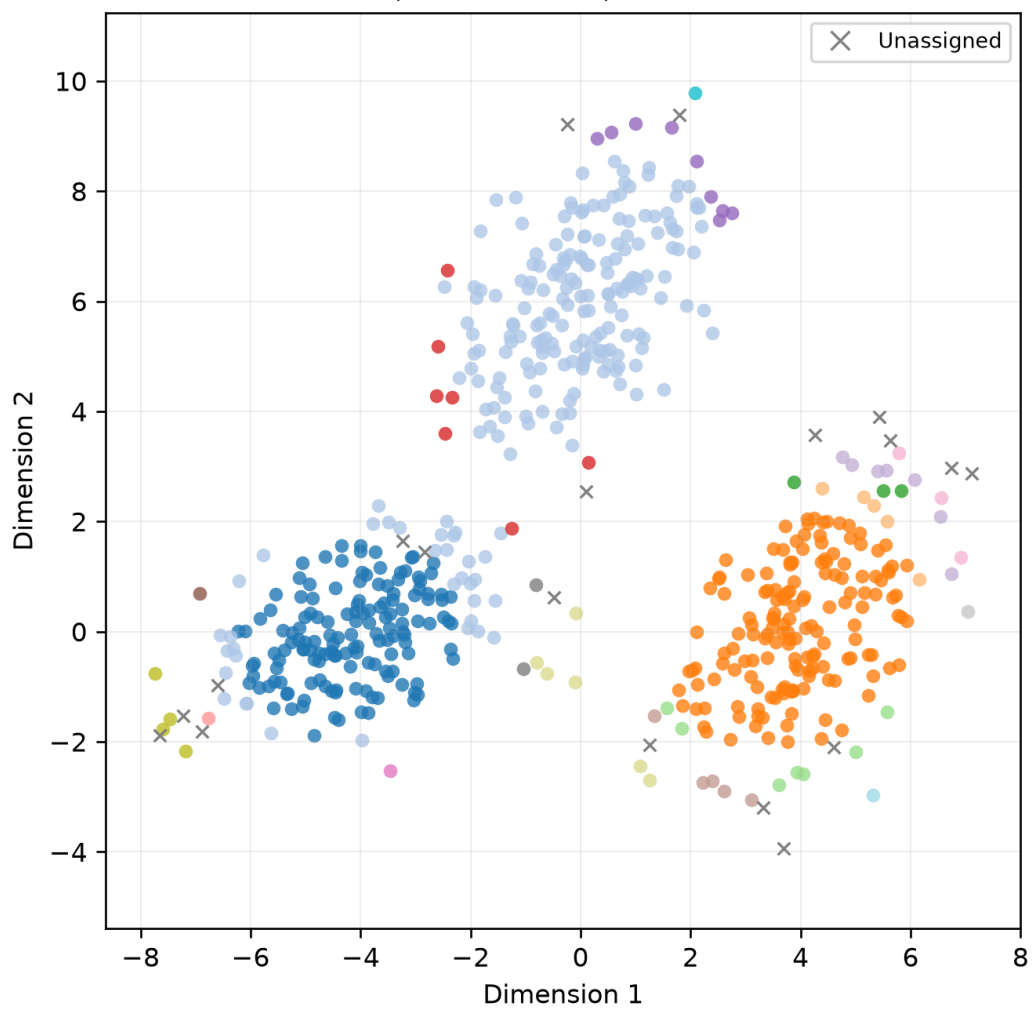
- **STC: ARI 0.6430, NMI 0.6449, K=20**
- **DBSCAN: ARI 0.9925, NMI 0.9857, K=3**
- HDBSCAN: ARI 0.9875, NMI 0.9762, K=3
- OPTICS: ARI 0.9900, NMI 0.9809, K=3

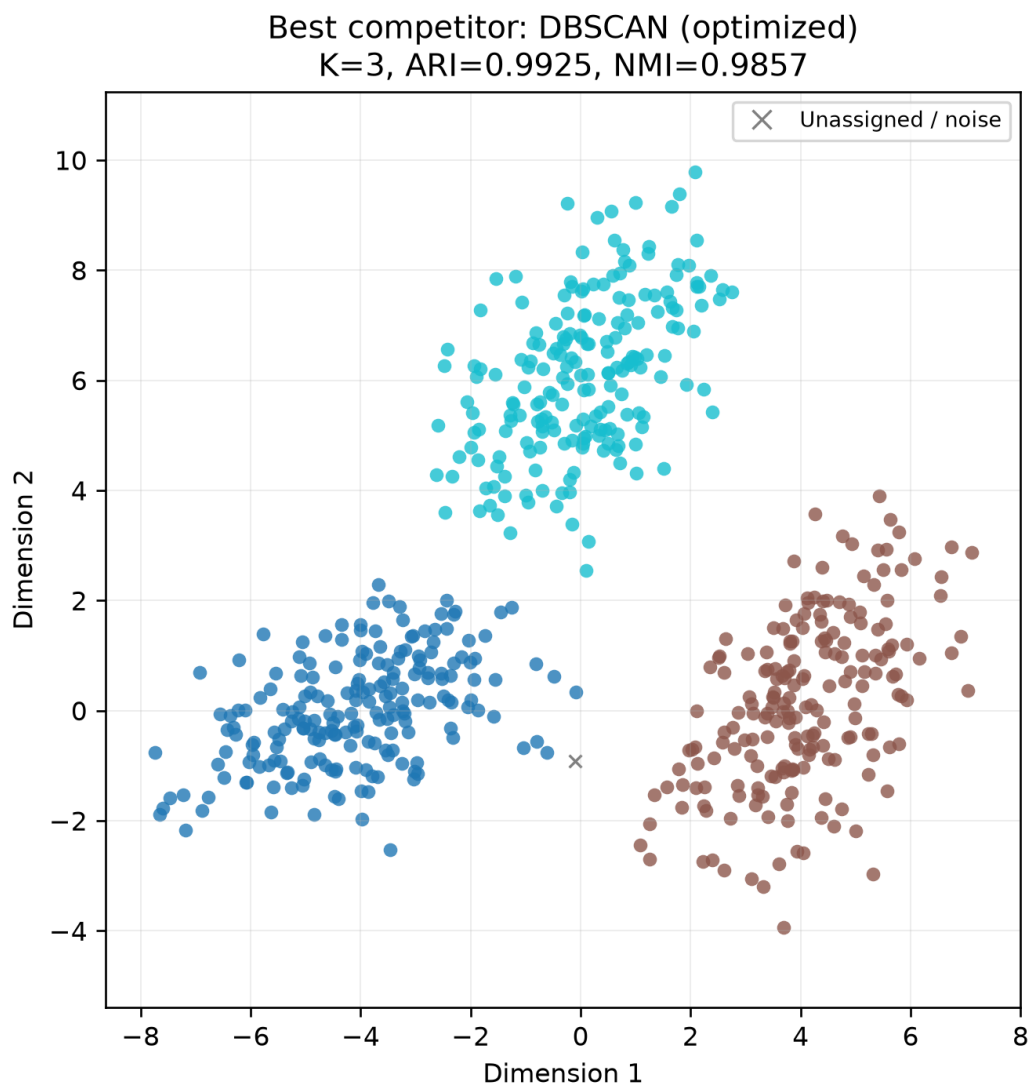
- k-means: ARI 0.9900, NMI 0.9809, K=3
- GMM-BIC: ARI 0.9900, NMI 0.9809, K=3
- UPGMA: ARI 0.9900, NMI 0.9809, K=3
- Ward: ARI 0.9900, NMI 0.9809, K=3

DBSCAN performs slightly best. STC falls well behind the baselines and strongly overclusters with **K=20 instead of K=3**.



SHATTER partition
K=20, ARI=0.6430, NMI=0.6449





3. High noise

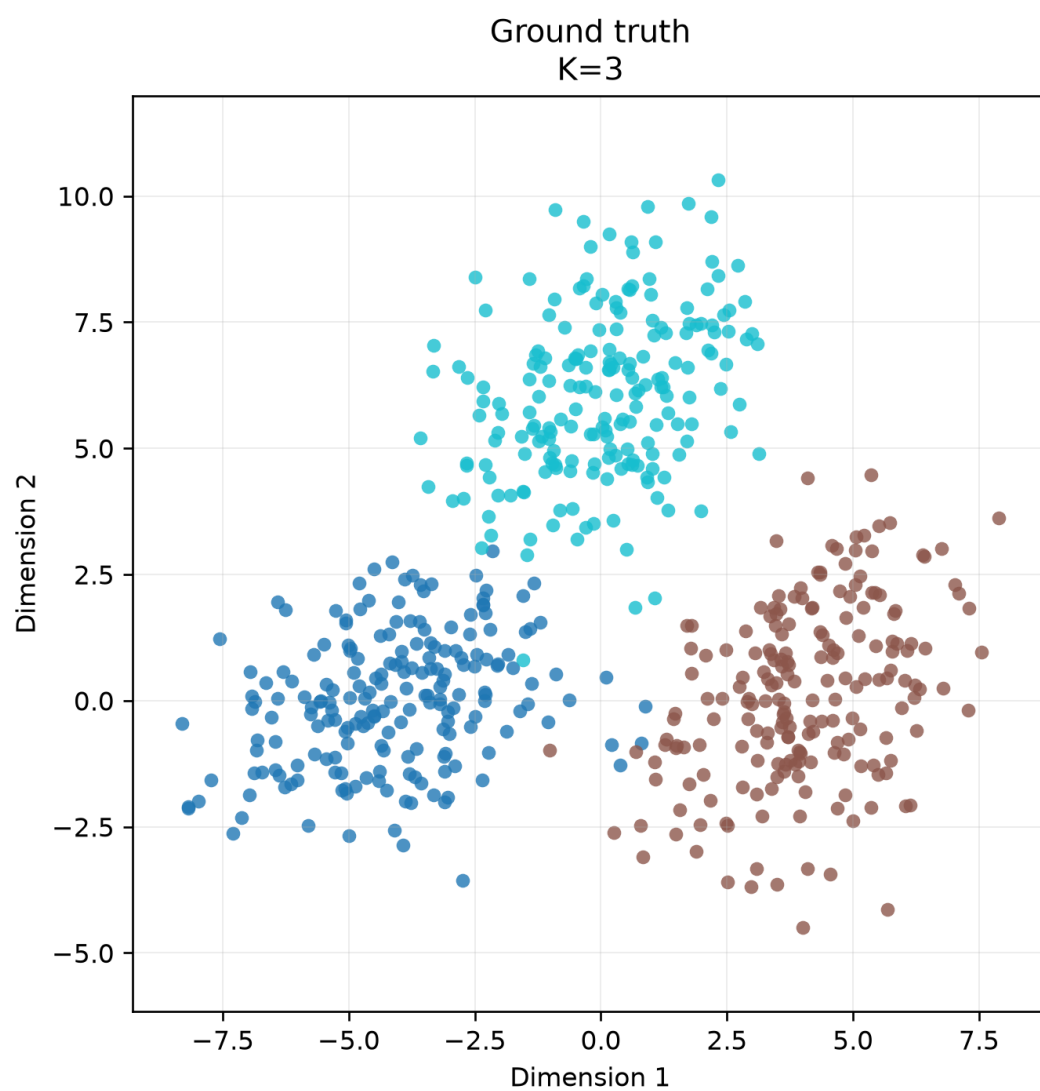
STC deteriorates further under high noise:

True K=3

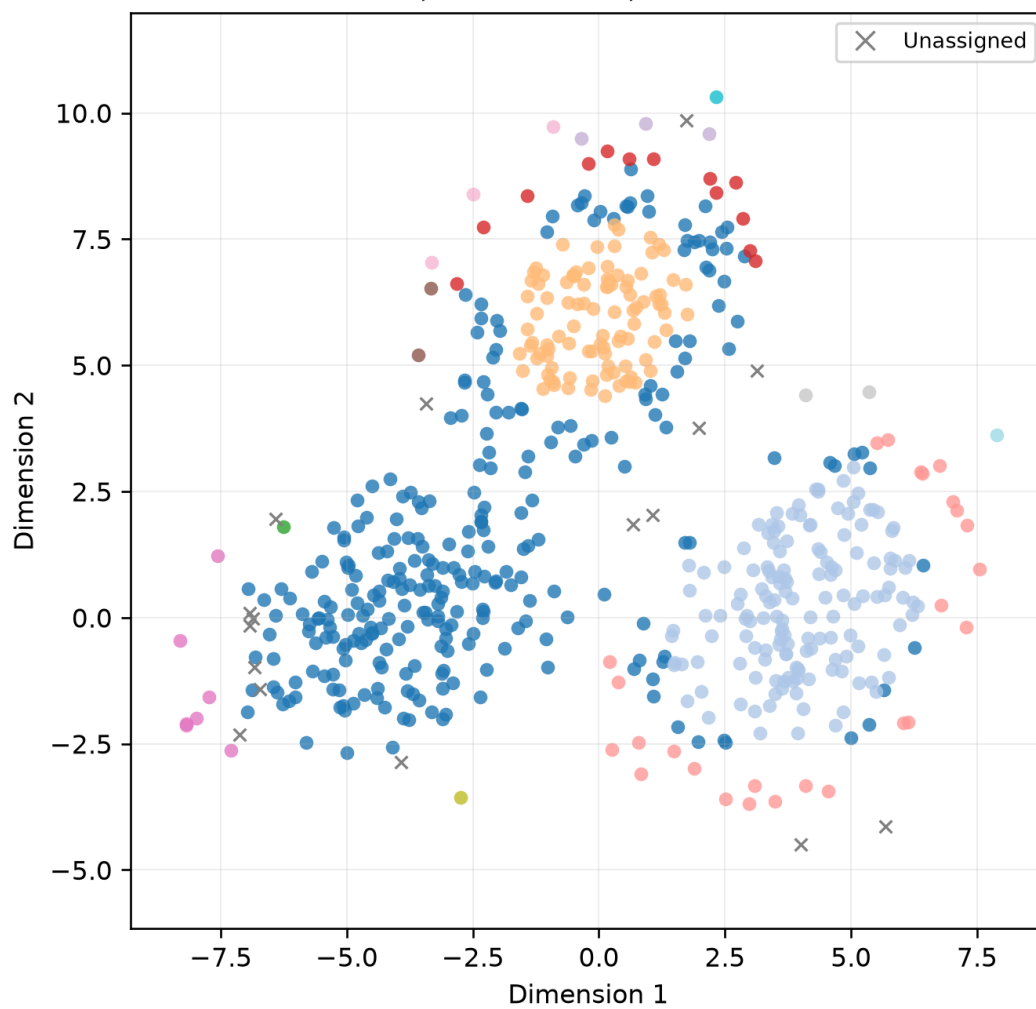
- **STC: ARI 0.4391, NMI 0.5179, K=14**
- DBSCAN: ARI 0.5171, NMI 0.6027, K=2
- HDBSCAN: ARI 0.5188, NMI 0.5975, K=2

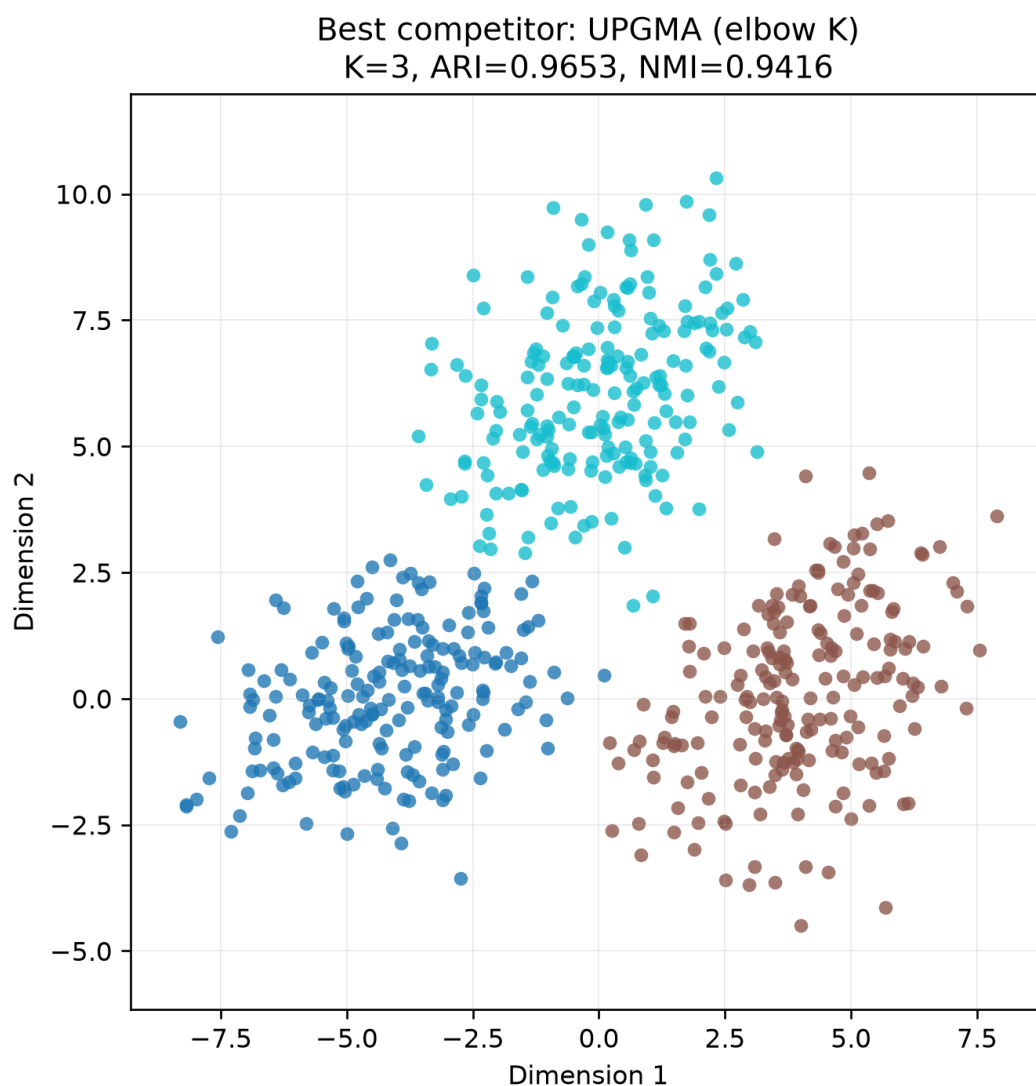
- OPTICS: ARI 0.5459, NMI 0.6212, K=3
- k-means: ARI 0.9504, NMI 0.9209, K=3
- GMM-BIC: ARI 0.9553, NMI 0.9246, K=3
- **UPGMA: ARI 0.9653, NMI 0.9416, K=3**
- Ward: ARI 0.9359, NMI 0.9027, K=3

UPGMA performs best. STC has relatively low agreement with the ground truth and severely overclusters with **K=14 instead of K=3**.



SHATTER partition
K=14, ARI=0.4391, NMI=0.5179





Report

STC underperforms all evaluated baseline methods on elliptical Gaussian clusters. Its mean ARI is 0.6396, while the strongest conventional methods achieve mean ARIs around 0.98. Even at low noise, STC overestimates the number of clusters, predicting 6 rather than the true 3. This fragmentation becomes more pronounced at intermediate noise, where STC predicts 20 clusters and ARI falls to 0.6430. At high noise, STC remains substantially below the centroid-, model-, and hierarchical baselines. Because coverage remains high, the principal failure mode is excessive subdivision rather than failure to assign points. These results indicate that STC is not well suited to elongated

Gaussian cluster structure, particularly when conventional methods already match the underlying geometry well.

Moon

1. Low noise

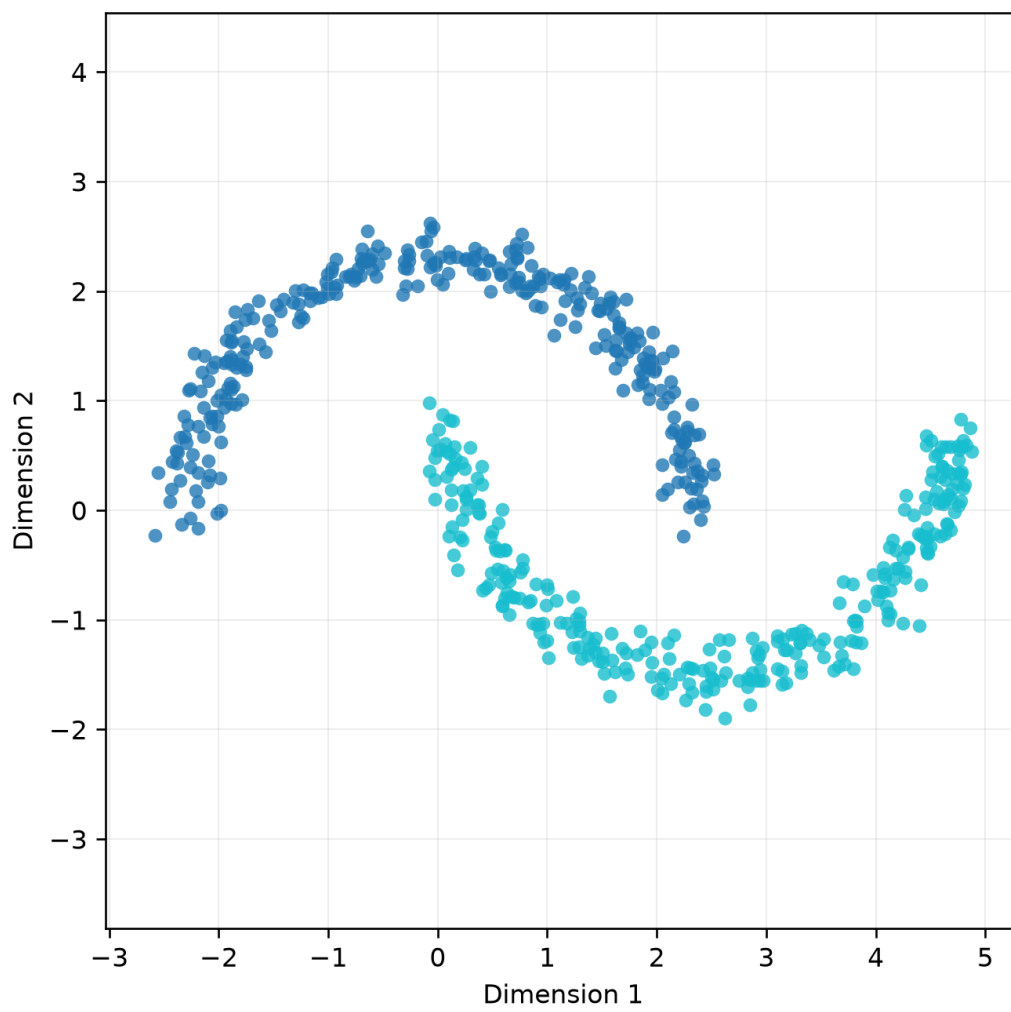
STC performs very well and correctly identifies the two moon-shaped clusters:

True K=2

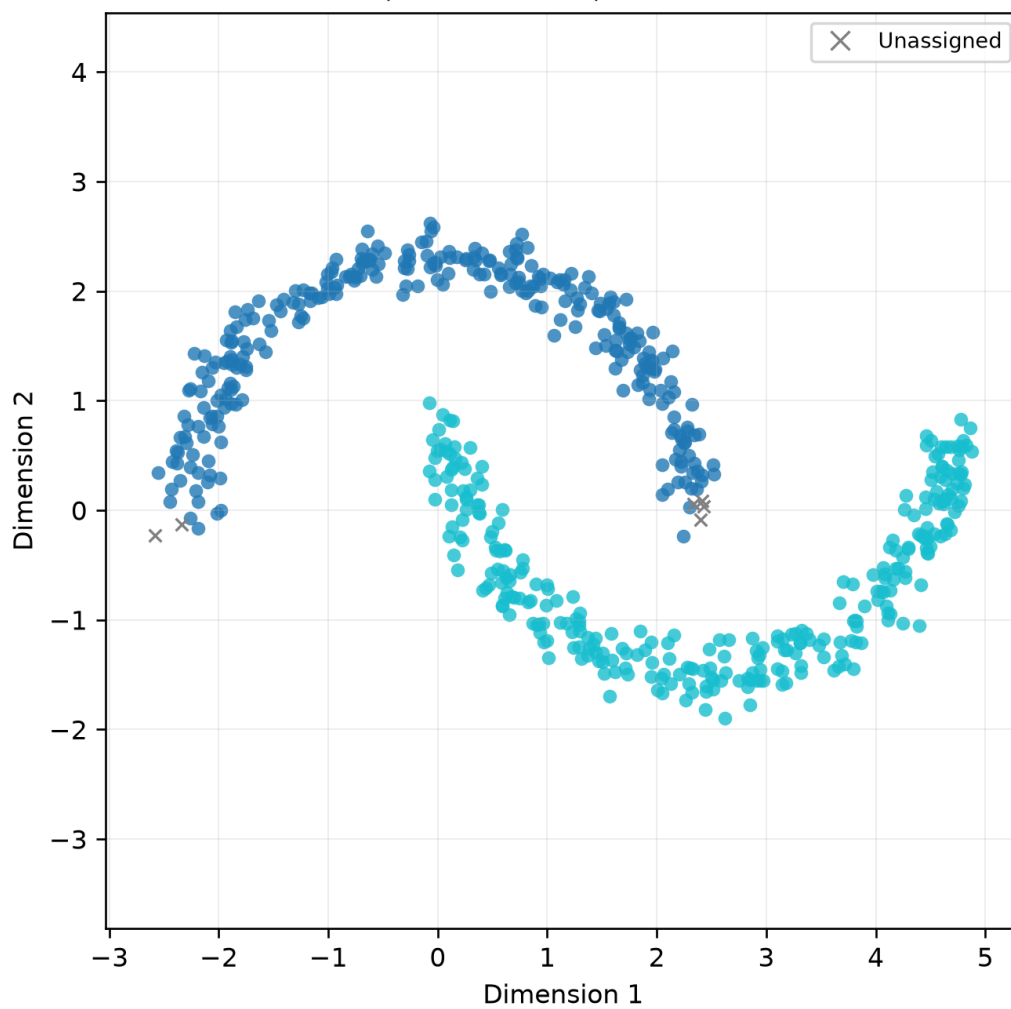
- **STC: ARI 0.9804, NMI 0.9658, K=2**
- DBSCAN: ARI 0.9967, NMI 0.9920, K=2
- **HDBSCAN: ARI 1.0000, NMI 1.0000, K=2**
- OPTICS: ARI 1.0000, NMI 1.0000, K=2
- k-means: ARI 0.3126, NMI 0.4067, K=4
- GMM-BIC: ARI 0.2106, NMI 0.4681, K=10
- UPGMA: ARI 0.3614, NMI 0.4783, K=4
- Ward: ARI 0.5317, NMI 0.6776, K=4

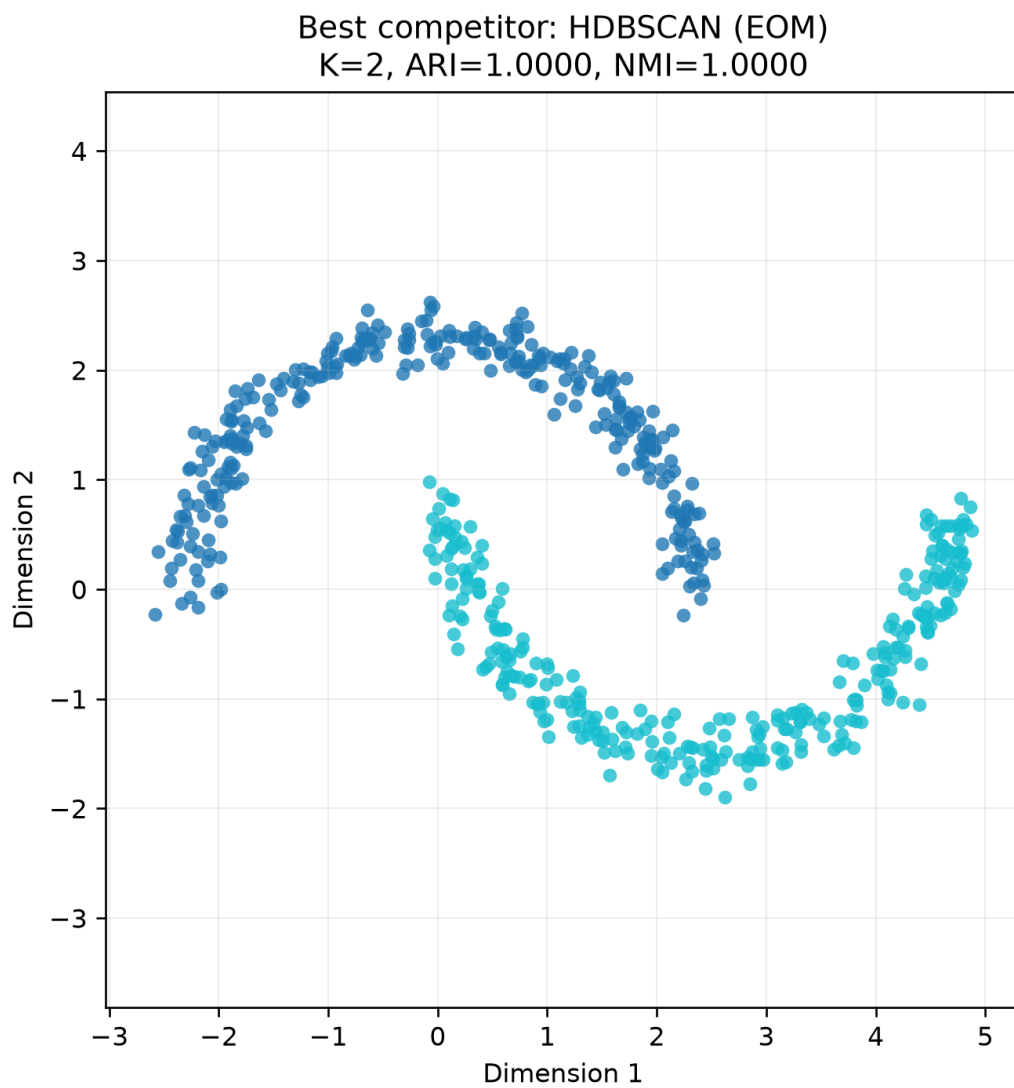
HDBSCAN and OPTICS perform perfectly. STC also performs very strongly and correctly recovers **K=2**.

Ground truth
K=2



SHATTER partition
K=2, ARI=0.9804, NMI=0.9658





2. Intermediate noise

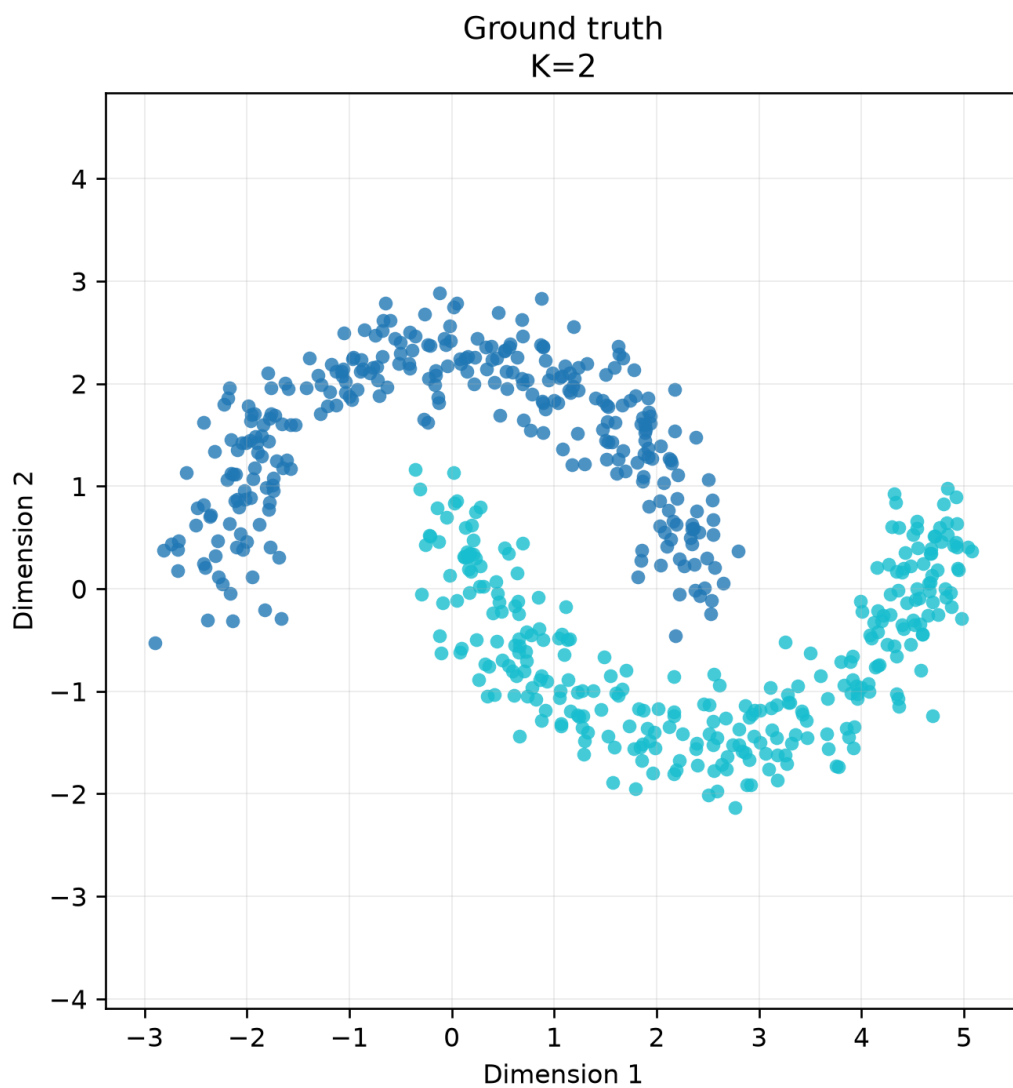
STC performance decreases and it begins to over cluster:

True K=2

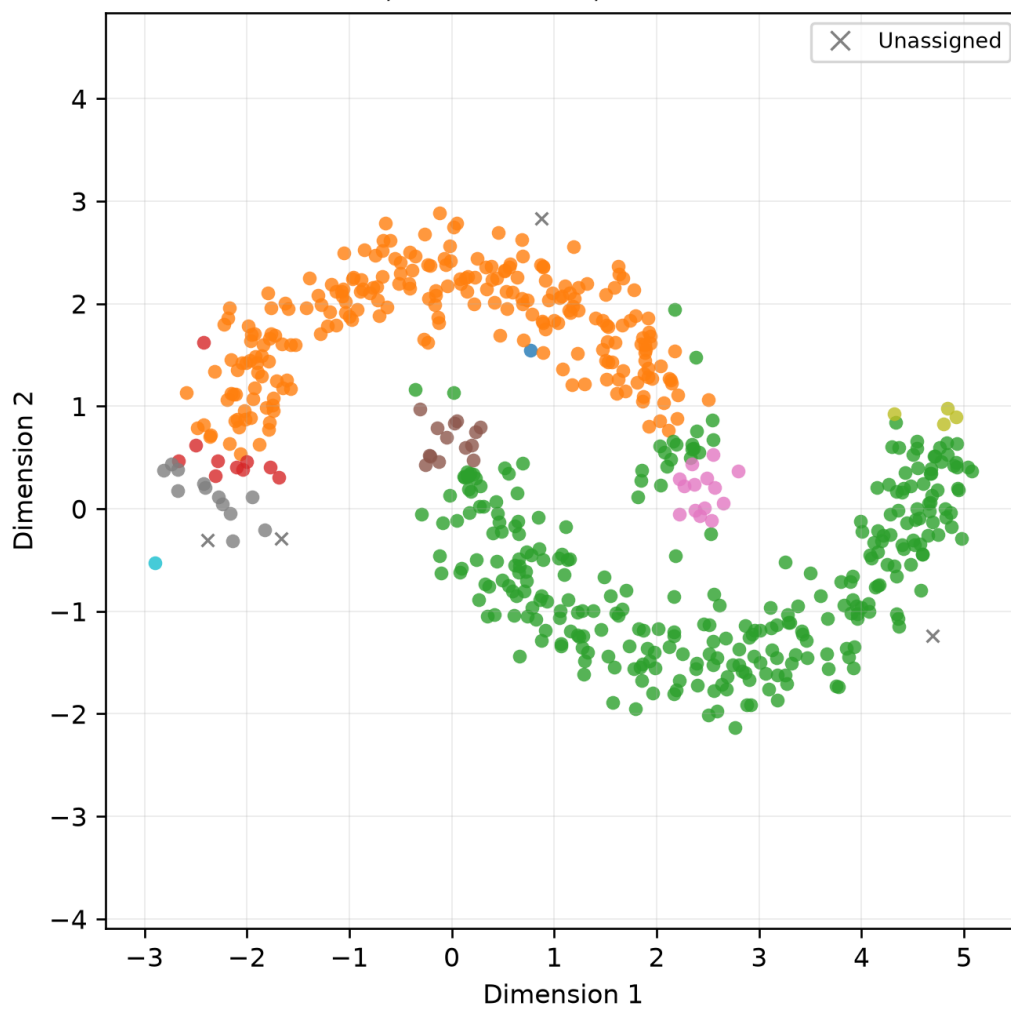
- **STC: ARI 0.6878, NMI 0.6148, K=9**

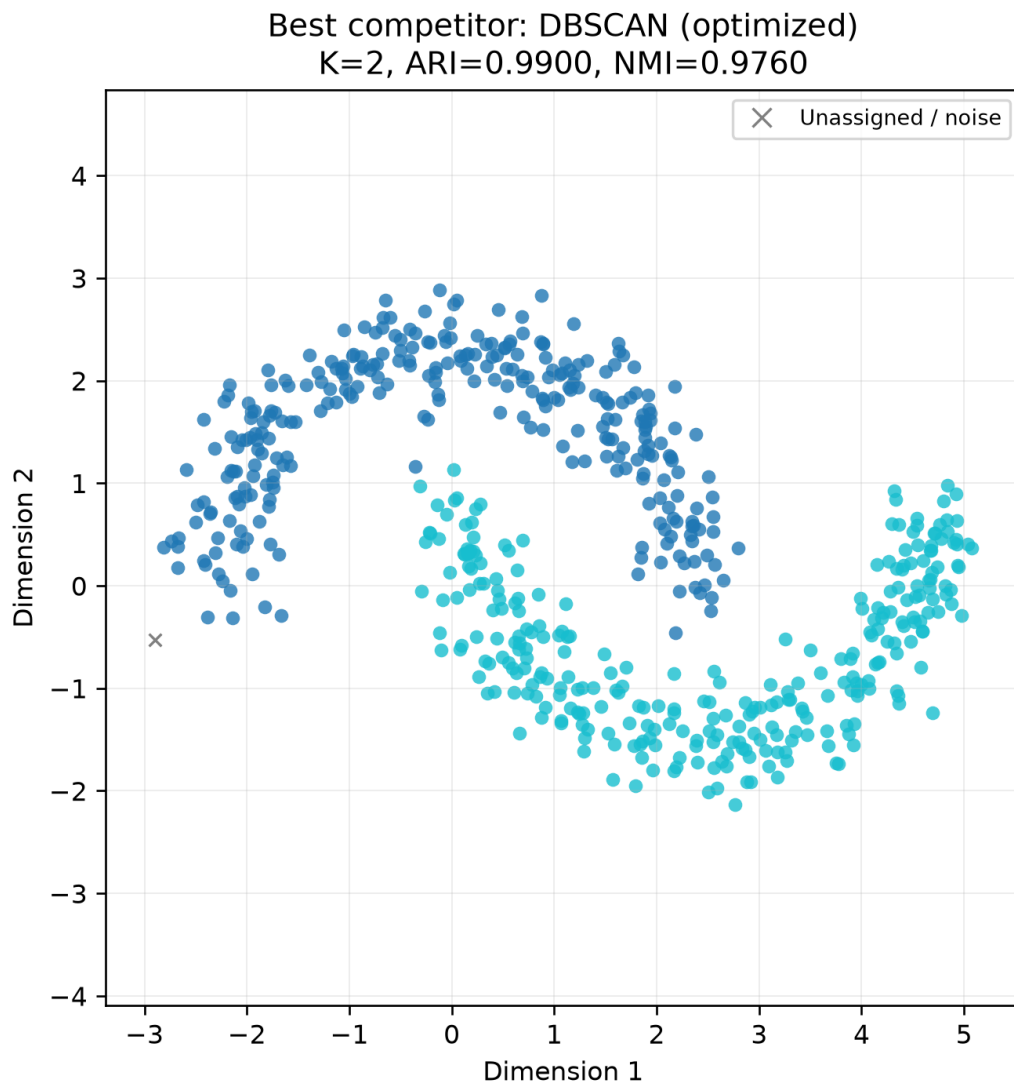
- **DBSCAN: ARI 0.9900, NMI 0.9760, K=2**
- HDBSCAN: ARI 0.9768, NMI 0.9468, K=2
- OPTICS: ARI 0.9768, NMI 0.9519, K=2
- k-means: ARI 0.3035, NMI 0.3840, K=4
- GMM-BIC: ARI 0.3347, NMI 0.5590, K=6
- UPGMA: ARI 0.2909, NMI 0.4170, K=3
- Ward: ARI 0.3300, NMI 0.4400, K=3

DBSCAN performs best. STC still captures part of the structure but strongly overclusters with **K=9 instead of K=2**.



SHATTER partition
K=9, ARI=0.6878, NMI=0.6148





3. High noise

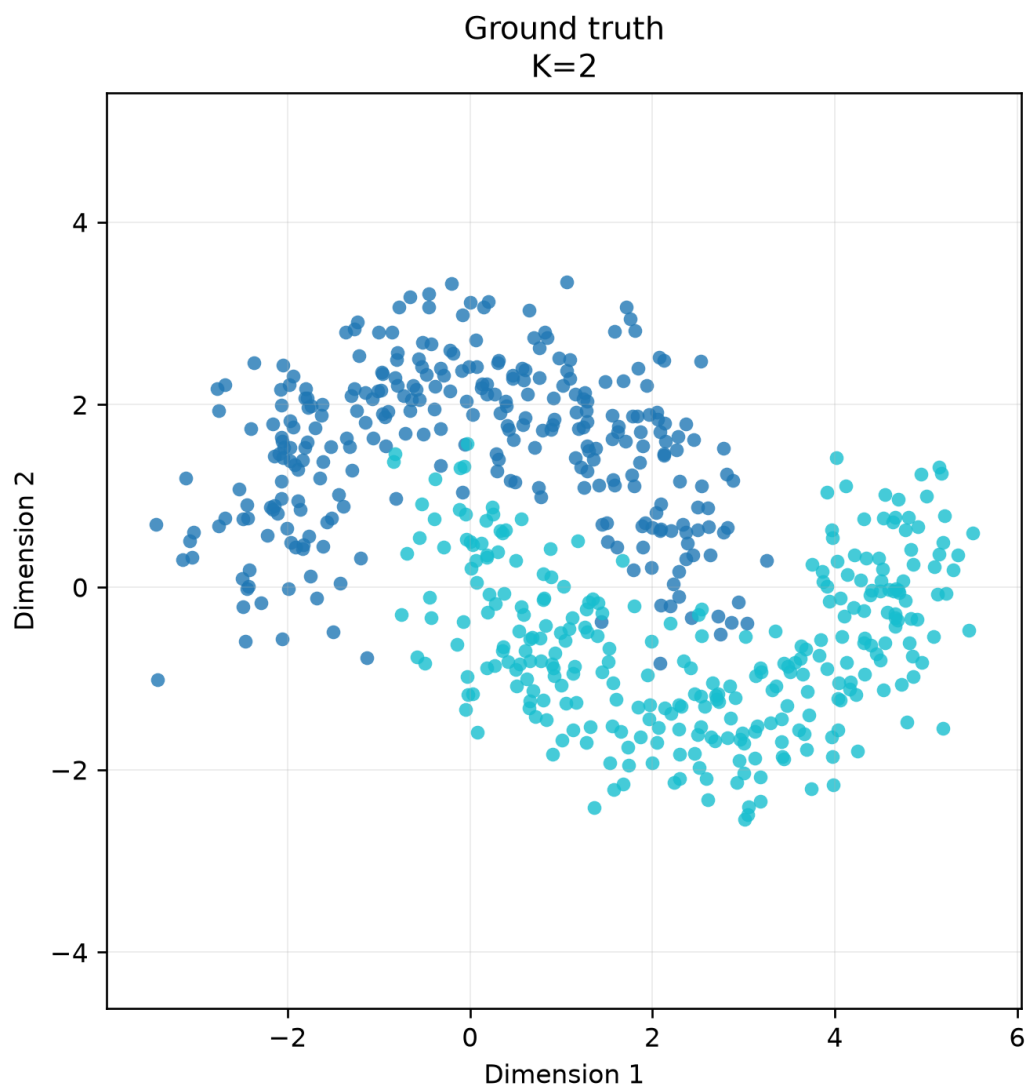
All methods struggle more, and STC severely overclusters:

True K=2

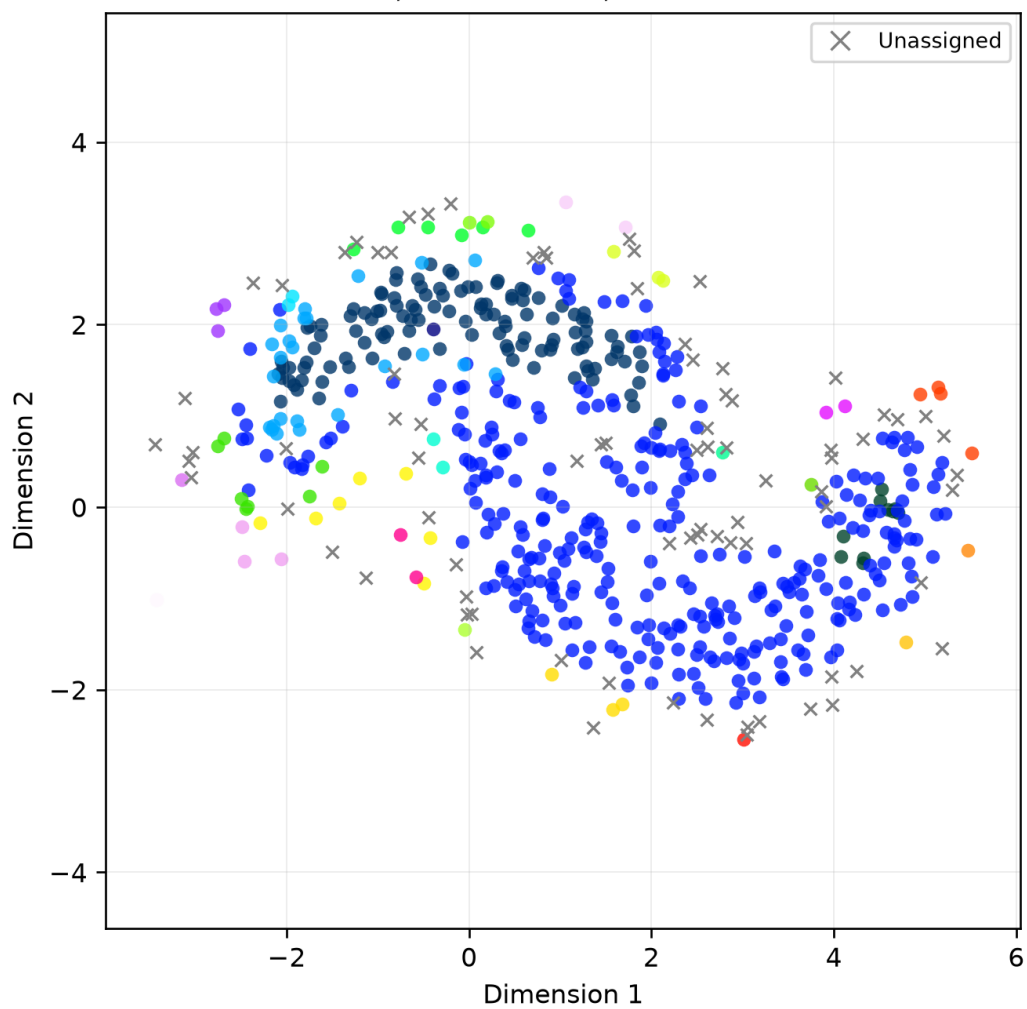
- **STC: ARI 0.2070, NMI 0.2564, K=27**
- DBSCAN: ARI 0.1936, NMI 0.1562, K=2
- HDBSCAN: ARI 0.0007, NMI 0.0153, K=2
- OPTICS: ARI 0.1000, NMI 0.2872, K=7

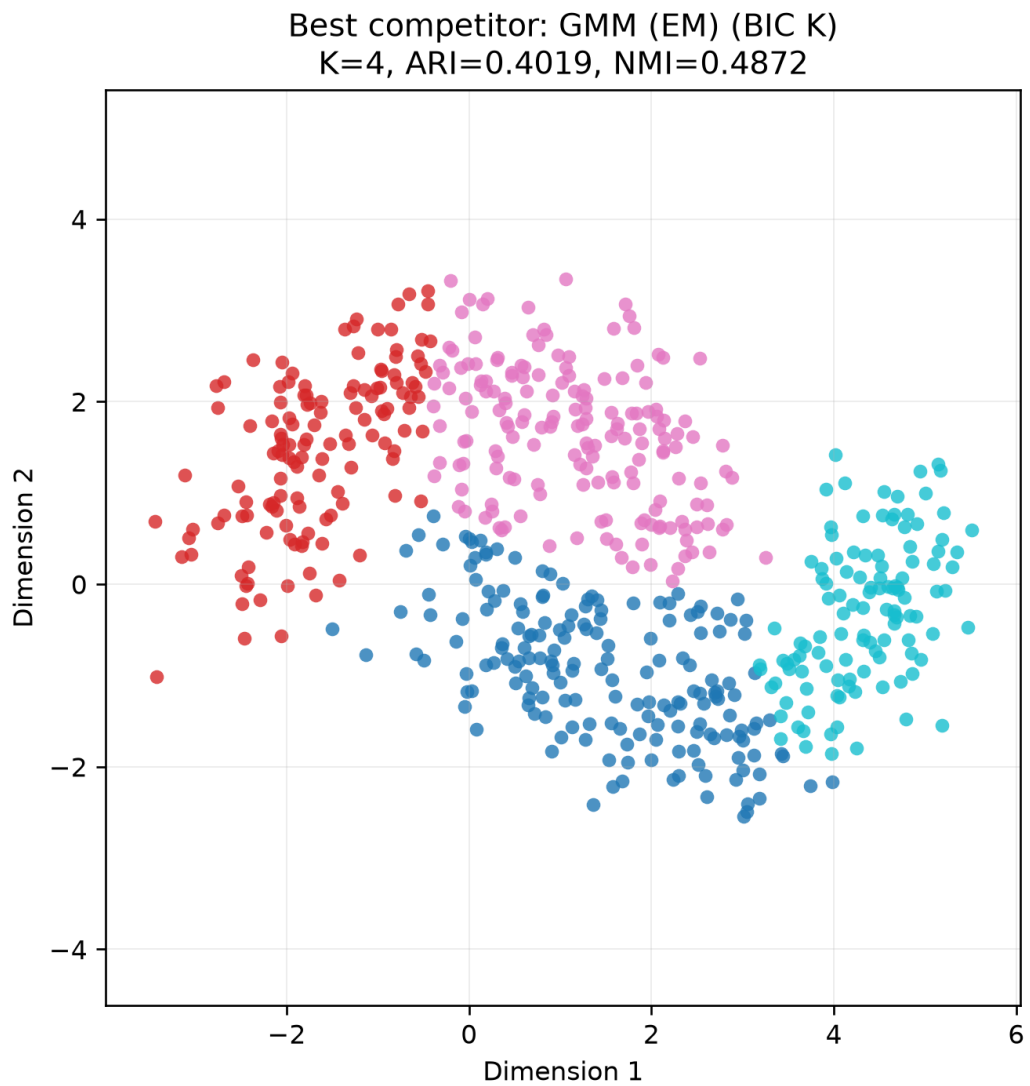
- k-means: ARI 0.3239, NMI 0.3699, K=4
- **GMM-BIC: ARI 0.4019, NMI 0.4872, K=4**
- UPGMA: ARI 0.2957, NMI 0.3749, K=4
- Ward: ARI 0.2908, NMI 0.3342, K=3

GMM-BIC performs best among the automatically selected-K baselines, although its result is still relatively weak. STC performs poorly and severely overclusters with **K=27** instead of K=2.



SHATTER partition
K=27, ARI=0.2070, NMI=0.2564





Report

STC performs very well on the non-convex moon geometry under low noise, achieving ARI 0.9804 and correctly recovering the two ground-truth clusters. At intermediate noise, STC remains substantially stronger than k-means, GMM, UPGMA, and Ward, indicating that its shape-sensitive clustering mechanism is advantageous for curved, non-convex structures. However, DBSCAN, HDBSCAN, and OPTICS are more robust at this noise level. As noise increases further, STC strongly over-fragments the data,

producing 27 clusters at high noise and reducing ARI to 0.2070. The moons benchmark therefore demonstrates a genuine geometric strength of STC but also reinforces the broader finding that its performance deteriorates when noise breaks the continuity of local density structure.

Nested Structure

1. Low noise

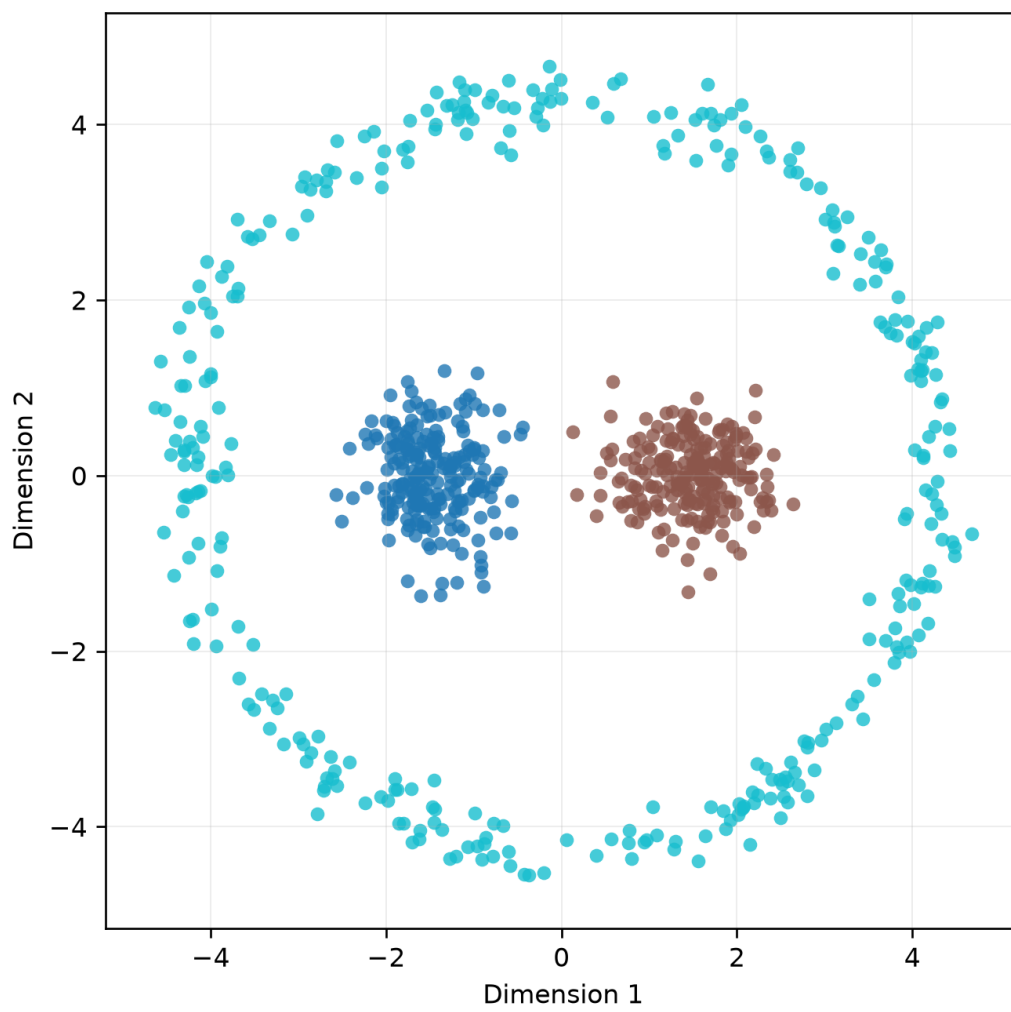
STC performs very strongly, although it overclusters:

True K=3

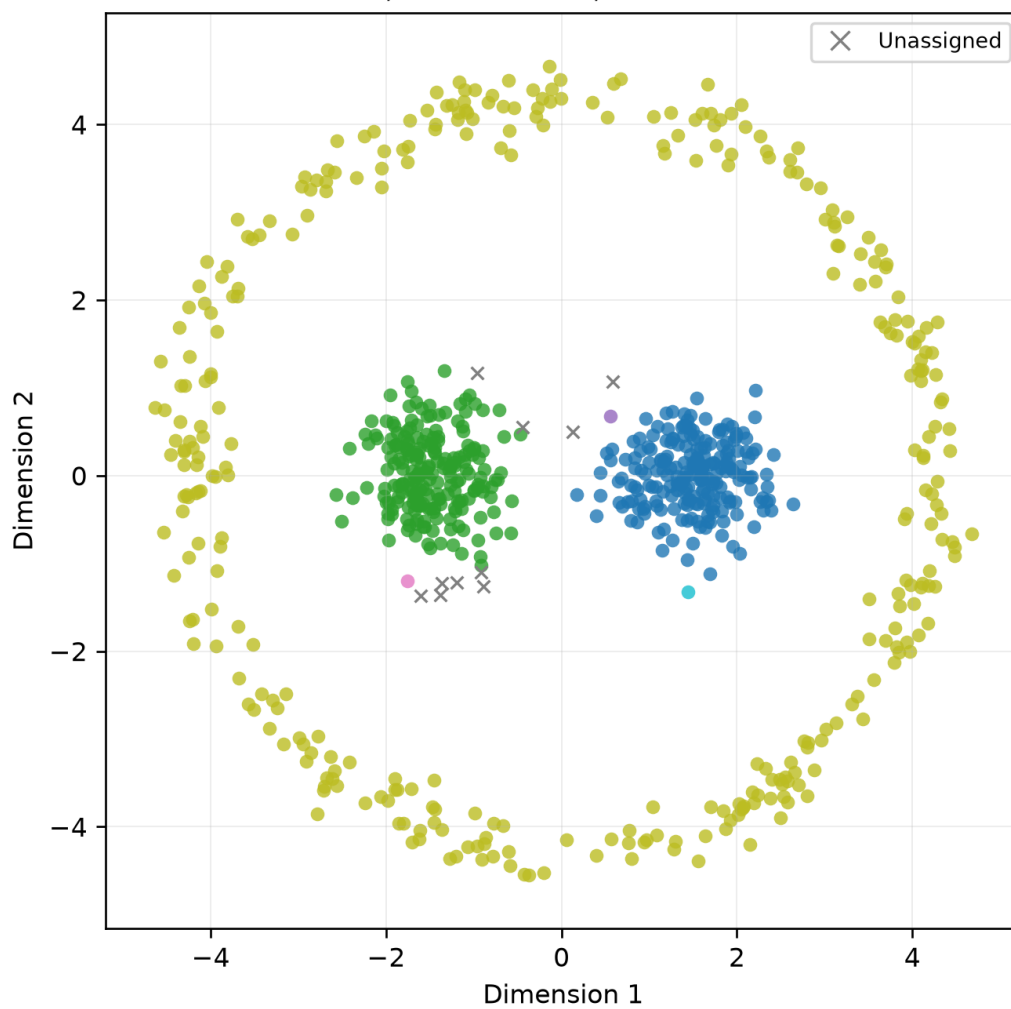
- **STC: ARI 0.9796, NMI 0.9598, K=6**
- DBSCAN: ARI 0.6313, NMI 0.7484, K=7
- **HDBSCAN: ARI 0.9984, NMI 0.9962, K=3**
- OPTICS: ARI 0.5853, NMI 0.6969, K=9
- k-means: ARI 0.4179, NMI 0.5434, K=4
- GMM-BIC: ARI 0.5912, NMI 0.7067, K=10
- UPGMA: ARI 0.3817, NMI 0.5125, K=5
- Ward: ARI 0.4164, NMI 0.5399, K=4

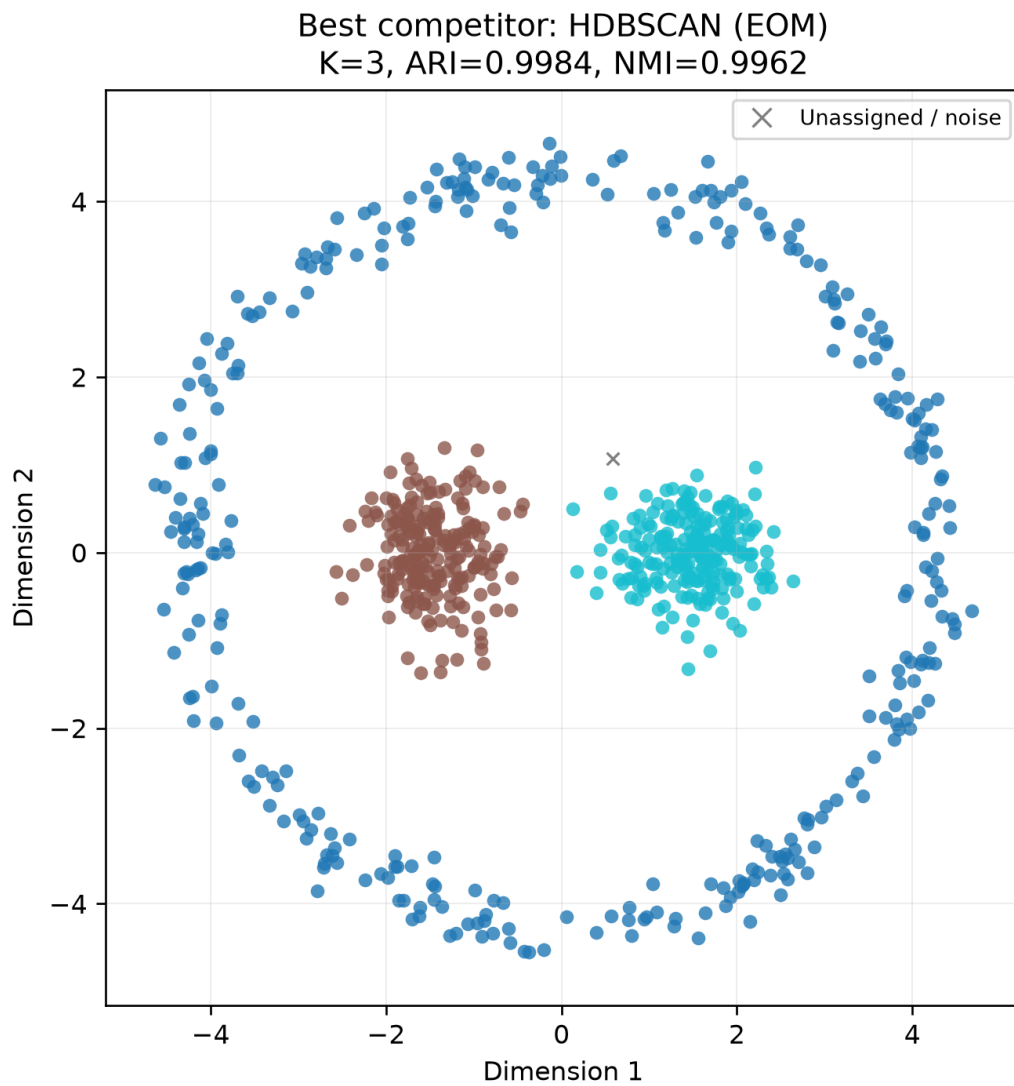
HDBSCAN performs best and correctly recovers **K=3**. STC is close in ARI/NMI but overclusters with **K=6 instead of K=3**.

Ground truth
K=3



SHATTER partition
K=6, ARI=0.9796, NMI=0.9598





2. Intermediate noise

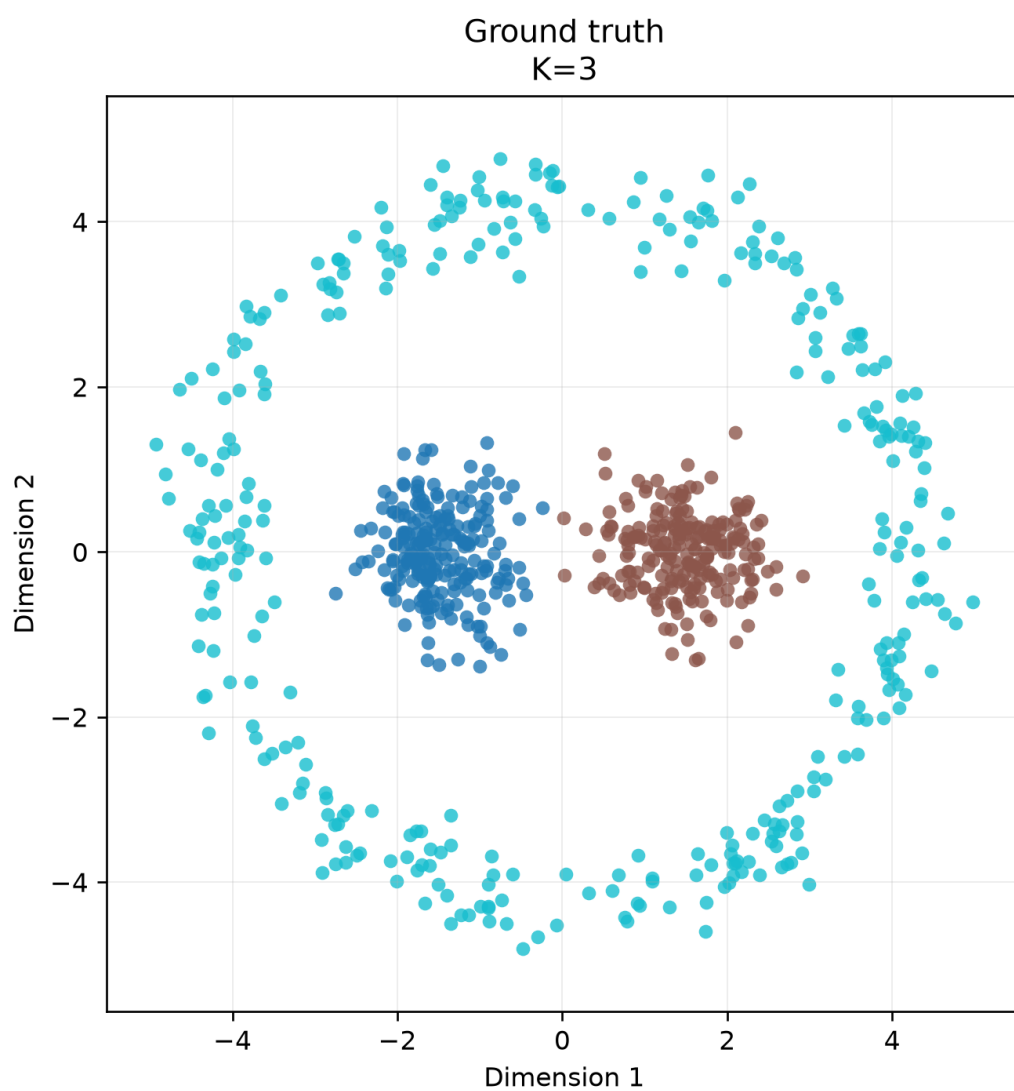
STC clearly outperforms the baselines, although overclustering increases:

True K=3

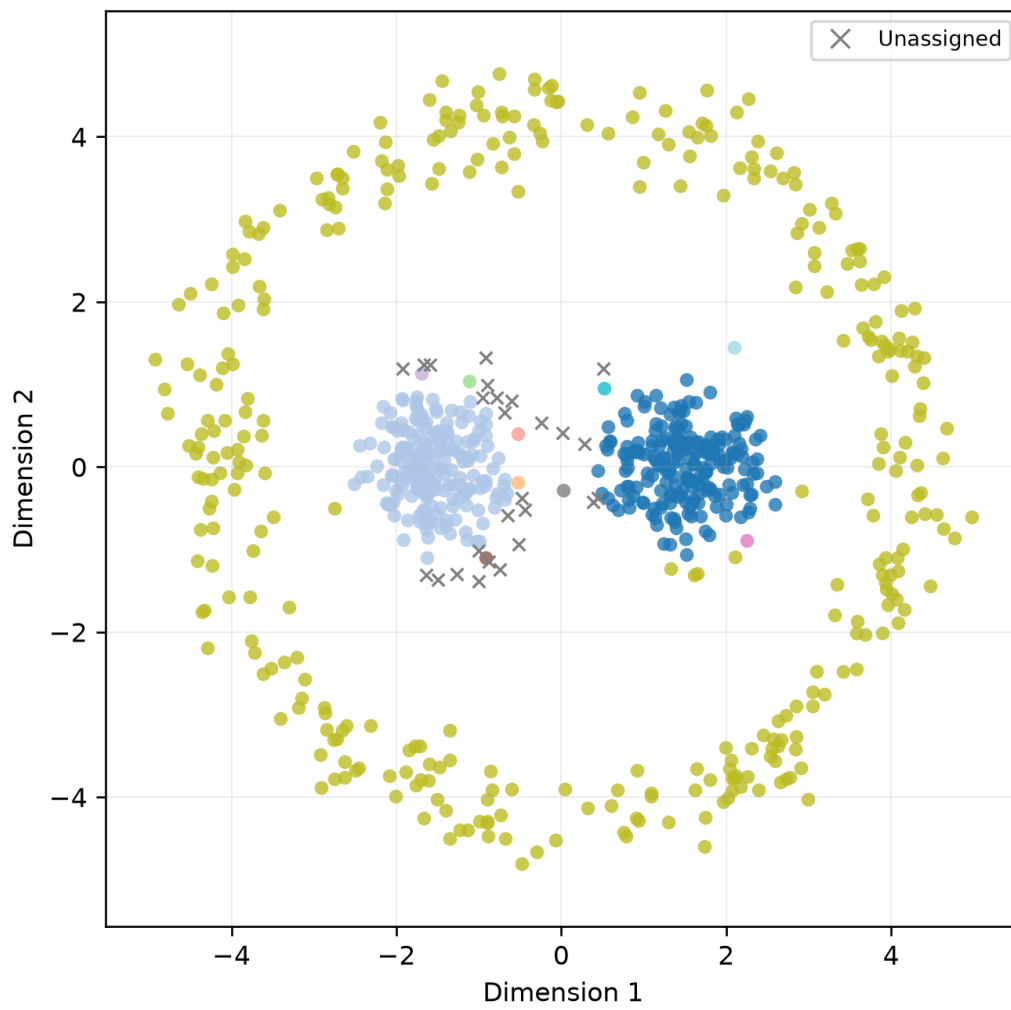
- **STC: ARI 0.9231, NMI 0.8755, K=12**
- DBSCAN: ARI 0.5889, NMI 0.6988, K=9
- HDBSCAN: ARI 0.5638, NMI 0.6489, K=16
- OPTICS: ARI 0.6043, NMI 0.7250, K=7

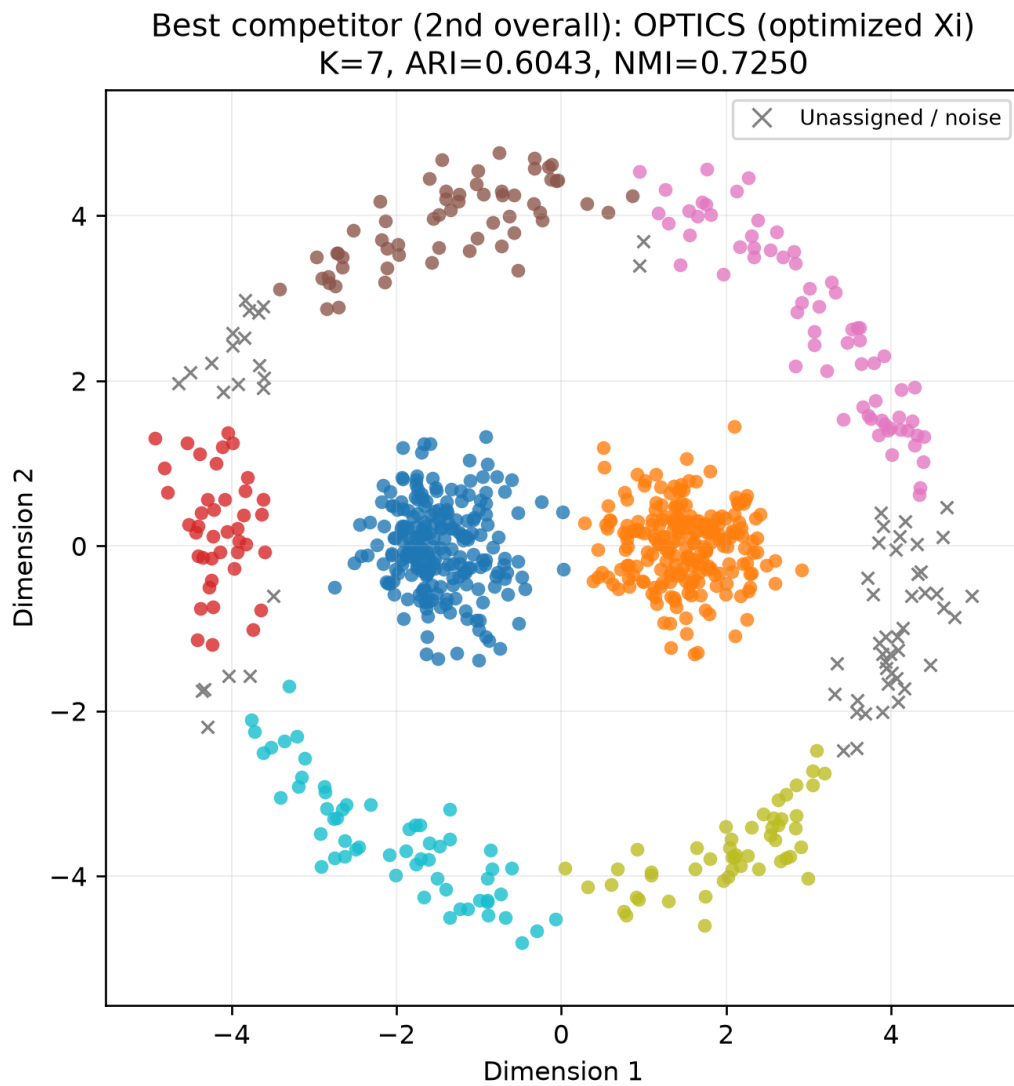
- k-means: ARI 0.4131, NMI 0.5321, K=4
- GMM-BIC: ARI 0.6005, NMI 0.7200, K=9
- UPGMA: ARI 0.3531, NMI 0.4840, K=4
- Ward: ARI 0.4147, NMI 0.5347, K=4

STC performs best by a large margin, but it substantially overclusters with **K=12 instead of K=3**.



SHATTER partition
K=12, ARI=0.9231, NMI=0.8755





3. High noise

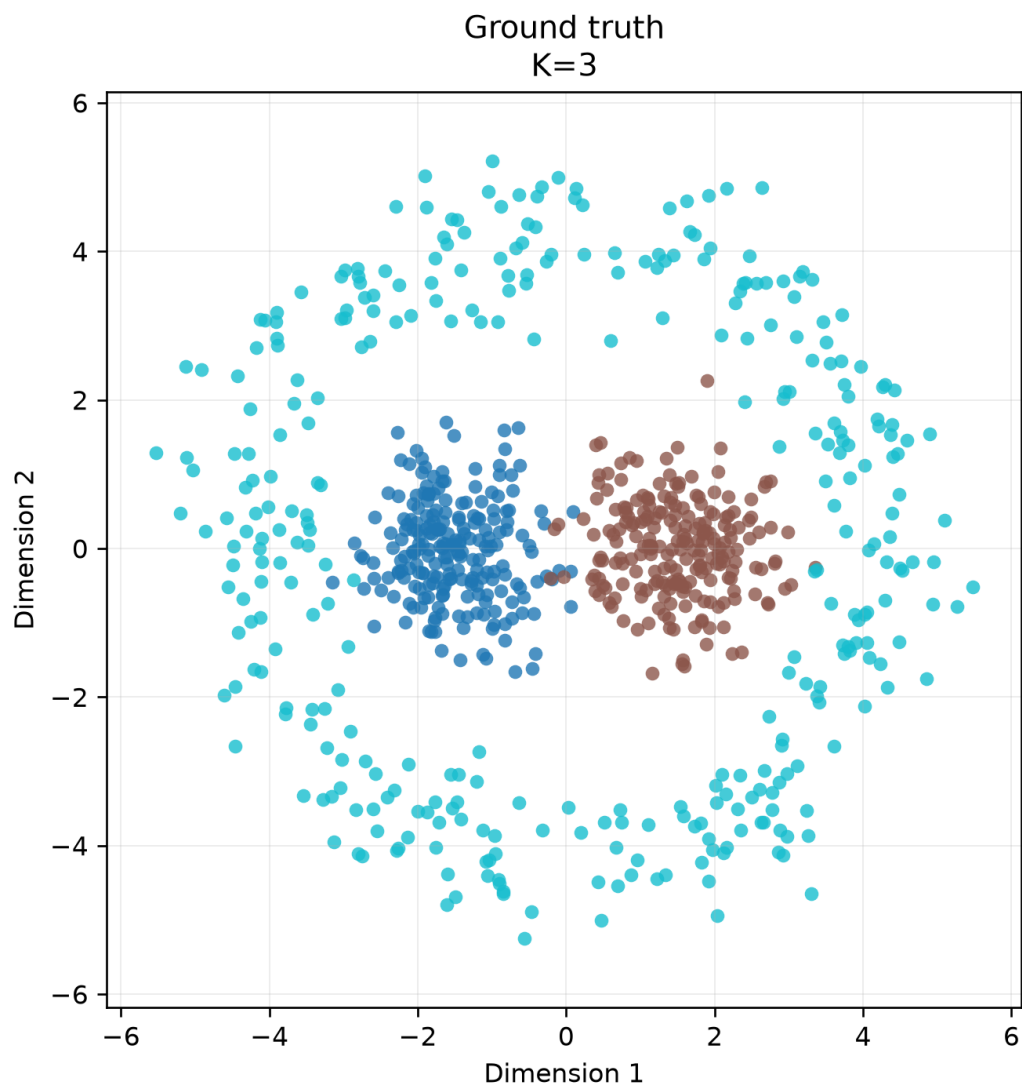
STC remains the strongest method despite further overclustering:

True K=3

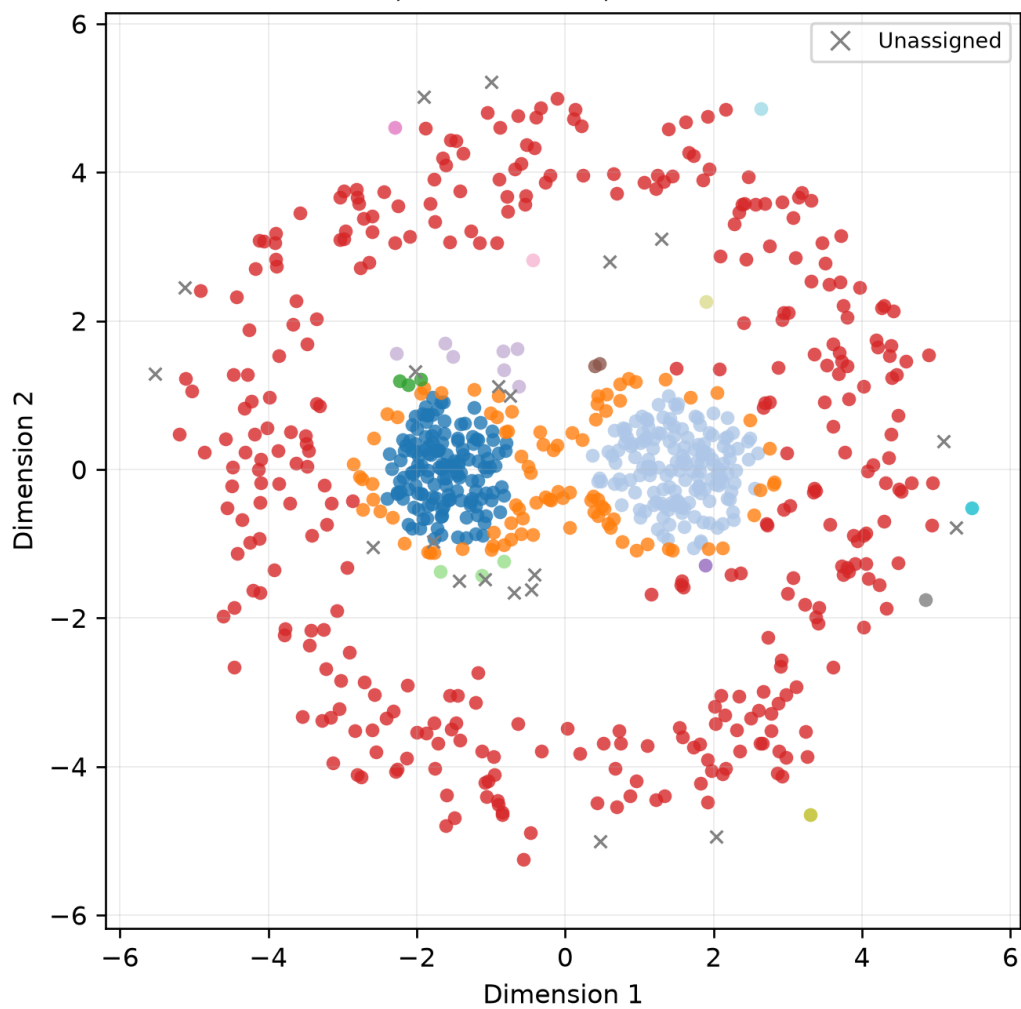
- **STC: ARI 0.7277, NMI 0.6838, K=16**
- DBSCAN: ARI -0.0219, NMI 0.0975, K=2
- HDBSCAN: ARI 0.2659, NMI 0.4412, K=14

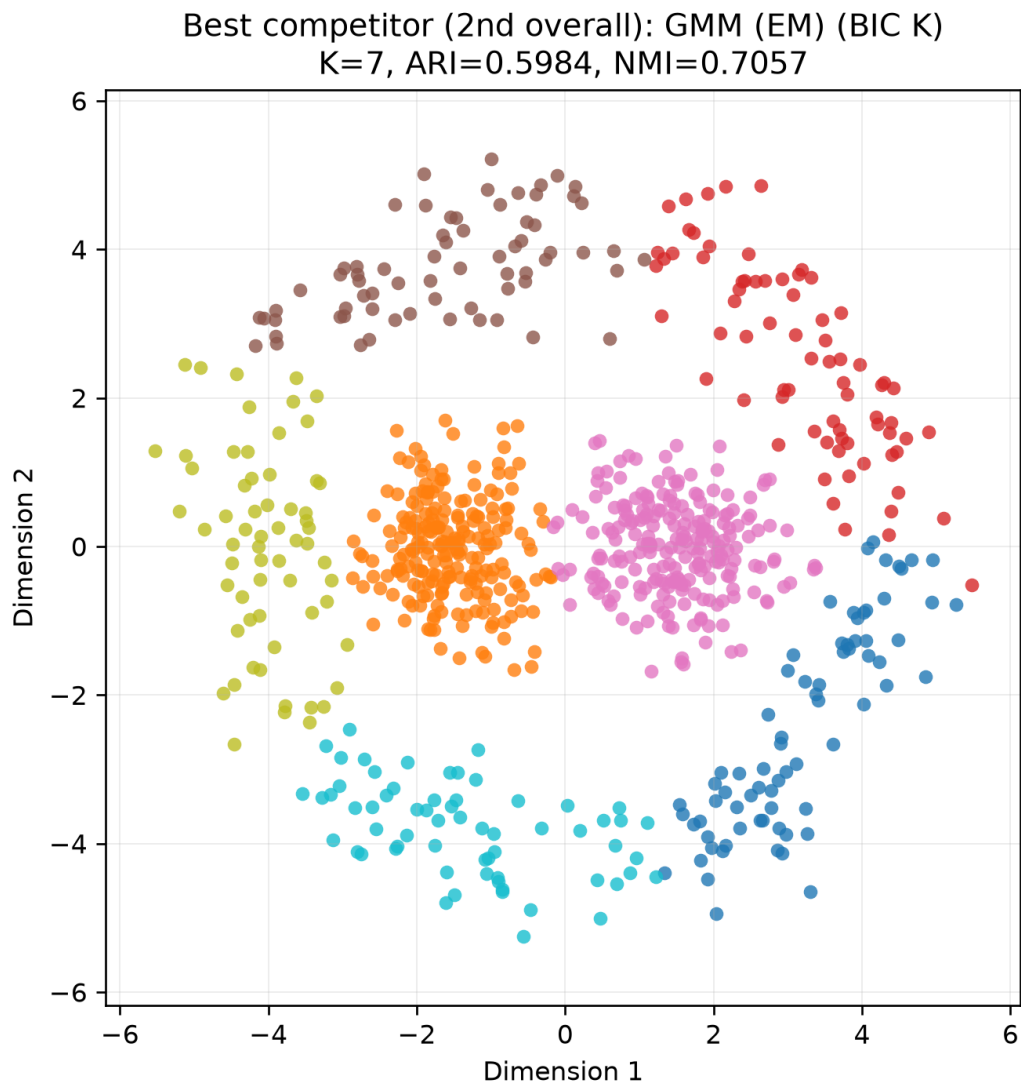
- OPTICS: ARI 0.3808, NMI 0.5610, K=4
- k-means: ARI 0.3973, NMI 0.5062, K=4
- GMM-BIC: ARI 0.5984, NMI 0.7057, K=7
- UPGMA: ARI 0.3352, NMI 0.4536, K=4
- Ward: ARI 0.3790, NMI 0.4842, K=4
-

STC achieves the highest ARI under high noise, although it severely overclusters with **K=16 instead of K=3**.



SHATTER partition
K=16, ARI=0.7277, NMI=0.6838





Report

STC performs substantially better than the evaluated baseline methods on nested cluster geometry. It achieves the highest mean ARI of all methods at 0.8768 and remains the strongest method under intermediate and high noise. Although STC overestimates the number of clusters, producing 6, 12, and 16 clusters for a ground-truth K of 3, the resulting partitions remain strongly aligned with the known structure. This indicates that STC is particularly effective at preserving complex nested geometry, even when its partition is more fragmented than the ground truth. The nested-structure benchmark therefore represents a clear strength of STC and provides strong evidence

that the method can outperform conventional clustering approaches on geometrically complex data.

Noise Outlier

1. Low noise

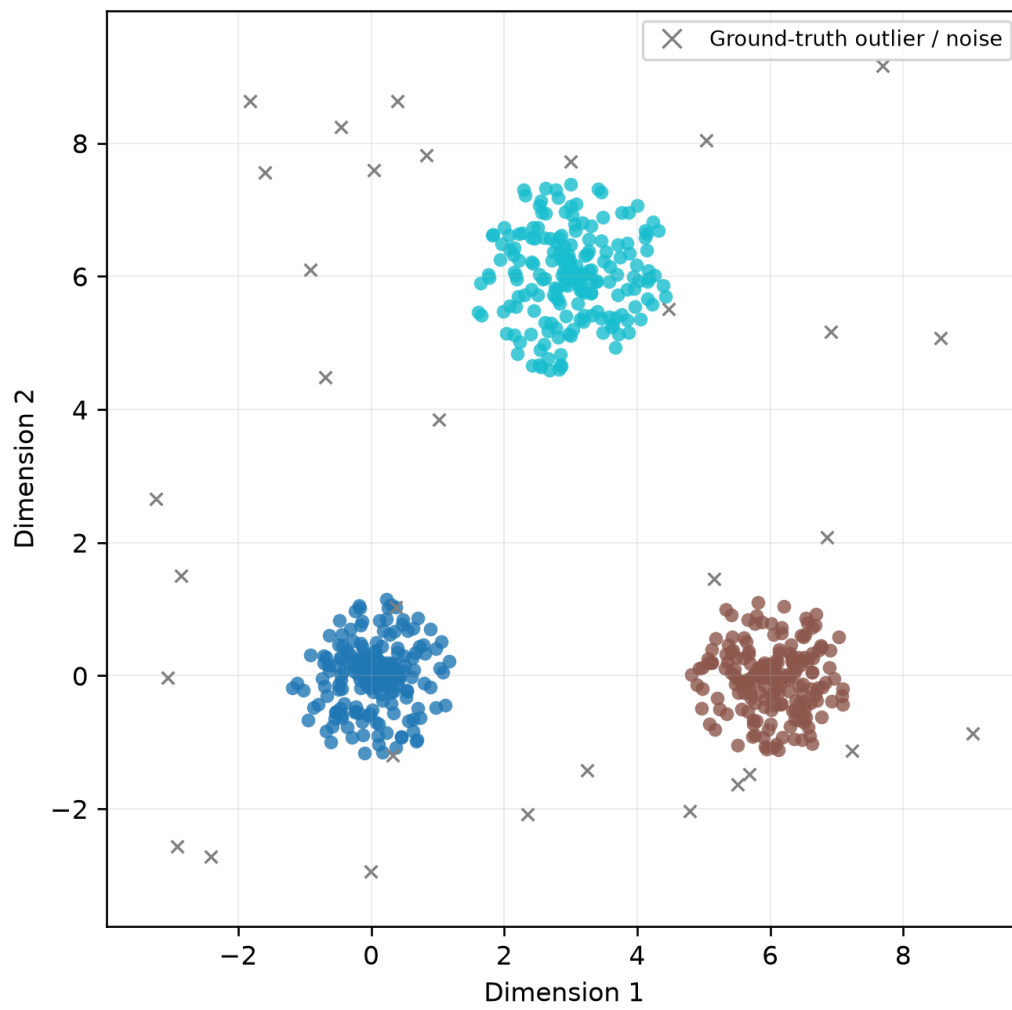
STC performs extremely well in ARI/NMI, but strongly overclusters:

True K=3

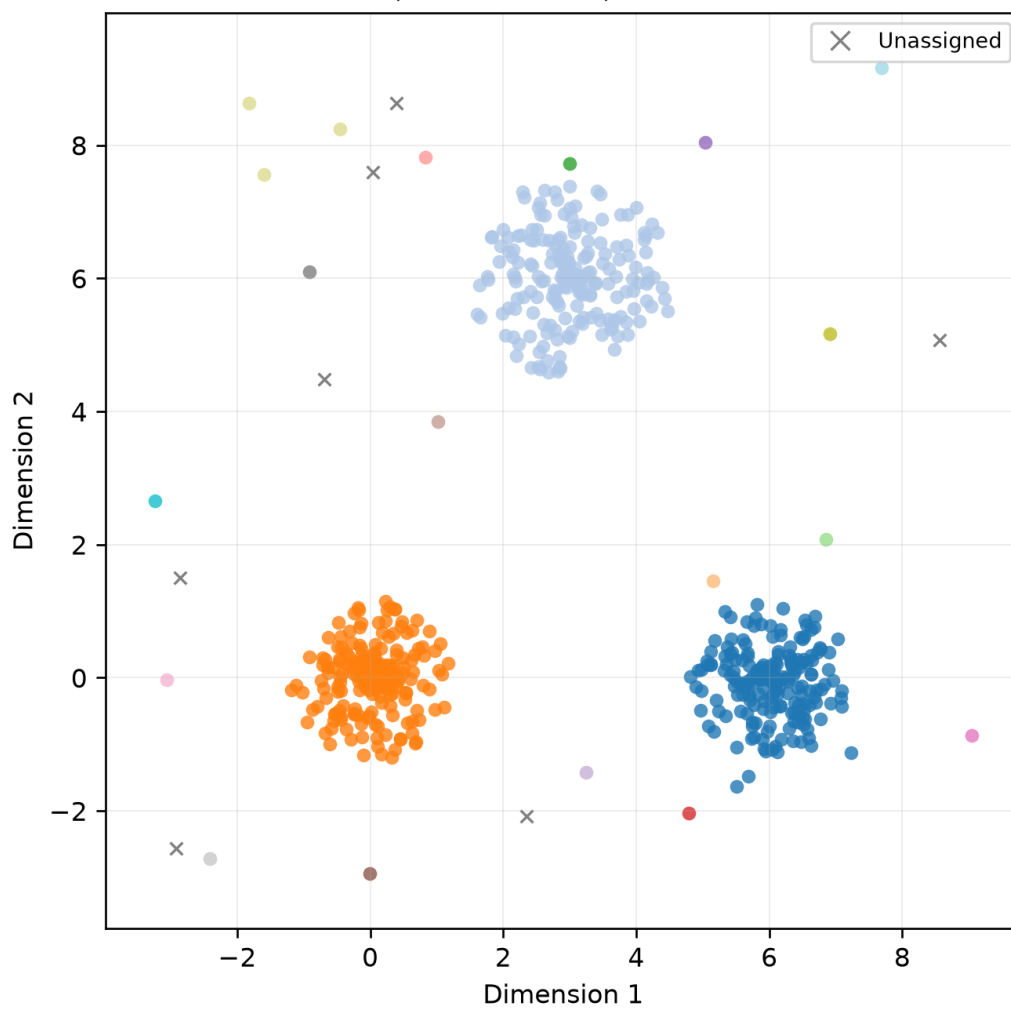
- **STC: ARI 0.9802, NMI 0.9248, K=20**
- DBSCAN: ARI 0.9786, NMI 0.9560, K=3
- HDBSCAN: ARI 0.9398, NMI 0.9069, K=3
- OPTICS: ARI 0.9353, NMI 0.9023, K=3
- k-means: ARI 0.9222, NMI 0.8924, K=3
- **GMM-BIC: ARI 0.9786, NMI 0.9560, K=4**
- UPGMA: ARI 0.9222, NMI 0.8927, K=3
- Ward: ARI 0.9222, NMI 0.8927, K=3

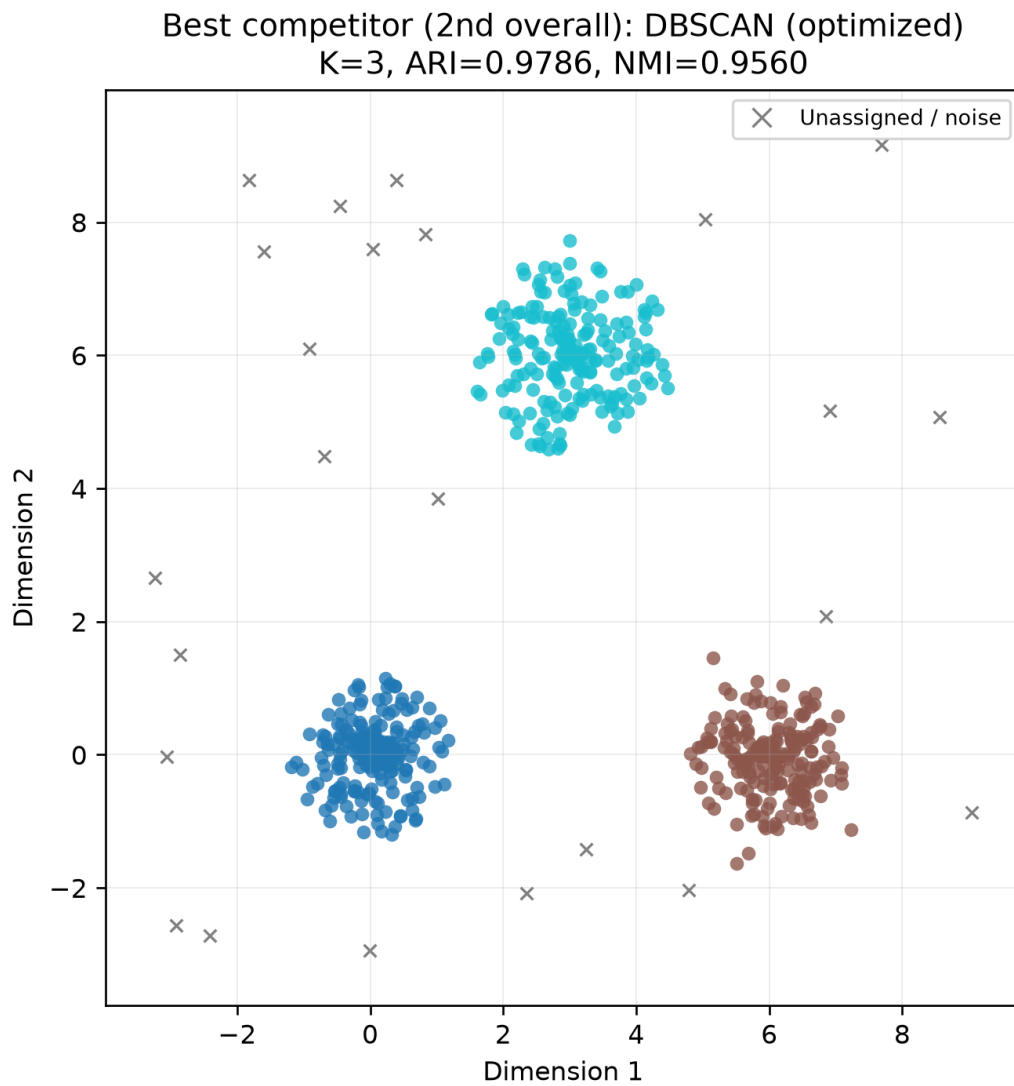
STC has the highest ARI, but its **K=20 instead of K=3** shows severe fragmentation despite the strong agreement score.

Ground truth
K=3



SHATTER partition
K=20, ARI=0.9802, NMI=0.9248





2. Intermediate noise

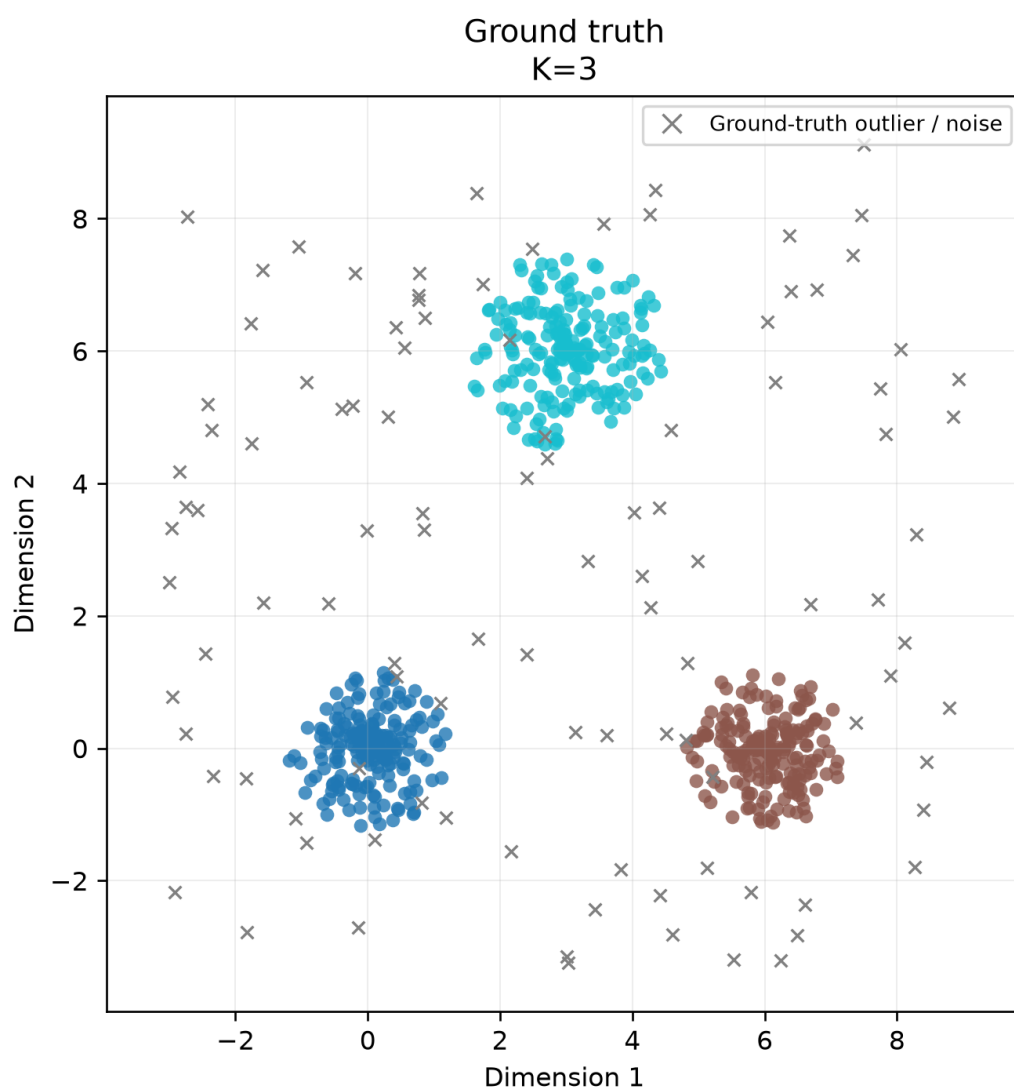
STC remains strong, but the overclustering becomes even more severe:

True K=3

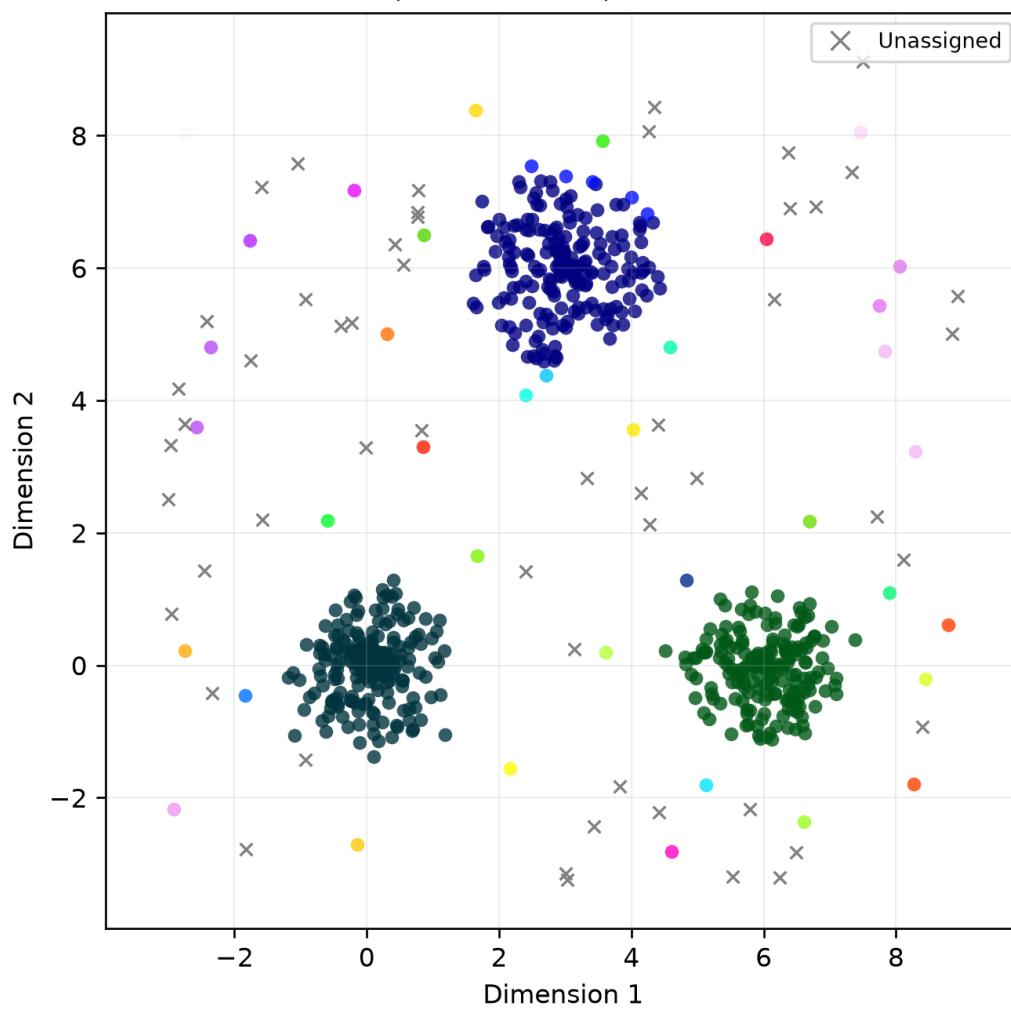
- **STC: ARI 0.9169, NMI 0.8374, K=37**
- DBSCAN: ARI 0.8282, NMI 0.8061, K=3
- HDBSCAN: ARI 0.8809, NMI 0.8349, K=7
- OPTICS: ARI 0.8634, NMI 0.8409, K=3

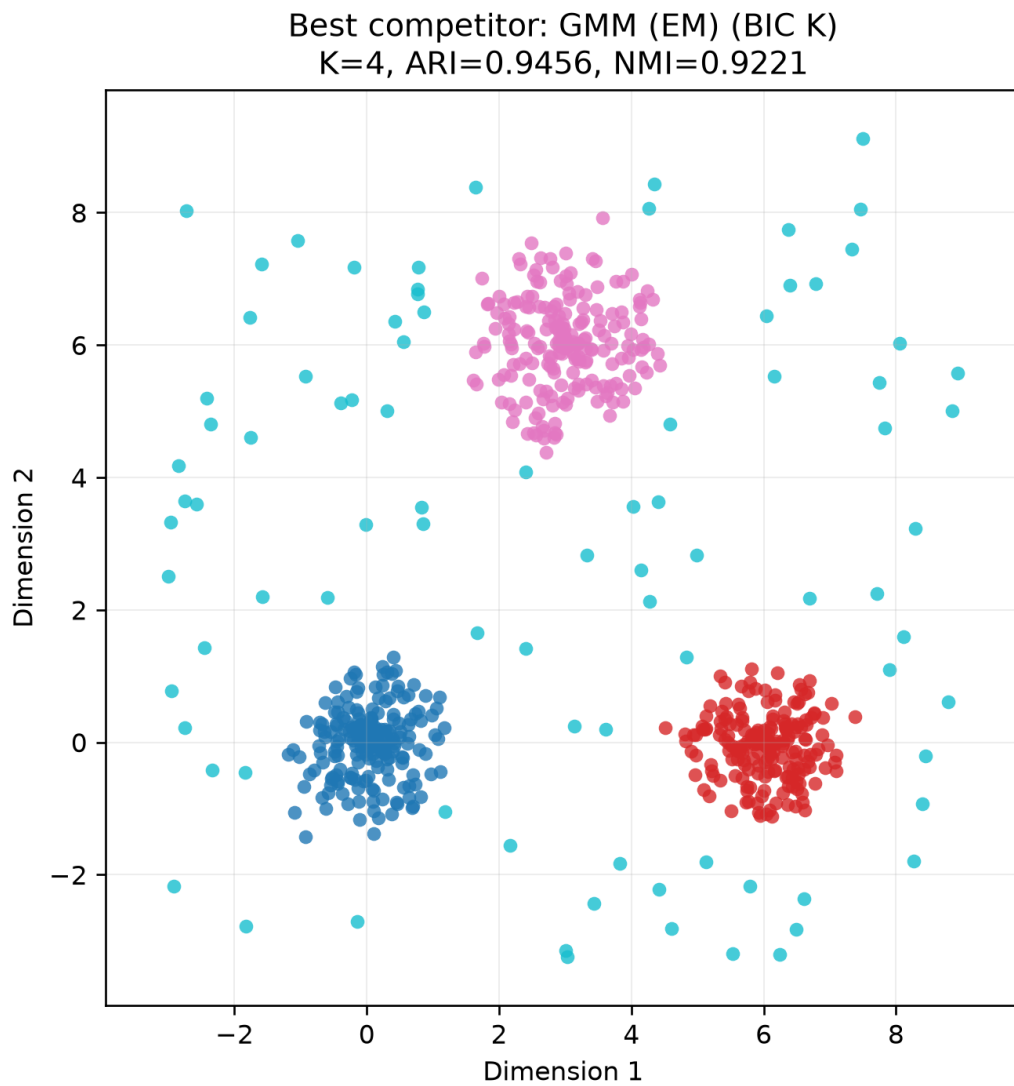
- k-means: ARI 0.7620, NMI 0.7618, K=3
- **GMM-BIC: ARI 0.9456, NMI 0.9221, K=4**
- UPGMA: ARI 0.7618, NMI 0.7612, K=3
- Ward: ARI 0.7660, NMI 0.7744, K=3

GMM-BIC performs best. STC still achieves high ARI/NMI, but severely overclusters with **K=37** instead of K=3.



SHATTER partition
K=37, ARI=0.9169, NMI=0.8374





3. High noise

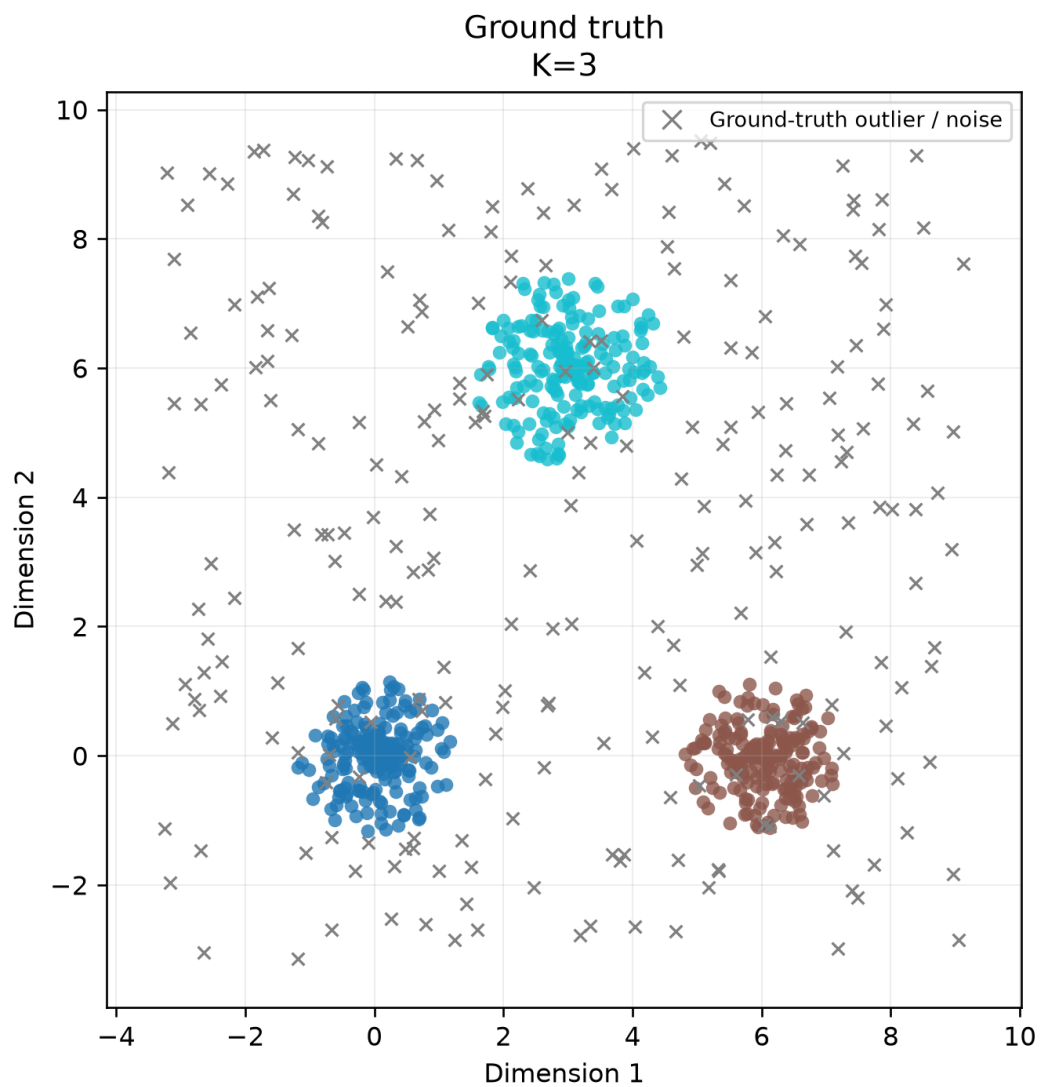
STC degrades under high outlier contamination but remains competitive:

True K=3

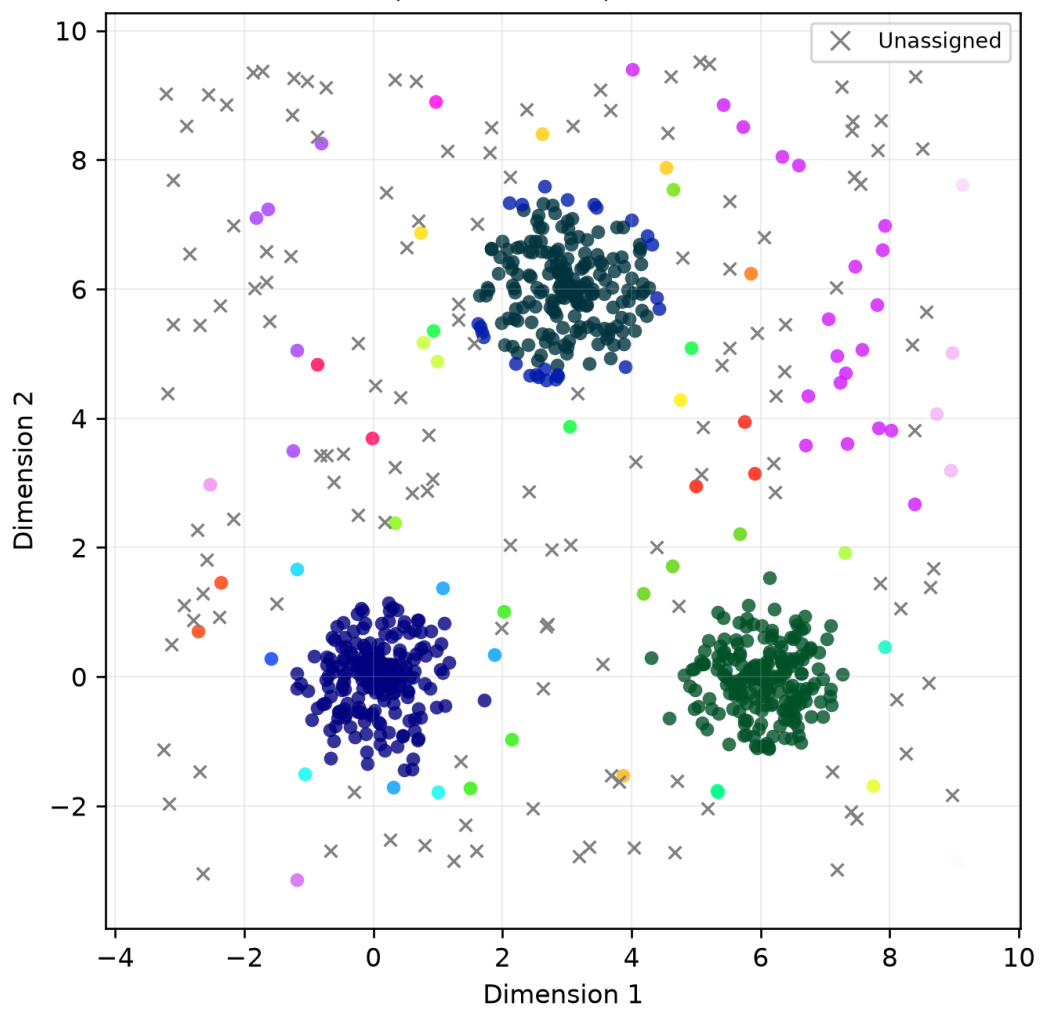
- **STC: ARI 0.7319, NMI 0.7293, K=34**
- DBSCAN: ARI 0.5927, NMI 0.6663, K=5
- HDBSCAN: ARI 0.6768, NMI 0.7087, K=10

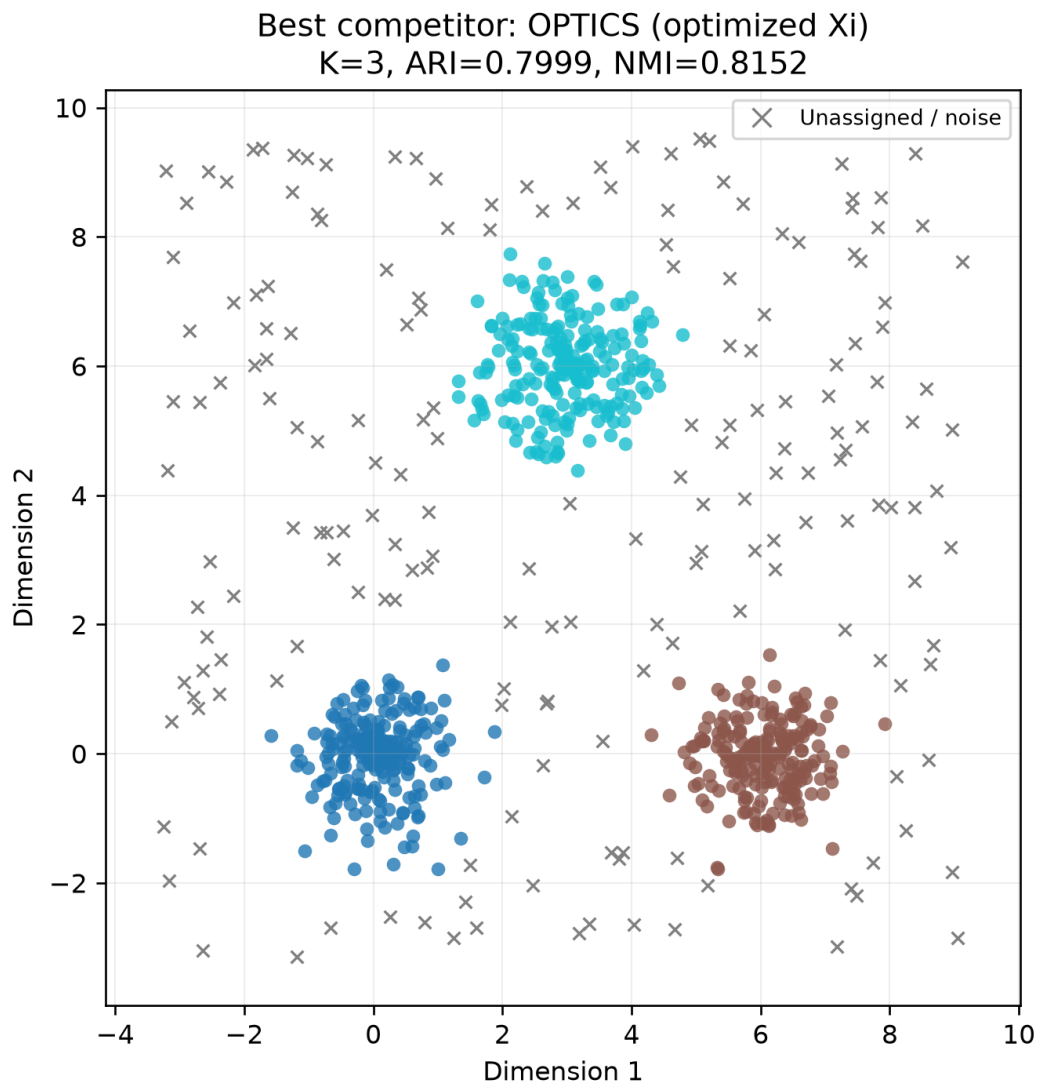
- **OPTICS: ARI 0.7999, NMI 0.8152, K=3**
- k-means: ARI 0.5269, NMI 0.6282, K=3
- GMM-BIC: ARI 0.7610, NMI 0.8032, K=5
- UPGMA: ARI 0.5445, NMI 0.6460, K=3
- Ward: ARI 0.5297, NMI 0.6310, K=3

OPTICS performs best under high noise. STC still retains relatively strong ARI/NMI, but severely overclusters with **K=34 instead of K=3**.



SHATTER partition
K=34, ARI=0.7319, NMI=0.7293





Report

STC demonstrates comparatively strong robustness to outlier contamination. It achieves a mean ARI of 0.8763, second only to GMM-BIC at 0.8951 and higher than all other evaluated primary baselines. STC is the best-performing primary method at low contamination and remains highly competitive at intermediate and high contamination. However, it substantially overestimates the number of clusters, producing 20, 37, and 34 clusters for a ground-truth K of 3. Despite this fragmentation, ARI remains high, indicating that the major underlying cluster relationships are largely preserved. The experiment therefore suggests that STC is robust to outliers in terms of partition

agreement, but that controlling the formation of many additional subclusters remains an important limitation.

Ring

1. Low noise

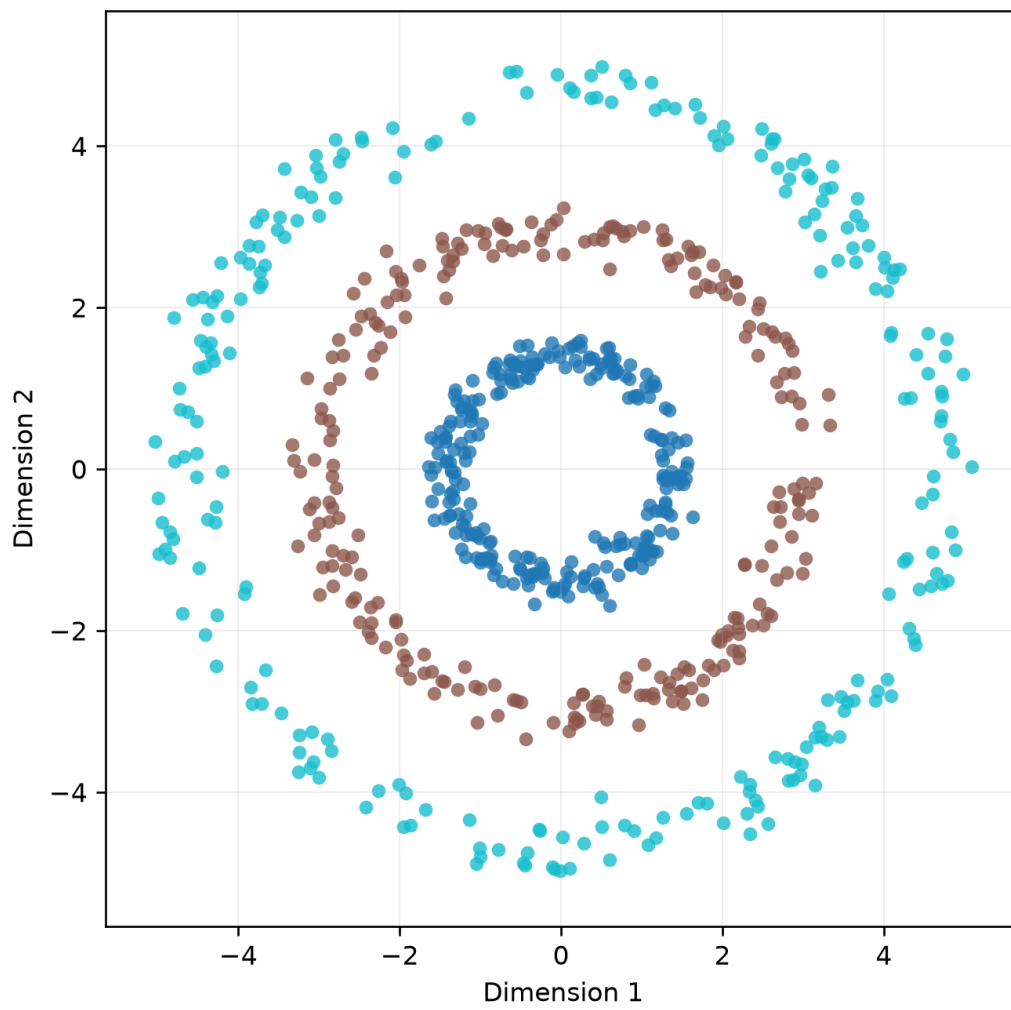
STC perfectly recovers the ring structure:

True K=3

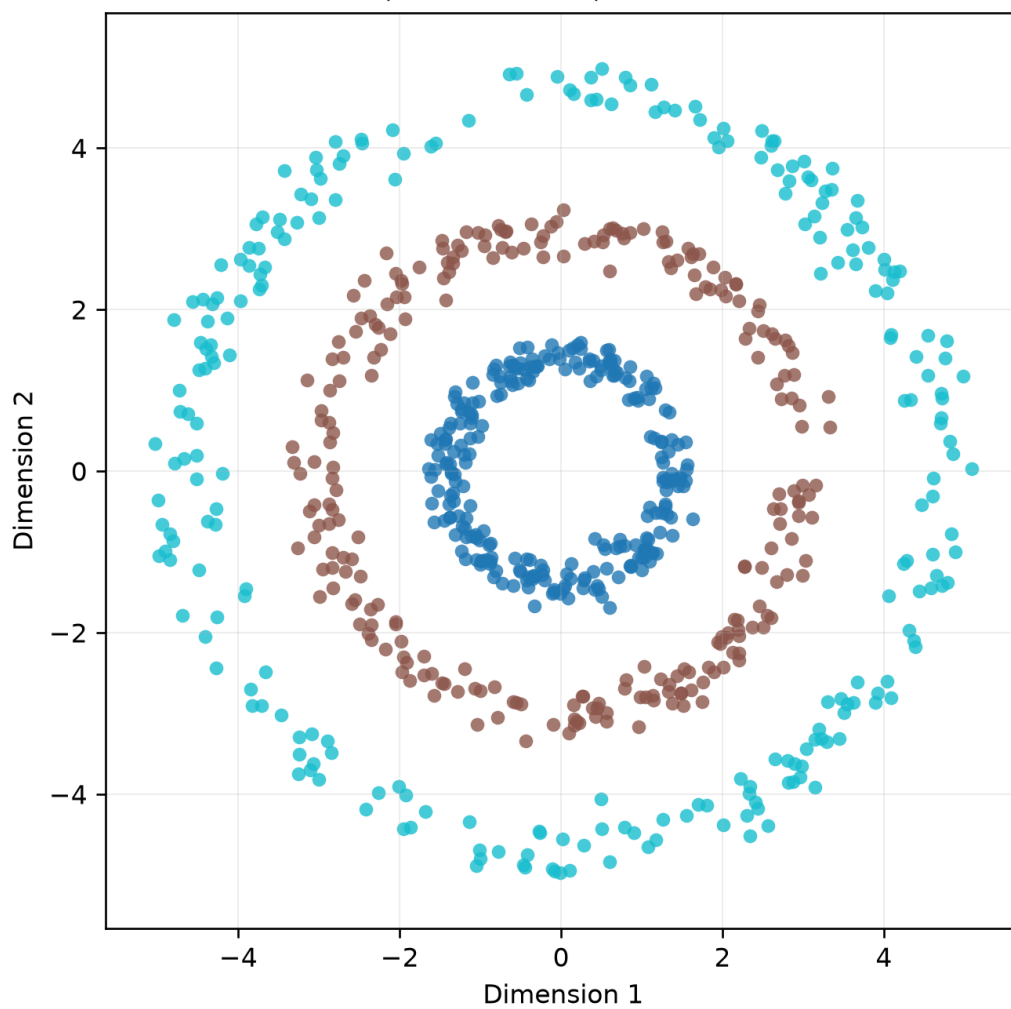
- **STC: ARI 1.0000, NMI 1.0000, K=3**
- DBSCAN: ARI 0.0576, NMI 0.2011, K=3
- HDBSCAN: ARI 0.7563, NMI 0.7286, K=15
- OPTICS: ARI 0.2121, NMI 0.4984, K=11
- k-means: ARI 0.0011, NMI 0.0045, K=4
- GMM-BIC: ARI 0.1993, NMI 0.4360, K=12
- UPGMA: ARI 0.3733, NMI 0.4003, K=6
- Ward: ARI 0.0769, NMI 0.1408, K=4

STC clearly performs best and perfectly recovers **K=3**. This is a particularly strong result because most conventional baselines struggle with the non-spherical ring structure.

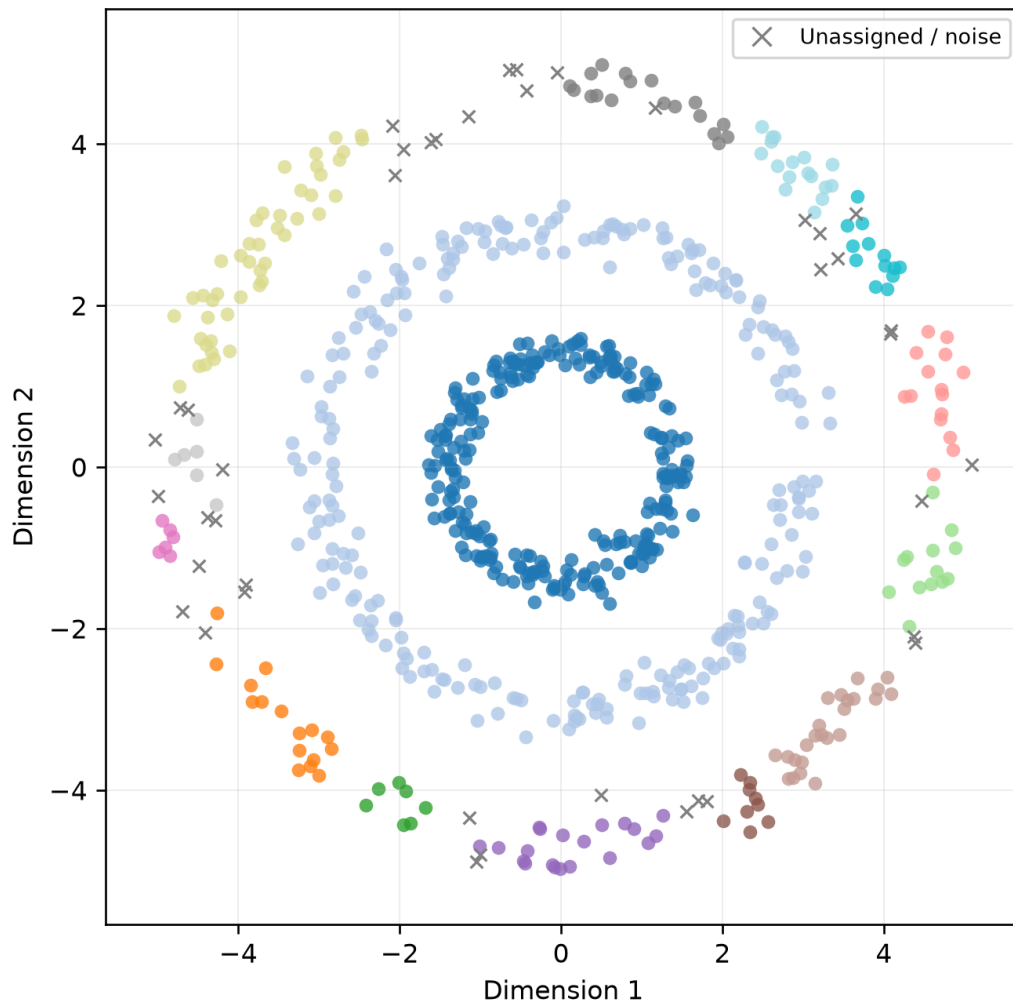
Ground truth
K=3



SHATTER partition
K=3, ARI=1.0000, NMI=1.0000



Best competitor (2nd overall): HDBSCAN (EOM)
K=15, ARI=0.7563, NMI=0.7286



2. Intermediate noise

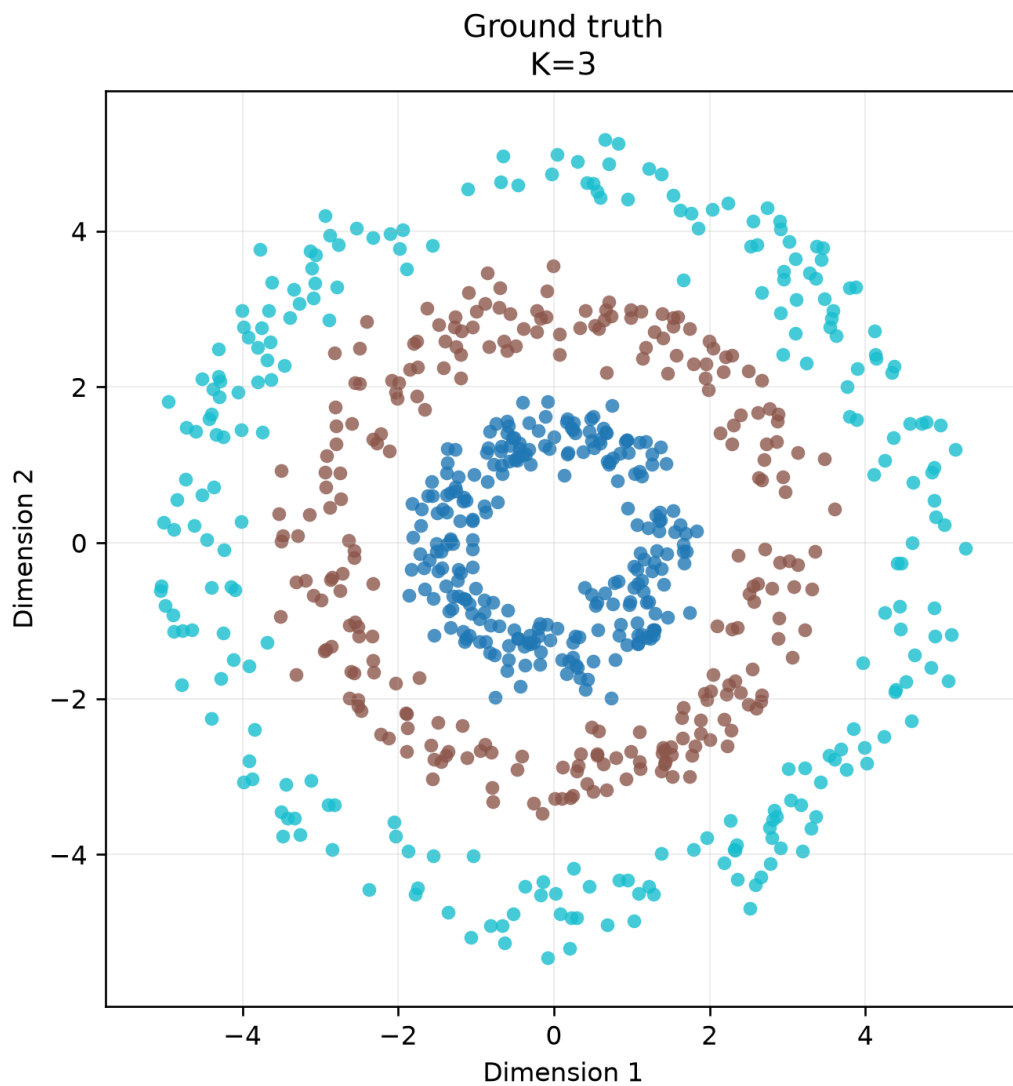
STC remains the strongest method, but overclustering appears:

True K=3

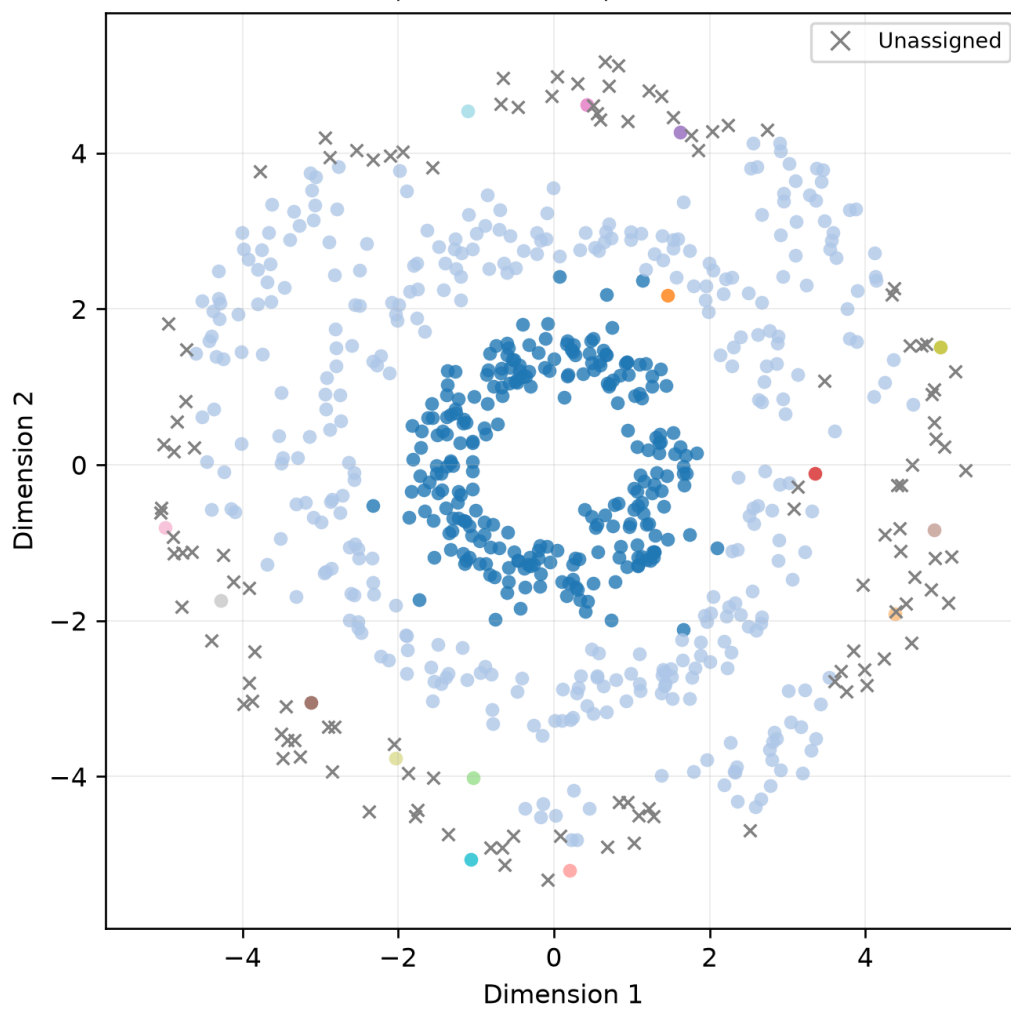
- **STC: ARI 0.6010, NMI 0.6512, K=17**
- **DBSCAN: ARI 0.0018, NMI 0.0419, K=2**

- HDBSCAN: ARI 0.4734, NMI 0.5593, K=21
- OPTICS: ARI 0.0752, NMI 0.3837, K=20
- k-means: ARI 0.0018, NMI 0.0051, K=4
- GMM-BIC: ARI 0.0000, NMI 0.0000, K=1
- UPGMA: ARI 0.3482, NMI 0.4109, K=6
- Ward: ARI 0.0151, NMI 0.0164, K=3
-

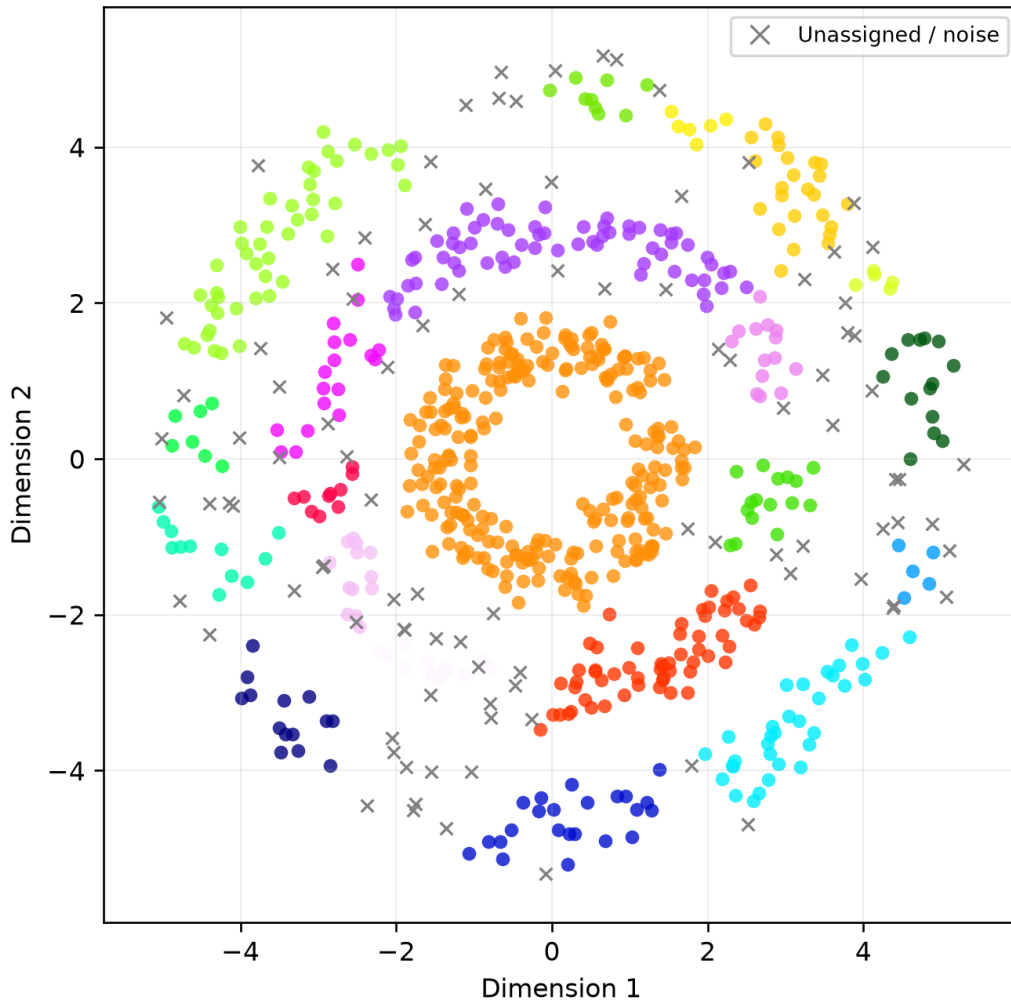
STC performs best, but the result becomes fragmented with **K=17 instead of K=3**.



SHATTER partition
K=17, ARI=0.6010, NMI=0.6512



K=21, ARI=0.4734, NMI=0.5593



3. High noise

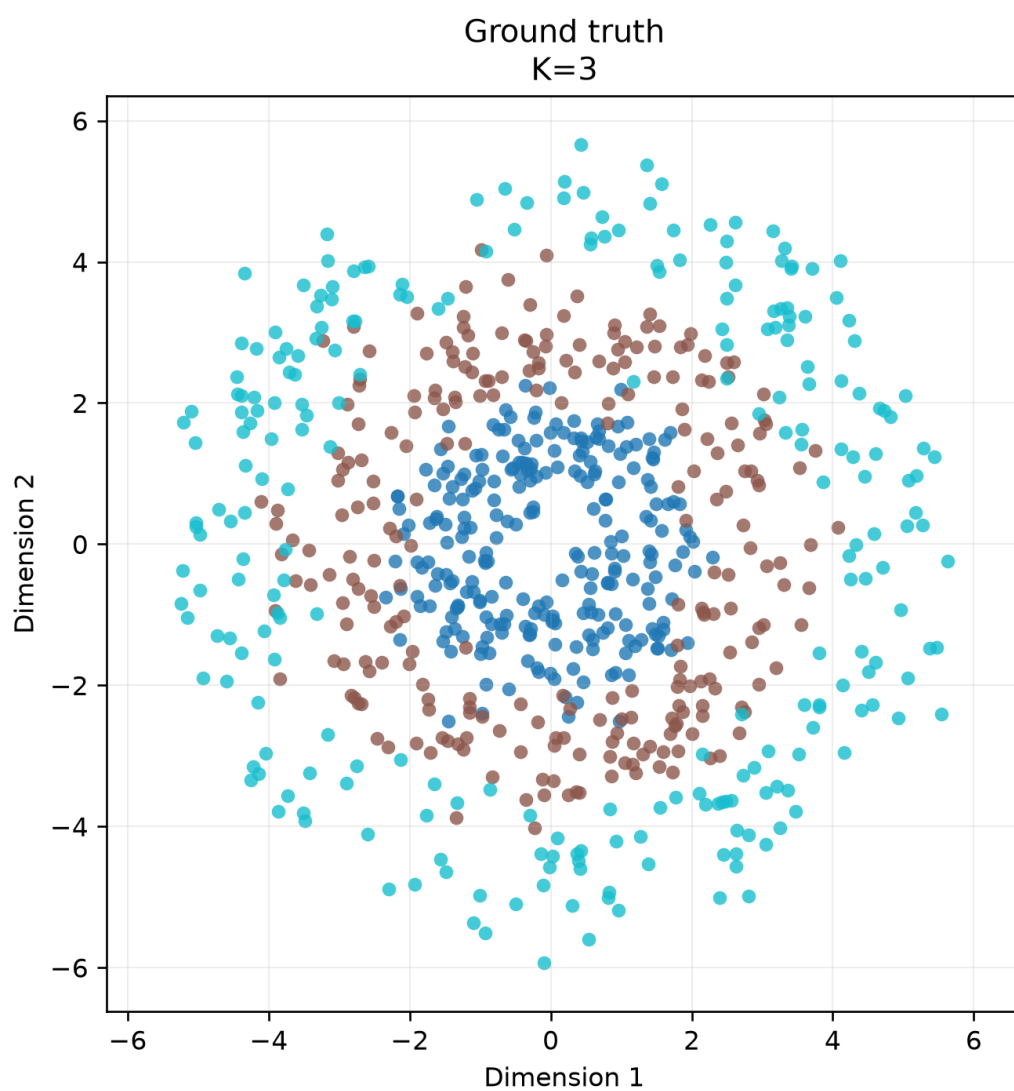
All methods struggle considerably, but STC still gives the strongest result:

True K=3

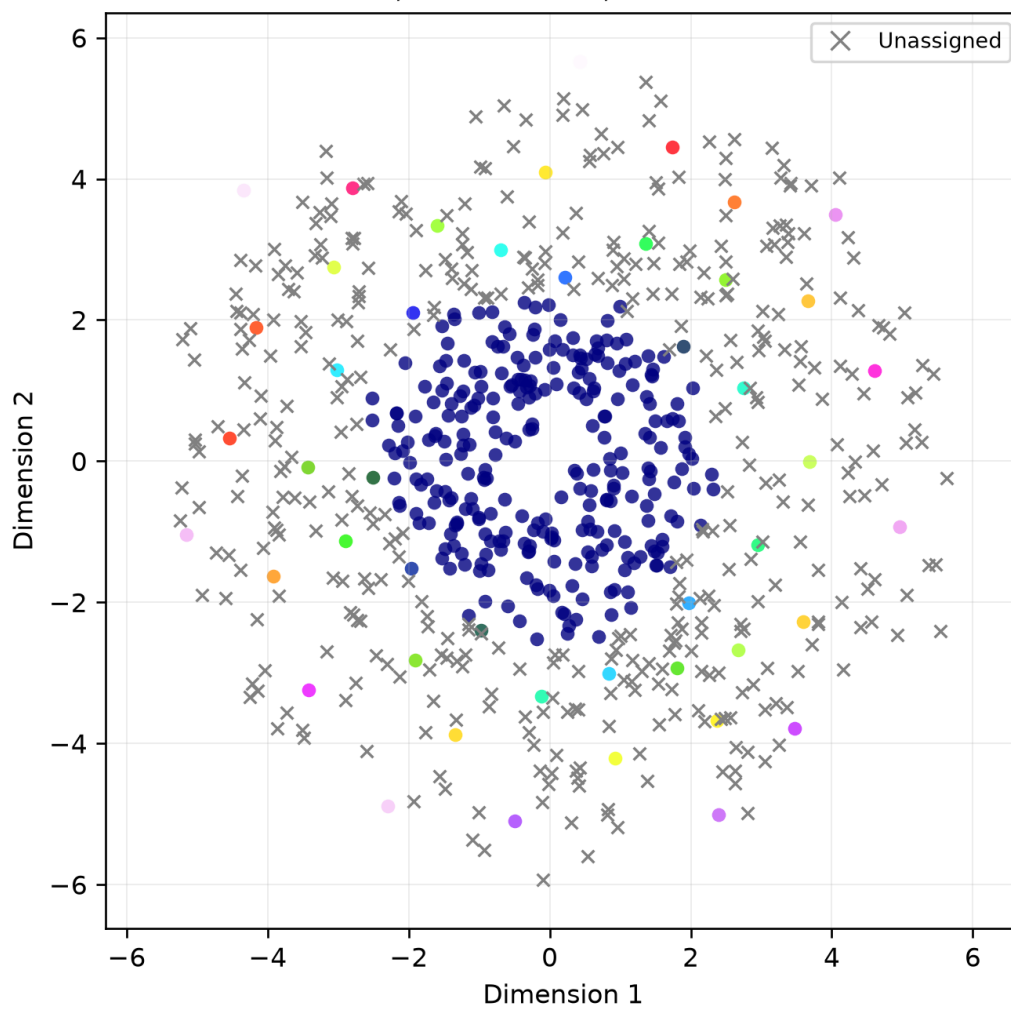
- **STC: ARI 0.4596, NMI 0.4848, K=47**
- DBSCAN: ARI 0.0917, NMI 0.2078, K=2
- HDBSCAN: ARI 0.0470, NMI 0.2485, K=32
- OPTICS: ARI 0.0613, NMI 0.2341, K=14

- k-means: ARI 0.0004, NMI 0.0037, K=4
- GMM-BIC: ARI 0.0000, NMI 0.0000, K=1
- UPGMA: ARI 0.0223, NMI 0.0570, K=4
- Ward: ARI 0.0536, NMI 0.0540, K=4
-

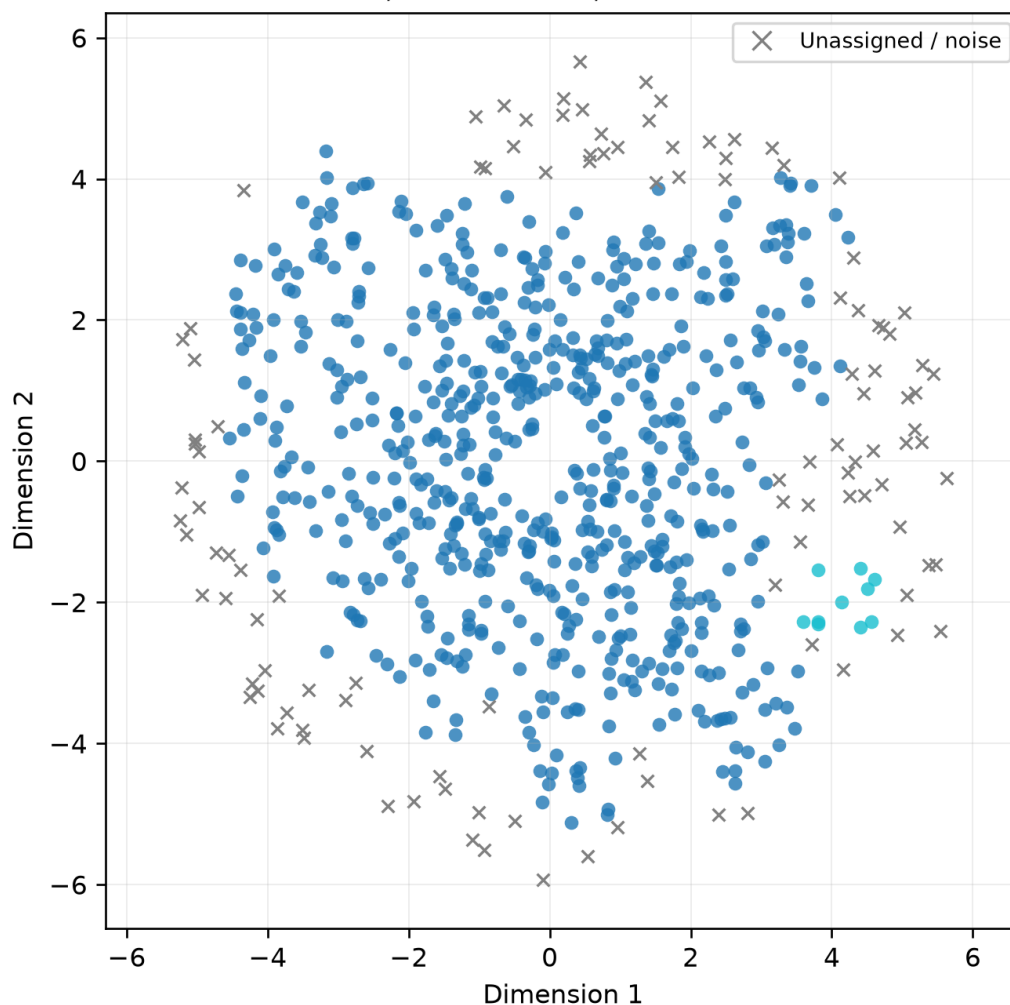
STC remains clearly better than the baselines, but severe overclustering occurs with **K=47** instead of **K=3**.



SHATTER partition
K=47, ARI=0.4596, NMI=0.4848



Best competitor (2nd overall): DBSCAN (optimized)
 $K=2$, ARI=0.0917, NMI=0.2078



Report

STC clearly outperforms all evaluated baseline methods on concentric ring geometry. It perfectly recovers the three ground-truth clusters under low noise and remains the strongest method at intermediate and high noise. The mean ARI of STC is 0.6869, compared with 0.4256 for HDBSCAN and substantially lower values for the remaining baselines. The main limitation is increasing over-fragmentation: the predicted cluster count rises from the correct $K = 3$ at low noise to 17 and 47 at intermediate and high noise. Despite this, STC preserves the ring structure much better than the competing methods, demonstrating a clear advantage on strongly non-convex geometry.

Spherical Blobs

1. Low noise

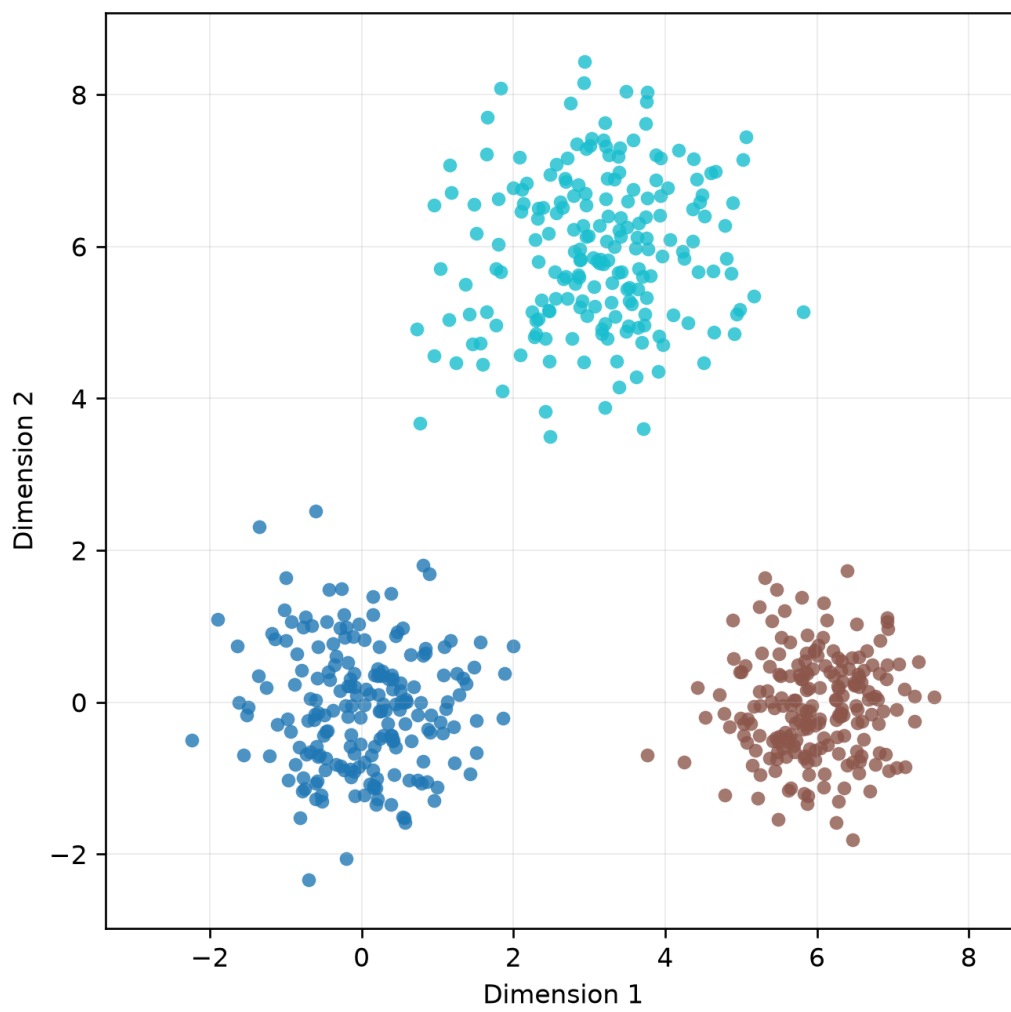
STC performs well, but already overclusters:

True K=3

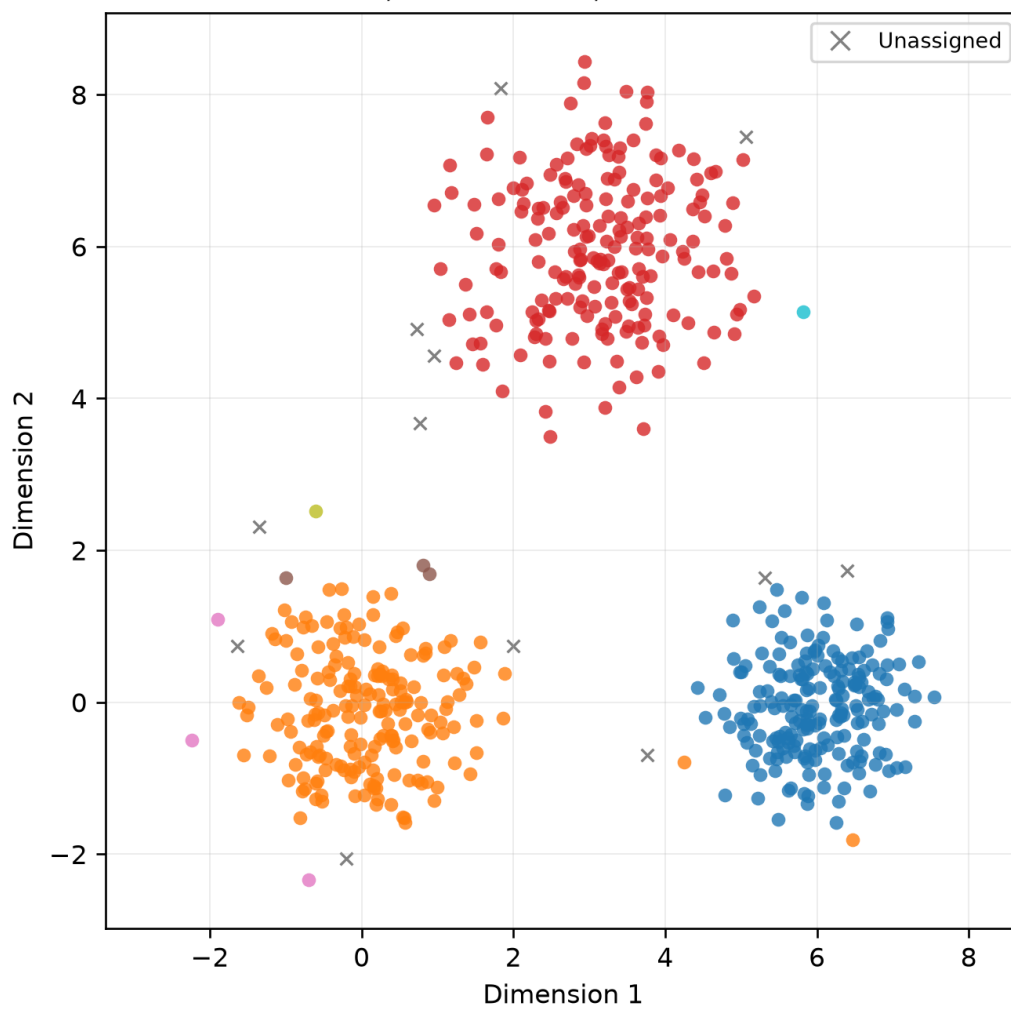
- **STC: ARI 0.9401, NMI 0.9025, K=7**
- DBSCAN: ARI 0.9950, NMI 0.9895, K=3
- HDBSCAN: ARI 1.0000, NMI 1.0000, K=3
- OPTICS: ARI 1.0000, NMI 1.0000, K=3
- **k-means: ARI 1.0000, NMI 1.0000, K=3**
- GMM-BIC: ARI 1.0000, NMI 1.0000, K=3
- UPGMA: ARI 1.0000, NMI 1.0000, K=3
- Ward: ARI 1.0000, NMI 1.0000, K=3

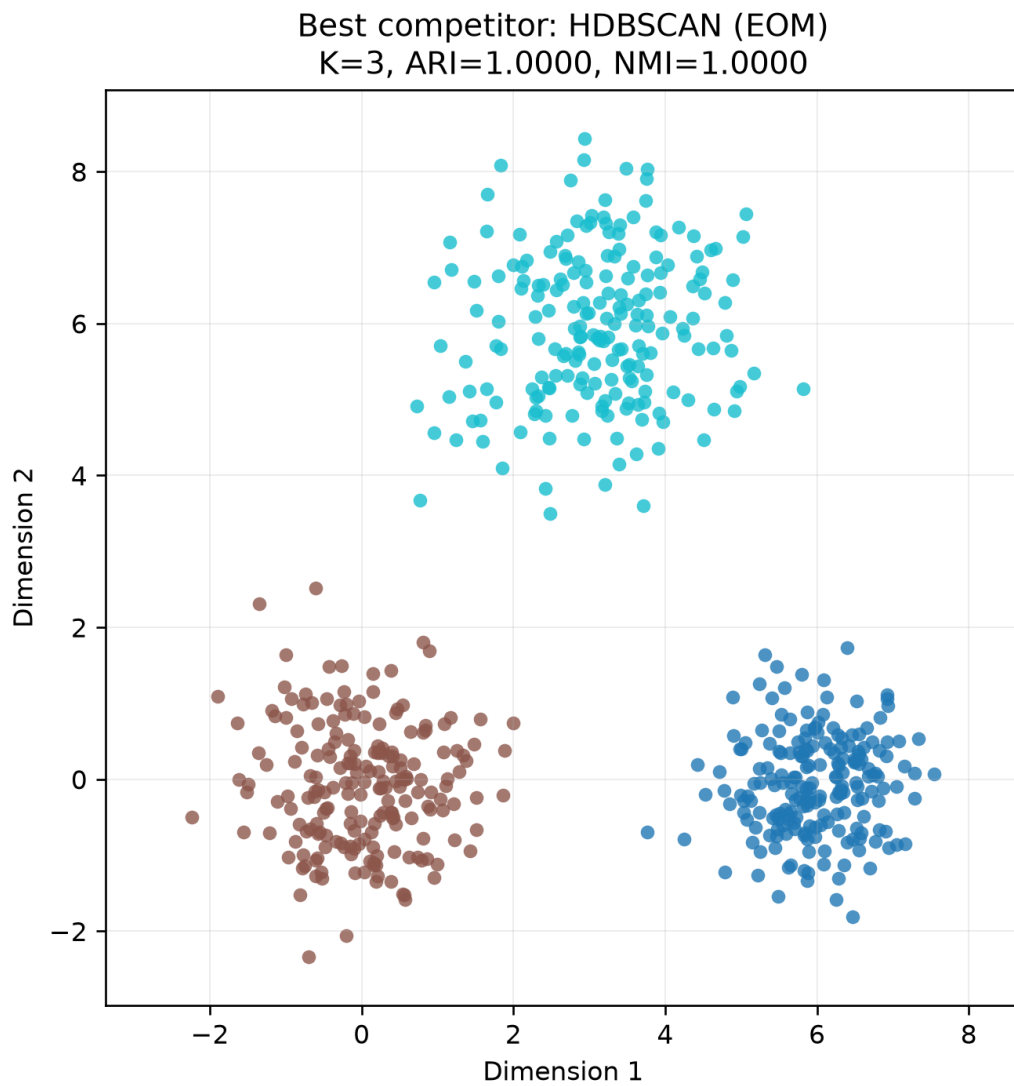
Several baselines recover the structure perfectly. STC also performs strongly but overclusters with **K=7 instead of K=3**.

Ground truth
K=3



SHATTER partition
K=7, ARI=0.9401, NMI=0.9025





2. Intermediate noise

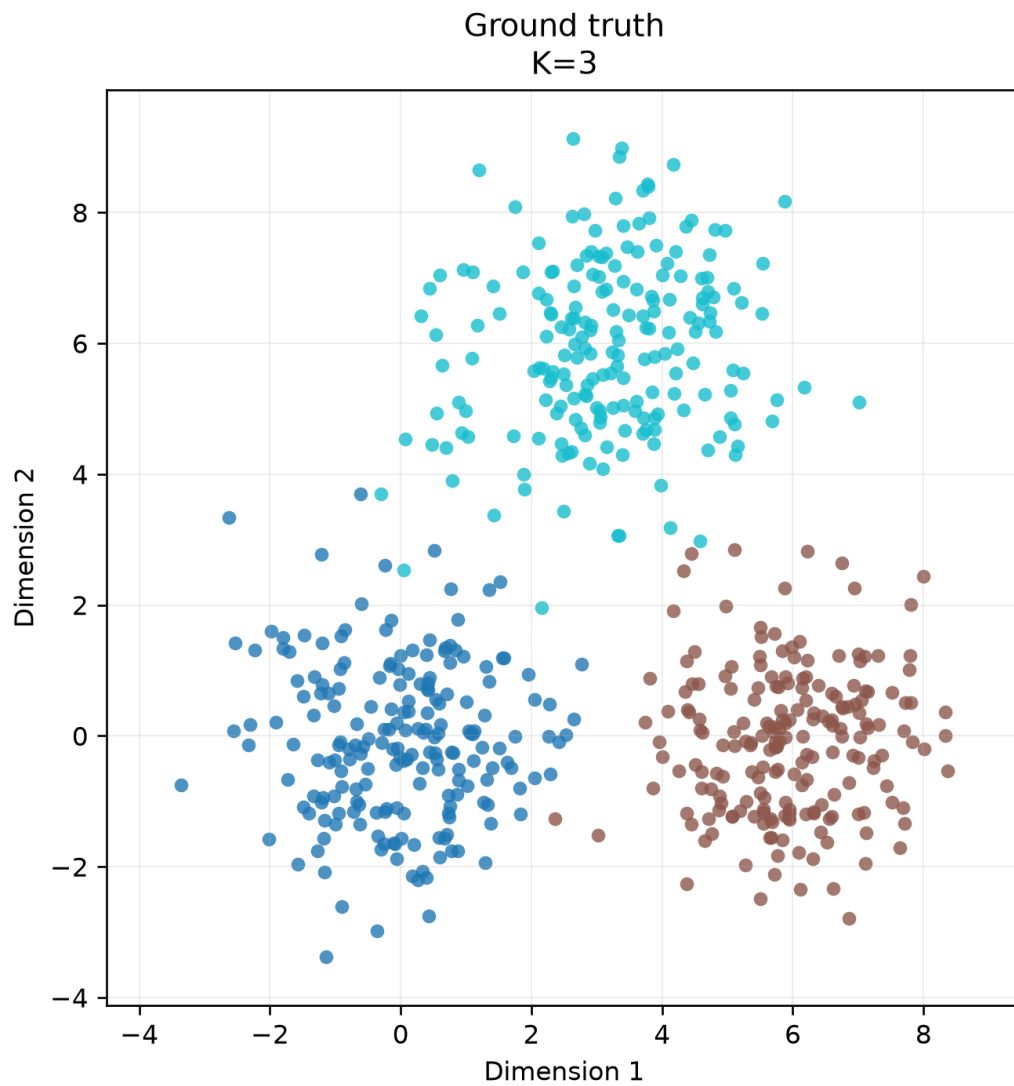
STC performance drops considerably and overclustering becomes severe:

True K=3

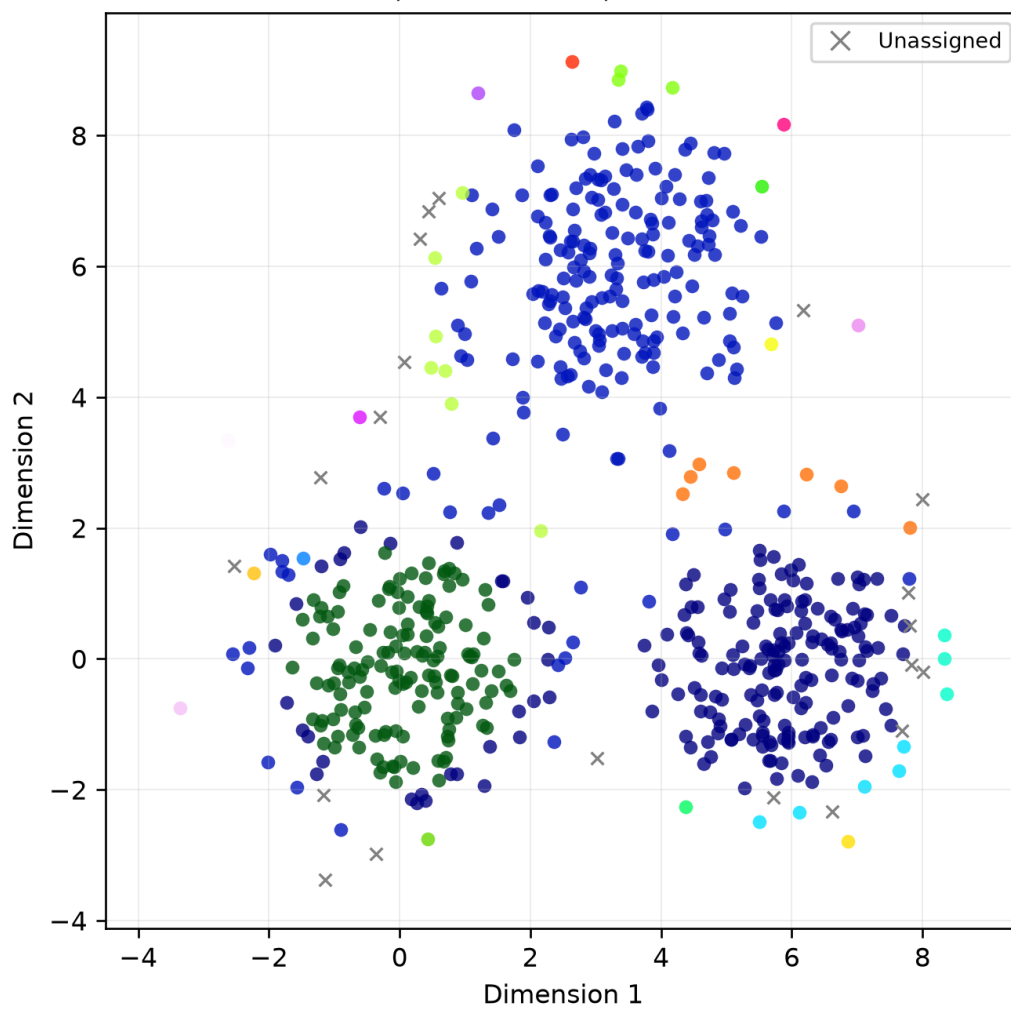
- **STC: ARI 0.6102, NMI 0.5781, K=22**
- DBSCAN: ARI 0.9650, NMI 0.9369, K=3
- HDBSCAN: ARI 0.8468, NMI 0.8020, K=4
- OPTICS: ARI 0.8237, NMI 0.8137, K=3
- **k-means: ARI 0.9800, NMI 0.9668, K=3**

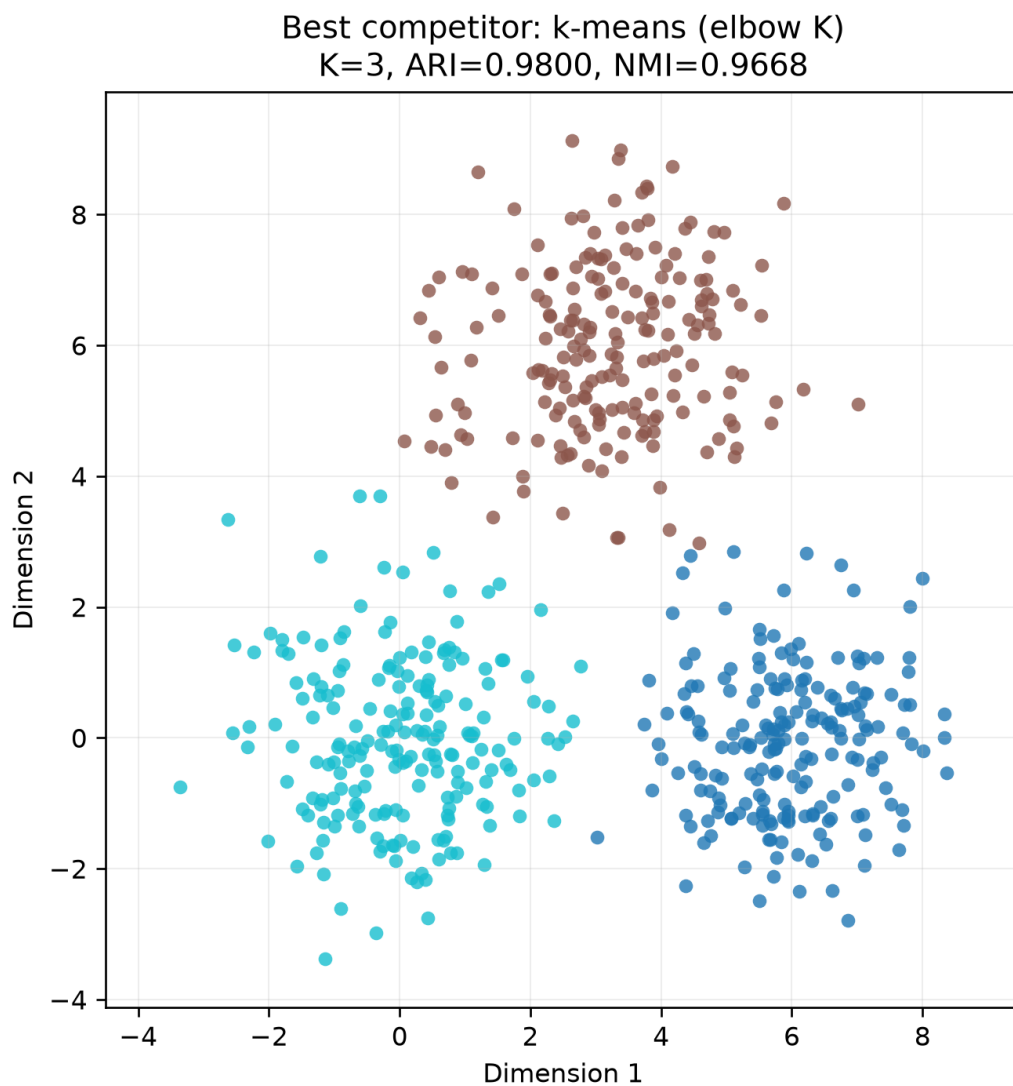
- GMM-BIC: ARI 0.9651, NMI 0.9423, K=3
- UPGMA: ARI 0.8665, NMI 0.8403, K=3
- Ward: ARI 0.9506, NMI 0.9238, K=3

K-means performs best. STC falls substantially behind and strongly overclusters with **K=22 instead of K=3**.



SHATTER partition
K=22, ARI=0.6102, NMI=0.5781





3. High noise

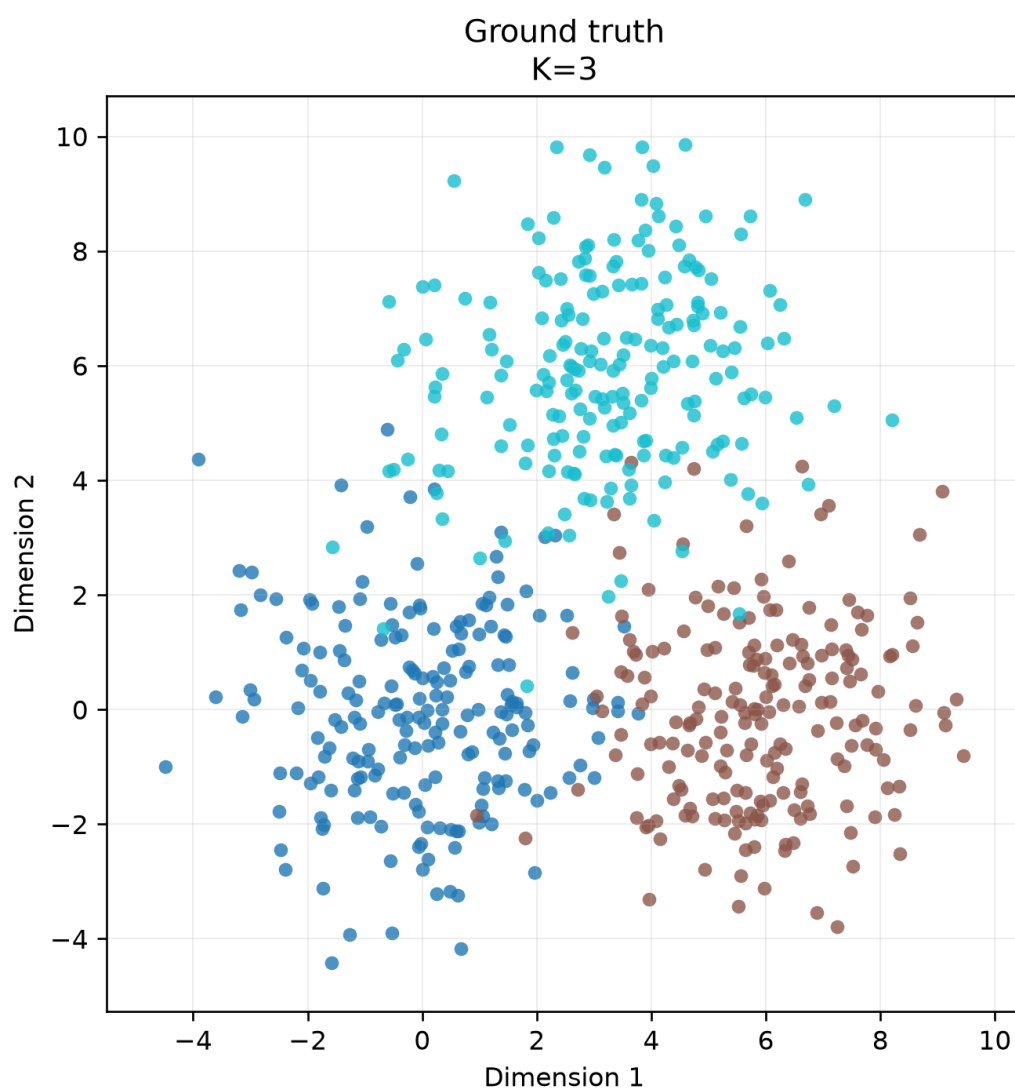
STC deteriorates strongly under high noise:

True K=3

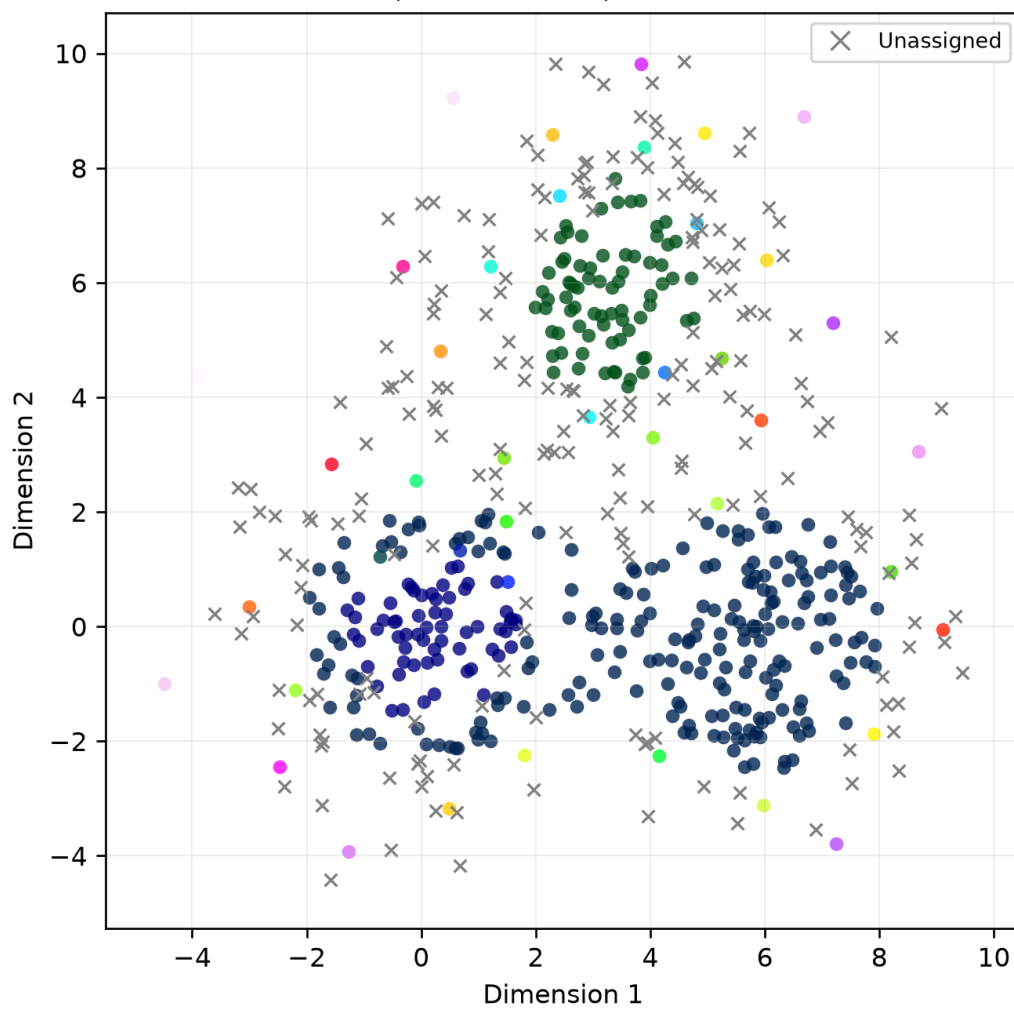
- **STC: ARI 0.2364, NMI 0.3548, K=44**
- DBSCAN: ARI 0.6506, NMI 0.6013, K=4
- HDBSCAN: ARI 0.3565, NMI 0.4018, K=3
- OPTICS: ARI 0.3310, NMI 0.4940, K=3

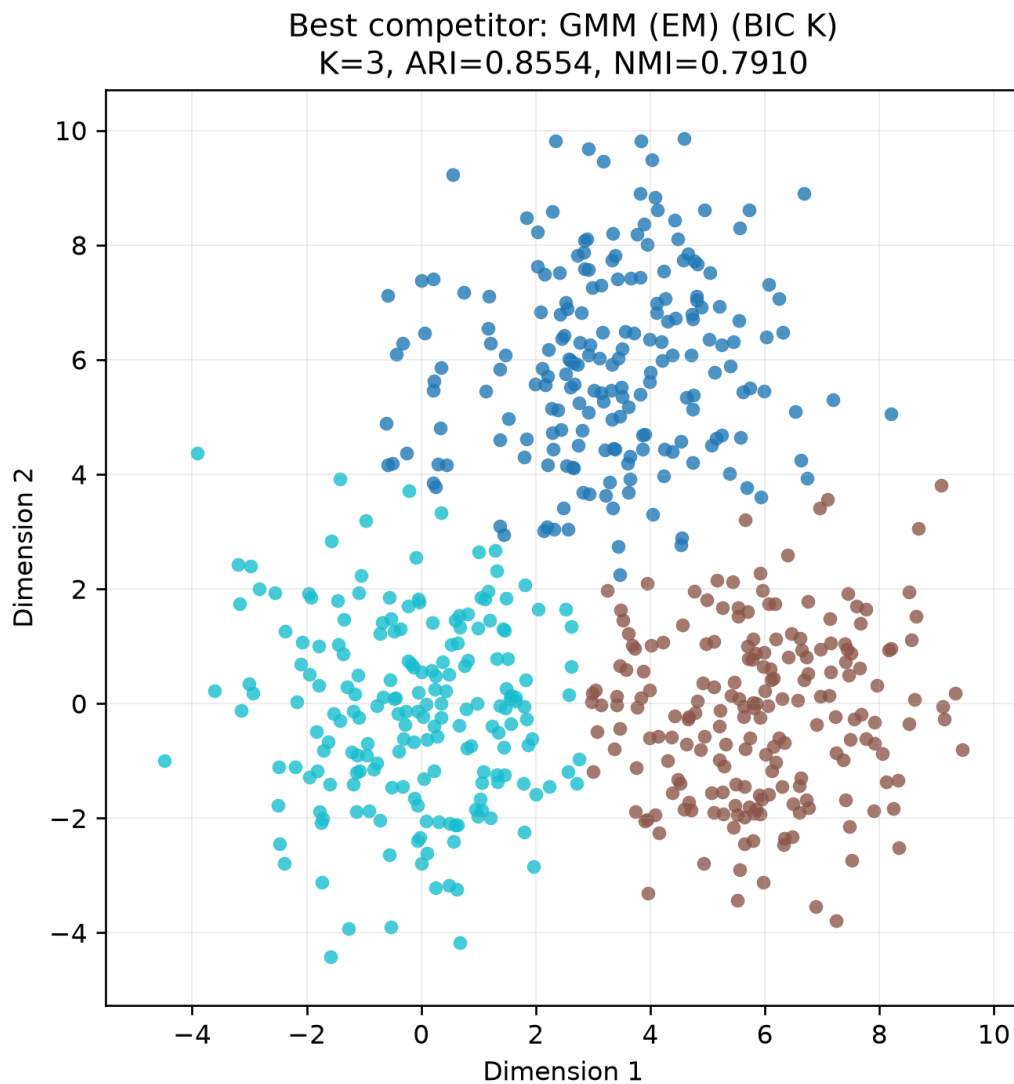
- k-means: ARI 0.8460, NMI 0.7783, K=3
- **GMM-BIC: ARI 0.8554, NMI 0.7910, K=3**
- UPGMA: ARI 0.7769, NMI 0.7176, K=4
- Ward: ARI 0.8135, NMI 0.7490, K=3

GMM-BIC performs best. STC performs poorly and severely overclusters with **K=44** instead of K=3.



SHATTER partition
K=44, ARI=0.2364, NMI=0.3548





Report

STC performs reasonably well under low noise but is consistently outperformed by conventional clustering methods on spherical cluster geometry. Its mean ARI is 0.5956, substantially below k-means, GMM, Ward, UPGMA, and DBSCAN. The main limitation is severe over-fragmentation: for a ground-truth $K = 3$, STC predicts 7, 22, and 44 clusters as noise increases. While the low-noise result remains strong, the ARI falls sharply at intermediate and high noise, indicating that STC is less stable than methods whose assumptions better match compact spherical clusters.

Spirals

1. Low noise

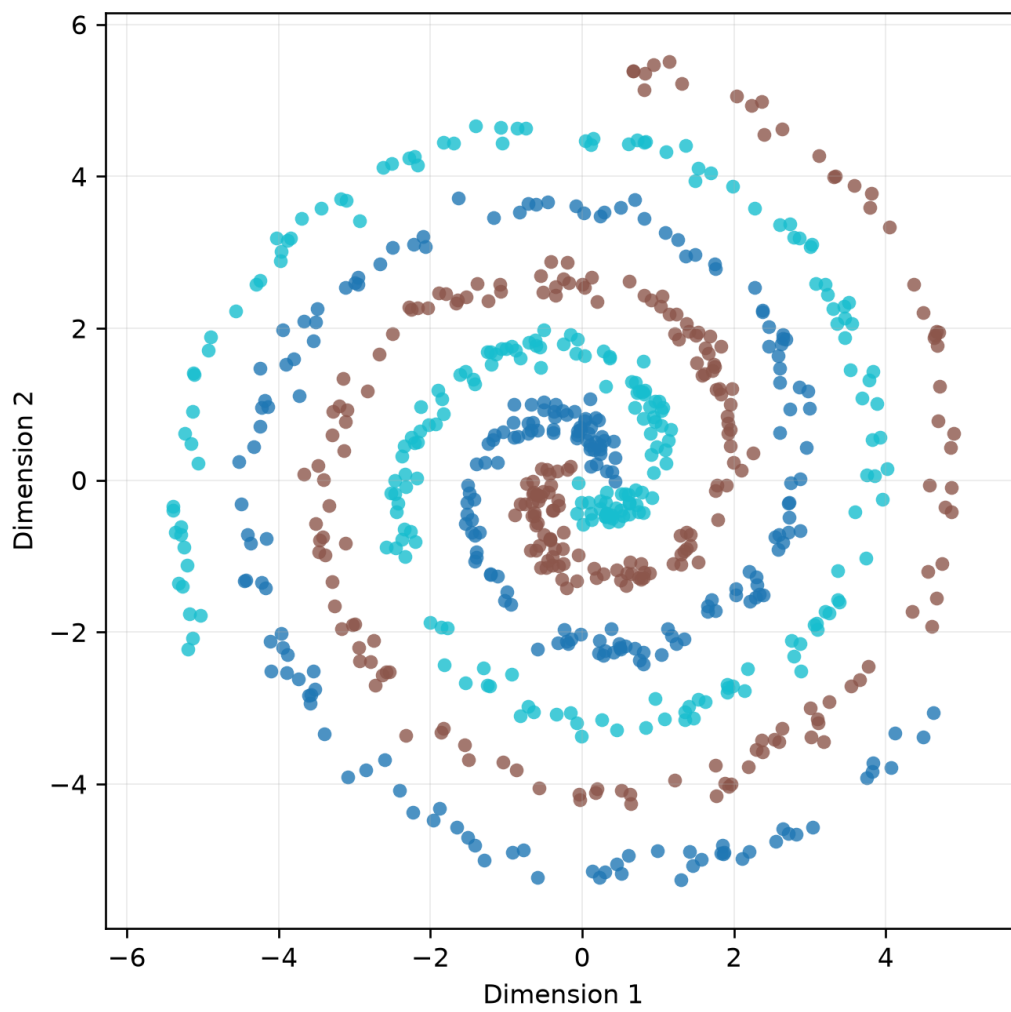
All methods perform poorly on the spiral structure:

True K=3

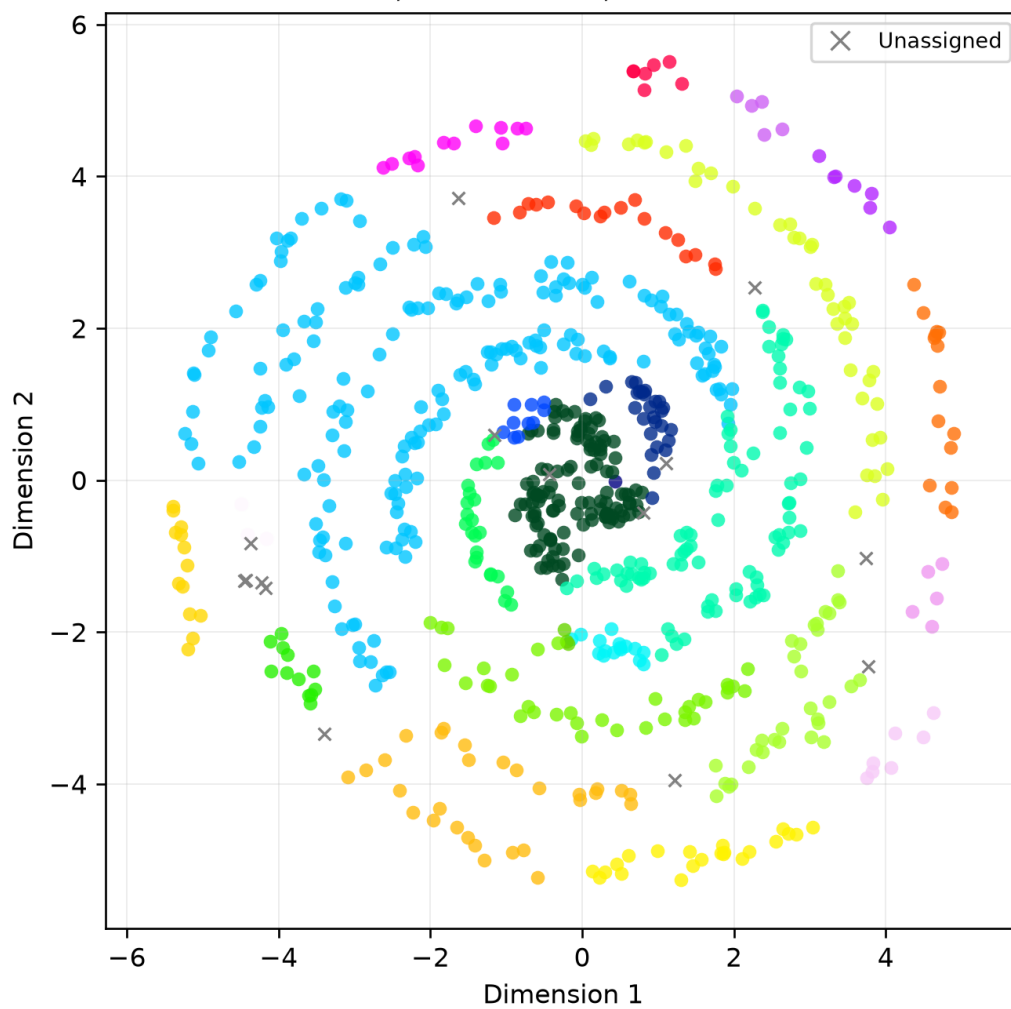
- **STC: ARI 0.0727, NMI 0.2761, K=25**
- DBSCAN: ARI 0.0424, NMI 0.3017, K=38
- **HDBSCAN: ARI 0.0770, NMI 0.3992, K=46**
- OPTICS: ARI 0.0446, NMI 0.3000, K=17
- k-means: ARI 0.0207, NMI 0.0252, K=4
- GMM-BIC: ARI 0.0000, NMI 0.0000, K=1
- UPGMA: ARI 0.0085, NMI 0.0112, K=5
- Ward: ARI 0.0165, NMI 0.0207, K=4

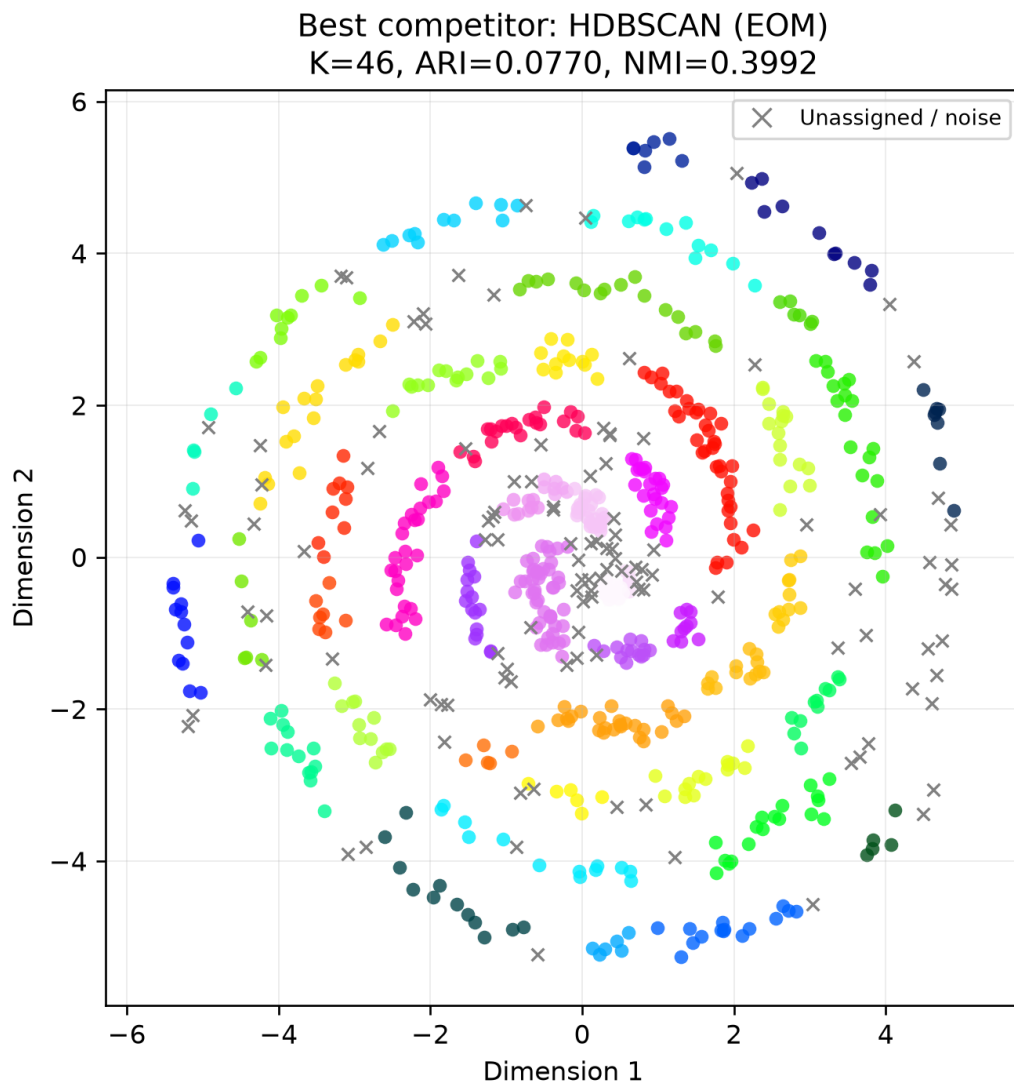
HDBSCAN has the highest ARI, but the result is still very poor. STC also performs poorly and strongly overclusters with **K=25 instead of K=3**.

Ground truth
K=3



SHATTER partition
K=25, ARI=0.0727, NMI=0.2761





2. Intermediate noise

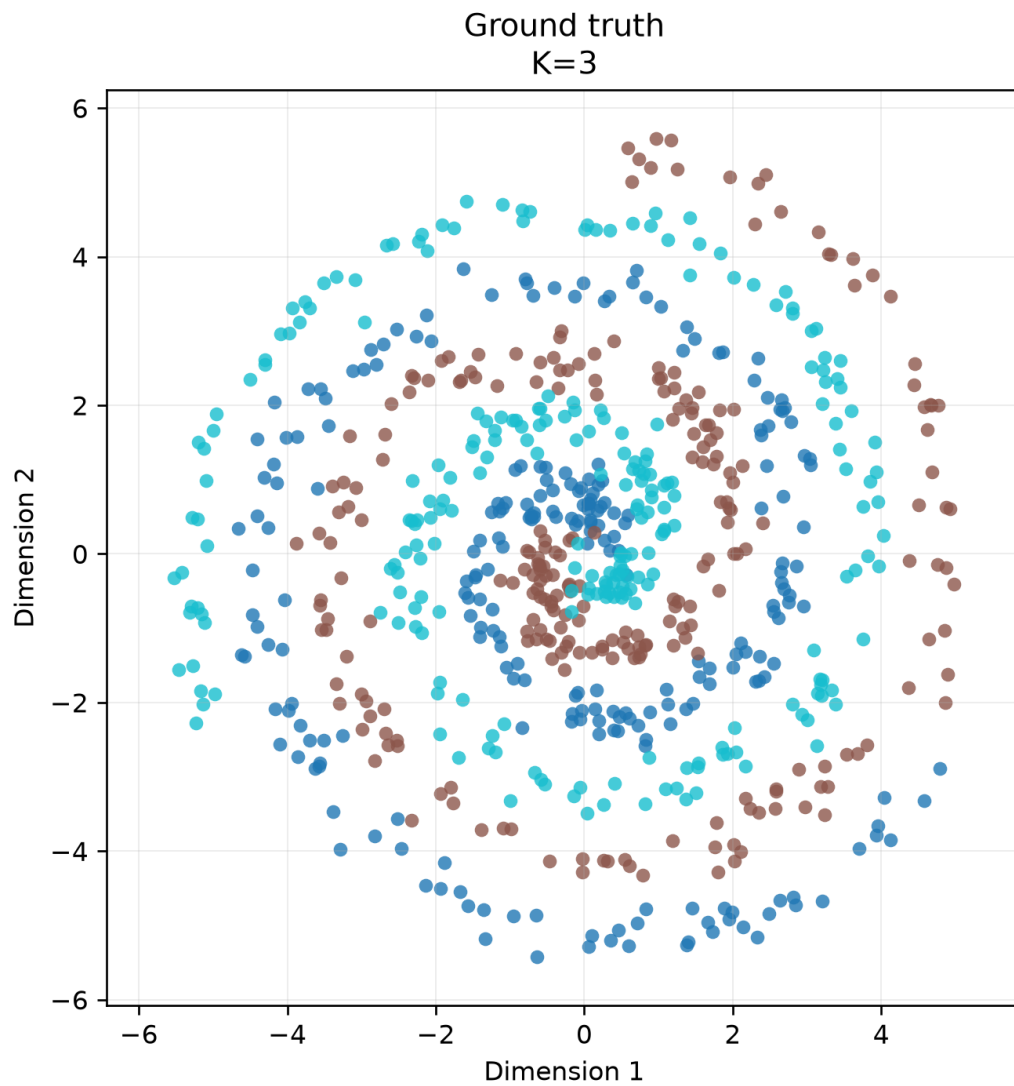
Performance deteriorates further for nearly every method:

True K=3

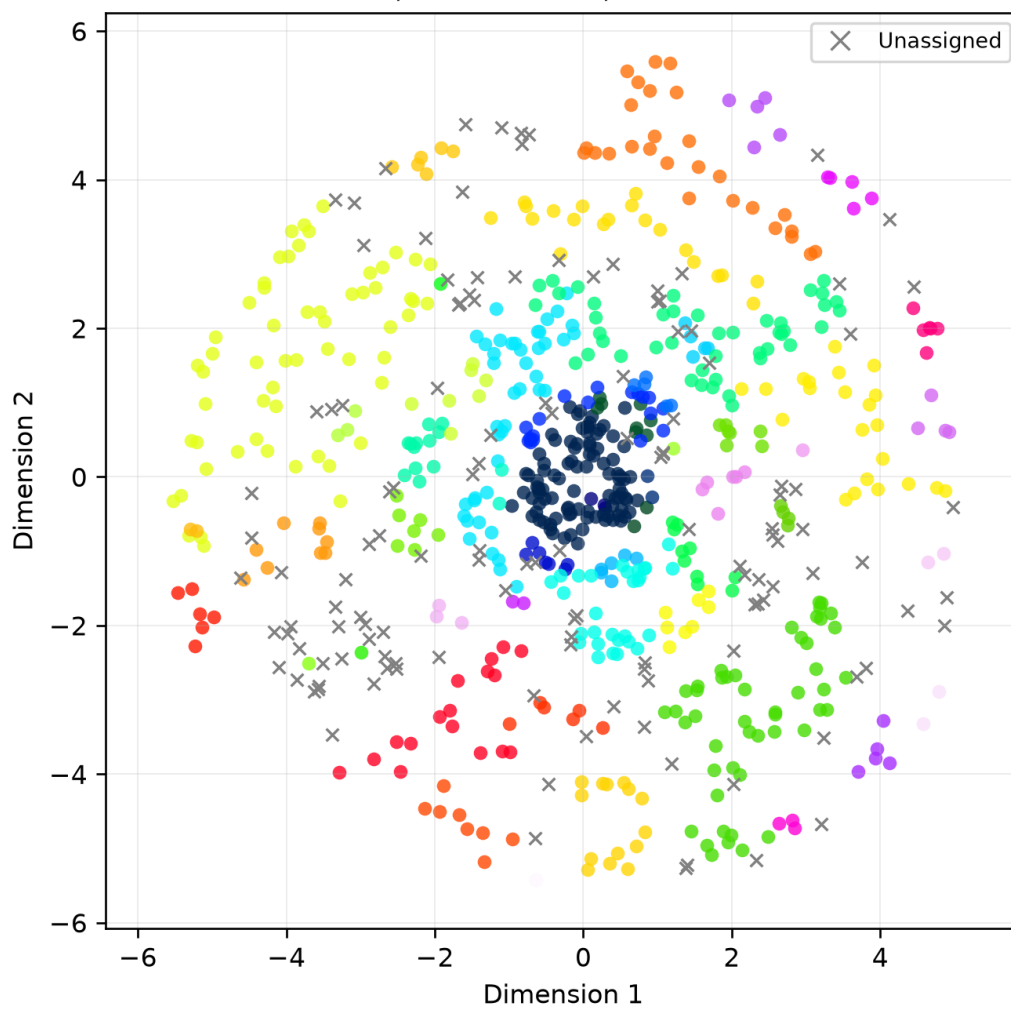
- **STC: ARI 0.0171, NMI 0.1644, K=52**
- DBSCAN: ARI 0.0001, NMI 0.0104, K=2
- **HDBSCAN: ARI 0.0349, NMI 0.2788, K=44**
- OPTICS: ARI 0.0230, NMI 0.2206, K=16

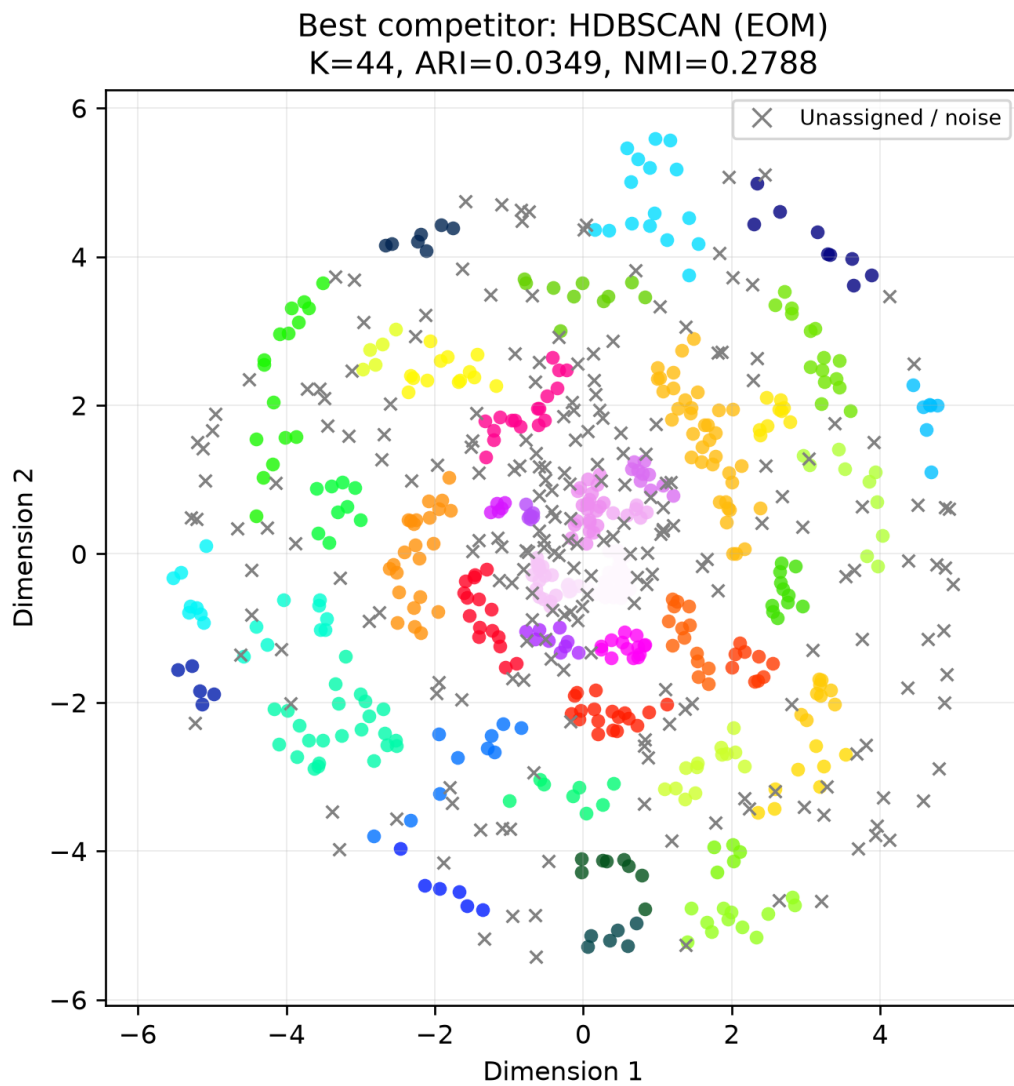
- k-means: ARI 0.0153, NMI 0.0224, K=5
- GMM-BIC: ARI 0.0000, NMI 0.0000, K=1
- UPGMA: ARI 0.0222, NMI 0.0249, K=5
- Ward: ARI 0.0125, NMI 0.0180, K=5

HDBSCAN has the highest ARI, but none of the methods recover the spiral structure well. STC severely overclusters with **K=52 instead of K=3**.



SHATTER partition
K=52, ARI=0.0171, NMI=0.1644





3. High noise

All methods essentially fail to recover the true spiral clusters:

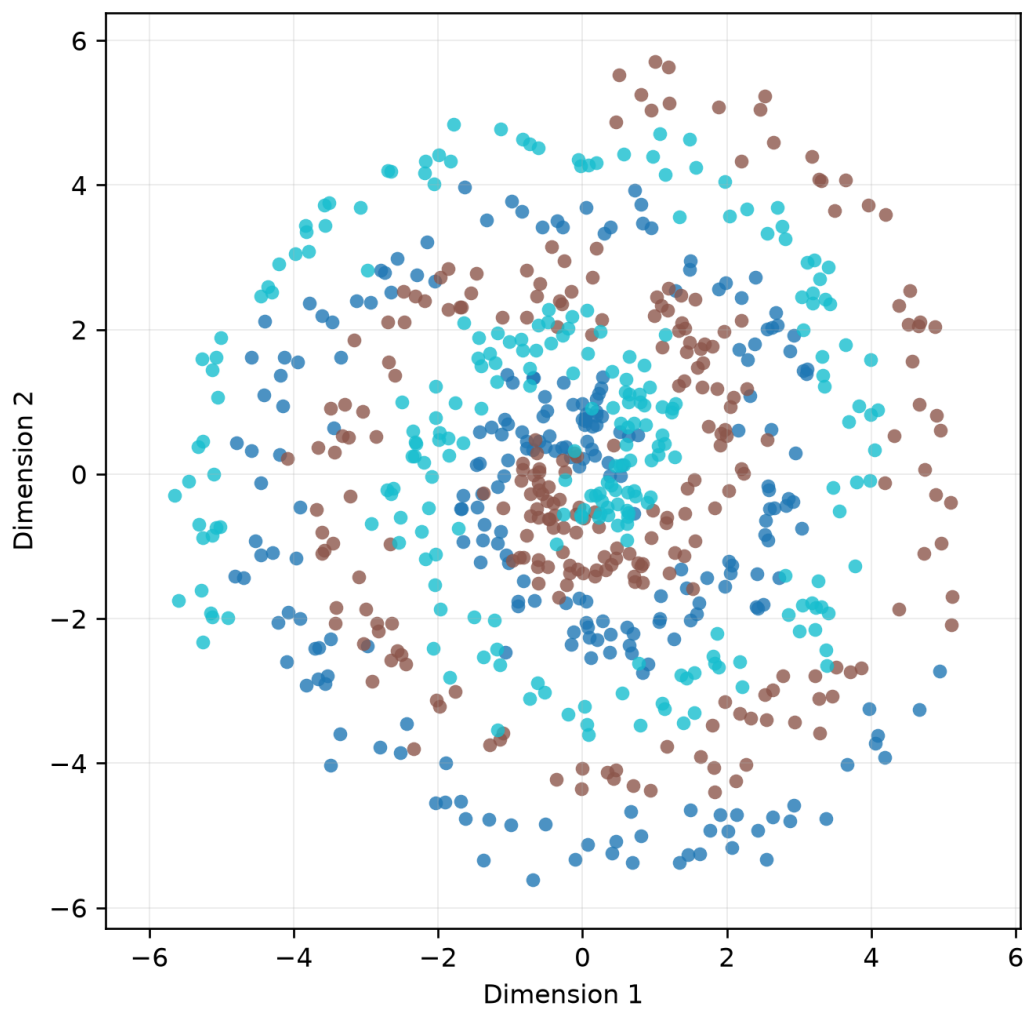
True K=3

- **STC: ARI 0.0122, NMI 0.0461, K=14**
- DBSCAN: ARI 0.0020, NMI 0.0663, K=5
- HDBSCAN: ARI 0.0103, NMI 0.1297, K=28
- **OPTICS: ARI 0.0153, NMI 0.1311, K=16**
- k-means: ARI 0.0136, NMI 0.0204, K=5

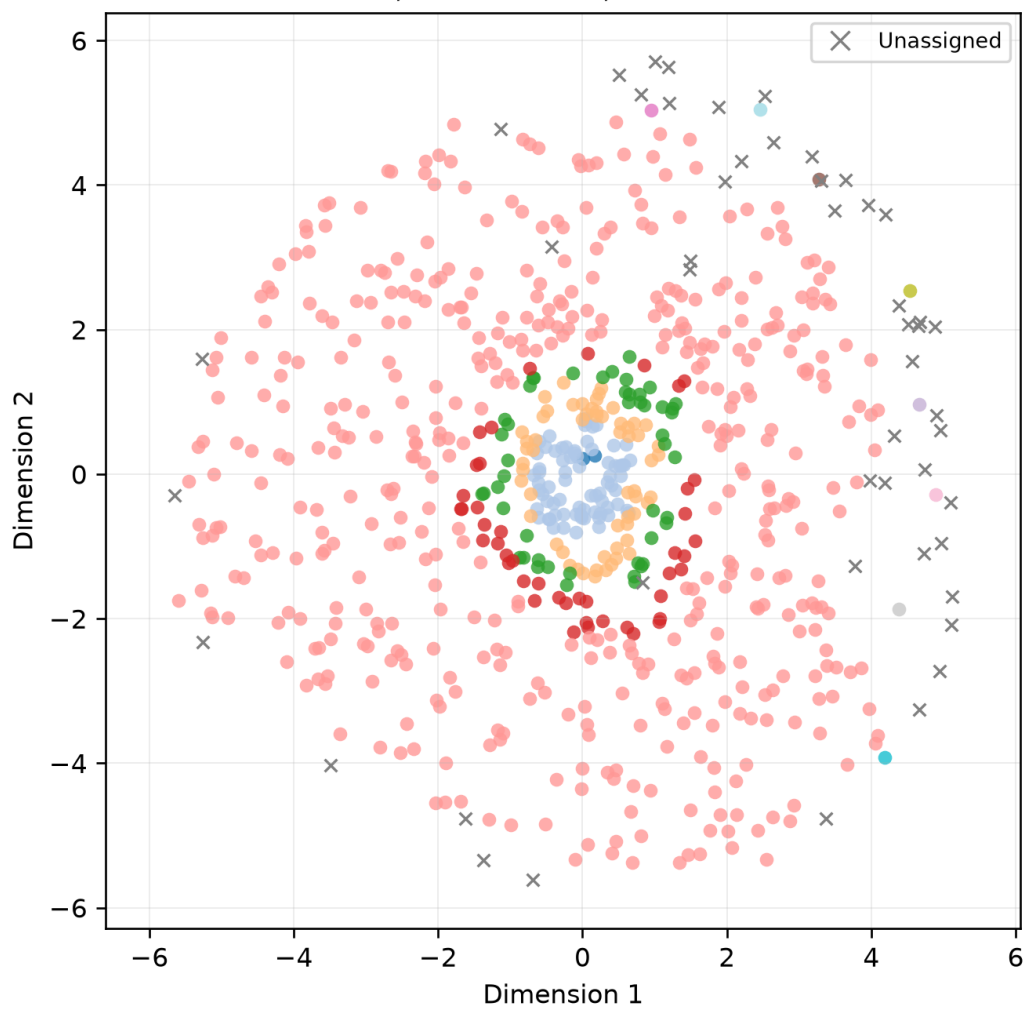
- GMM-BIC: ARI 0.0000, NMI 0.0000, K=1
- UPGMA: ARI 0.0102, NMI 0.0190, K=5
- Ward: ARI 0.0029, NMI 0.0073, K=4

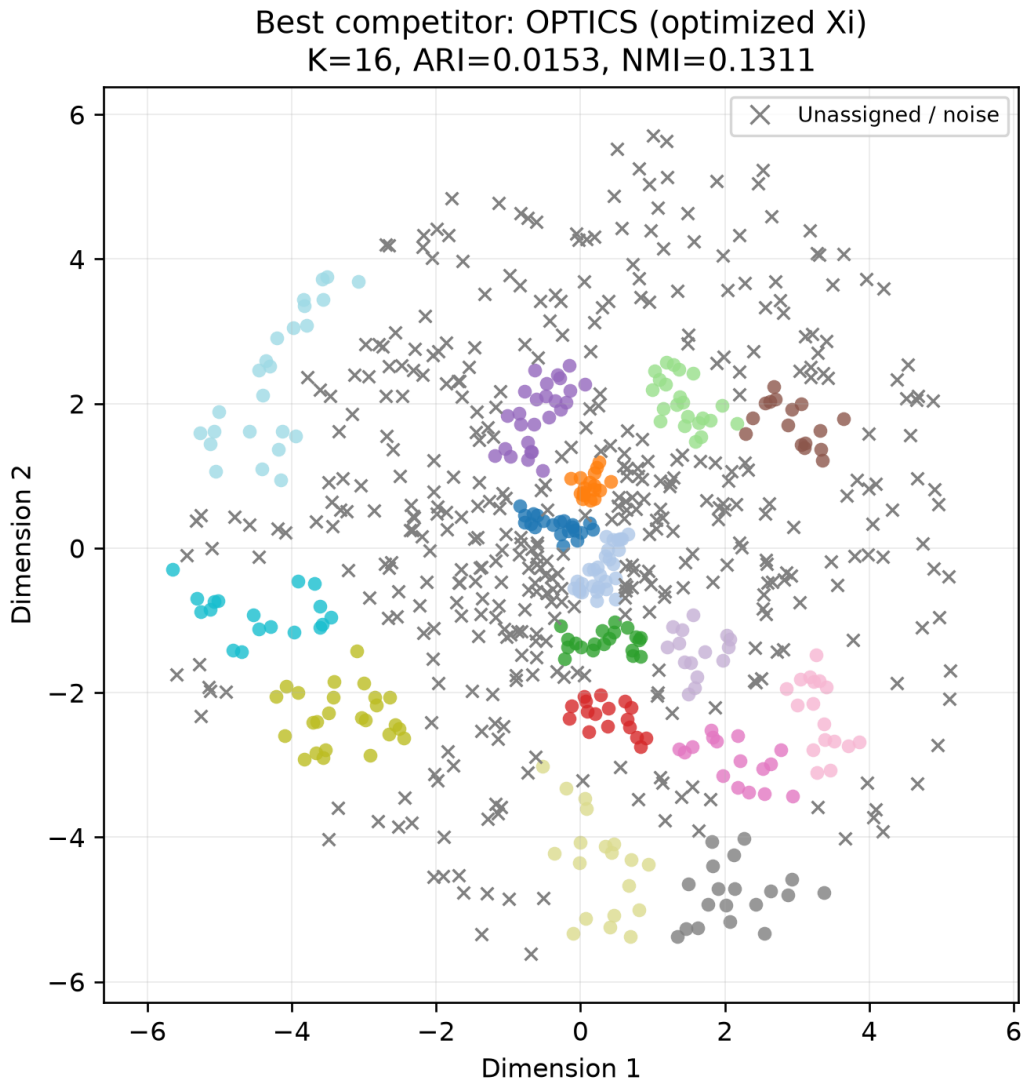
OPTICS has the highest ARI, but the differences are very small, and **all methods perform extremely poorly**. STC also overclusters with **K=14 instead of K=3**.

Ground truth
K=3



SHATTER partition
K=14, ARI=0.0122, NMI=0.0461





Report

STC does not successfully recover the intertwined spiral structure. Its mean ARI is 0.0340, and performance remains close to zero across all three noise levels. However, the same is true for the evaluated baseline methods, none of which achieves meaningful agreement with the known three-cluster partition. STC additionally over-fragments the data, producing 25, 52, and 14 clusters for a ground-truth $K = 3$. The spiral benchmark should therefore be regarded as a common failure case rather than evidence of a specific disadvantage relative to the baselines. The results also show that high internal density coherence can coexist with very low external clustering accuracy, emphasizing the need to evaluate STC with ground-truth metrics such as ARI and NMI.

